

# Benchmark Calculation for the Watts Bar Unit 1 Cycles 1–3 Using the SCALE 6.3/Polaris–PARCS v3.4.2 Code Package



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**December 2024**

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Nuclear Energy and Fuel Cycle Division

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## ABBREVIATIONS

|         |                                                            |
|---------|------------------------------------------------------------|
| BA      | burnable absorber                                          |
| BOC     | beginning of cycle                                         |
| CASL    | Consortium for Advanced Simulation of Light Water Reactors |
| CR      | control rods                                               |
| EFPD    | effective full power day                                   |
| ENDF    | Evaluated Nuclear Data File                                |
| HFP     | hot full power                                             |
| HZP     | hot zero power                                             |
| IFBA    | integral fuel burnable absorber                            |
| Inc-718 | Inconel-718                                                |
| ITC     | isothermal temperature coefficient                         |
| ORNL    | Oak Ridge National Laboratory                              |
| PWR     | pressurized water reactor                                  |
| SS-304  | stainless steel 304                                        |
| TPBAR   | tritium-producing burnable absorber rod                    |
| TVA     | Tennessee Valley Authority                                 |
| WABA    | wet annual burnable absorber                               |
| WBN1    | Watts Bar Unit 1                                           |
| Zr-4    | Zircaloy-4                                                 |

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## ABSTRACT

Benchmark calculations for the Watts Bar Unit 1 cycles 1–3 were performed to validate the SCALE 6.3/Polaris–PARCS v3.4.2 with the ENDF/B-VII.1 56-group library by comparing the simulated results with the measured data. The benchmark results are used in evaluating uncertainties of the SCALE/Polaris–PARCS code package for pressurized water reactor physics analysis for key nuclear parameters such as reactivity, control bank worth, temperature coefficients, and pin and assembly power peaking factors. This report details plant and fuel design specifications and input data for SCALE/Polaris, GenPMAXS, and PARCS. Additional details are provided for the input and output files produced for the benchmark calculations. The benchmark results were summarized such that they can be used in evaluating uncertainties for key nuclear parameters with other benchmark results.

## 1. INTRODUCTION

Watts Bar Unit 1 (WBN1) [1, 2] is a Westinghouse Electric Company four-loop, multicycle, full-core pressurized water reactor (PWR) operated by the Tennessee Valley Authority (TVA). The reactor began operation at 3,411 MW<sub>th</sub> power in 1996. The startup physics testing and power operation for WBN1 has provided valuable benchmarking data that TVA has made publicly available, contributing to the extensive validation basis developed for the Consortium for Advanced Simulation of Light Water Reactors (CASL) [3] VERA core simulator [4]. The benchmark provides measured reactor data for hot zero power (HZP) physics tests, boron letdown curves, and 3D in-core flux maps from 58 instrumented assemblies.

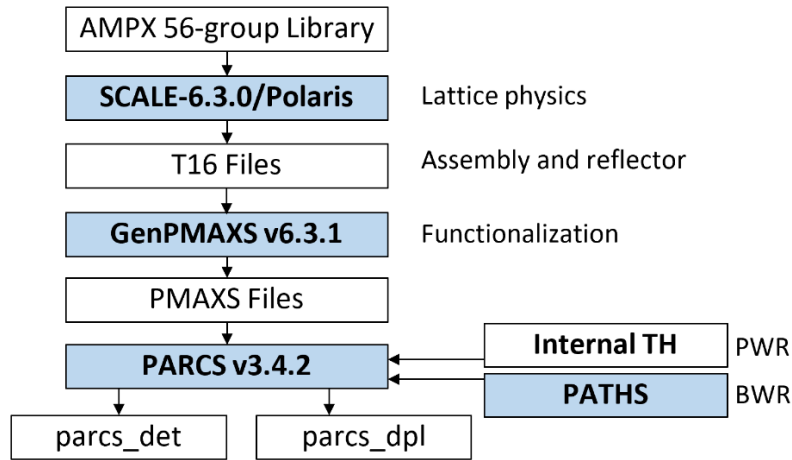
This study performed the benchmark calculations for the WBN1 benchmark problems to validate the SCALE 6.3/Polaris [5, 6]–GenPMAXS v6.3.1 [7]–PARCS v3.4.2 [8] code package for PWR analysis with the ENDF/B-VII.1 AMPX 56-group library [9]. The simulated benchmark results were compared with the measured data, and they are used in the validation of the SCALE 6.3/PARCS–GenPMAXS–PARCS v3.4.2 code package [10] by evaluating uncertainties for key nuclear parameters such as reactivity, control bank work, temperature coefficients, and pin and assembly power peaking factors.

The Polaris–GenPMAXS–PARCS code package is briefly overviewed, and detailed information is provided for the plant design data and measured data in Section 2. Since the measured data were obtained for HZP and hot full power (HFP), the design data need to be thermally expanded to prepare input data for Polaris, GenPMAXS, and PARCS because the code package does not support internal thermal expansion. However, cold dimensions without thermal expansion were used for the HZP and HFP calculations in this study. In Section 3 the simulated benchmark results are summarized and compared with the measured data. Additional details are provided for the input and output files generated from the benchmark calculations. Section 4 discusses the pending issues in Polaris and PARCS, and Section 5 presents the summary and concluding remarks.

## 2. CODE PACKAGE AND DESIGN SPECIFICATION

### 2.1 PROCEDURE AND CODE PACKAGE

Benchmark calculations for the WBN1 cycles 1–3 were performed using the SCALE 6.3/Polaris–GenPMAXS v6.3.1–PARCS v3.4.2 code package with the ENDF/B-VII.1 AMPX 56-group library. Figure 2.1 is a flowchart of the Polaris–GenPMAXS–PARCS code package to simulate PWRs. The SCALE/Polaris code, developed by Oak Ridge National Laboratory, is a 2D lattice physics code to generate 2-group homogenized cross sections and pin power form factors for fuel assemblies and reflectors. The Polaris depletion calculations were performed at the reference states with various branch calculations to cover all the probable states in the reactor. The GenPMAXS code, developed by the University of Michigan, prepares a cross-section table-set for each type of fuel assembly and reflector in which 2-group macroscopic cross sections are tabulated as a function of burnup, boron concentration, moderator and fuel temperatures, moderator density, and control rod insertion. Once cross-section table-sets are prepared for all types of fuel assemblies and radial and axial reflectors, whole-core nodal diffusion calculations are performed using the PARCS code. Typically, a simplified internal thermal-hydraulics model can be used for the PWR physics analysis.



**Figure 2.1. Flowchart of the Polaris–GenPMAXS–PARCS procedure.**

The SCALE 6.3/Polaris, GenPMAXS v6.3.1, and PARCS v3.4.2 code package with the ENDF/B-VII.1 AMPX 56-group library were used in these benchmark calculations. Table 2.1 provides detailed information on the programs and data used in the calculations.

**Table 2.1. Programs used in the benchmark calculations.**

| Program                          | File name                                      | Version | md5sum <sup>a</sup>              | Reference |
|----------------------------------|------------------------------------------------|---------|----------------------------------|-----------|
| SCALE/Polaris                    | scalerte                                       | 6.3.0   | c30aa9d4841d51636ea08f2aa71cf1e5 | [5, 6]    |
| ENDF/B-VII.1<br>56-group library | scale.rev00.xn56g19v7.1                        | 6.3.0   | 2a1ab889c81354758bd85b3c0593db7a | [9]       |
| GenPMAXS                         | genpmaks-v6.3.1-linux2-<br>intel-x64-release.x | 6.3.1   | 5a0428587c6ba65e2028e6dc98d29488 | [7]       |
| PARCS                            | parcs-v342-linux2-intel-<br>x64-release.x      | 3.4.2   | fd005b709d0b597dd6910129d03fe8f5 | [8]       |

<sup>a</sup>Unix program to ensure file integrity

## 2.2 SPECIFICATION OF THE WBN1 CORE

### 2.2.1 Design Data

WBN1 [1, 2] is a Westinghouse Electric Company four-loop, multicycle, full-core PWR operated by the TVA that began operation at 3,411 MW<sub>th</sub> power in 1996. The startup physics testing and power operation for WBN1 has provided valuable benchmarking data that TVA has made publicly available, contributing to the extensive validation basis developed for the CASL [3] VERA core simulator [4]. The benchmark provides measured reactor data for HZP physics tests, boron letdown curves, and 3D in-core flux maps from 58 instrumented assemblies.

Table 2.2 provides key design specifications of the WBN1 core for cycles 1–3. Table 2.3 provides specifications of the WBN1 fuel assemblies and radial and axial reflectors. Since the plant measured data is for the HZP and HFP conditions, the geometry and composition specifications were thermally expanded at the HZP temperature using the thermal expansion coefficients provided in Table 2.4. These thermally expanded data can be used in the Polaris, GenPMAXS, and PARCS inputs. The thermally expanded dimension ( $l_T$ ) at temperature  $T$  can be calculated using

$$l_T = l_{25} \times (1.0 + \Delta T \times \varepsilon), \quad (2.1)$$

where  $\Delta T$  is the temperature difference between 25°C and actual temperature ( $T$ ) and  $\varepsilon$  denotes the thermal expansion coefficient. However, thermal expansion was not considered in this study.

**Table 2.2. Specification of the WBN1 core.**

| Parameter                    | Design data                                                                              | Hot dimension            |
|------------------------------|------------------------------------------------------------------------------------------|--------------------------|
| Core power                   | 3411 MW <sub>th</sub>                                                                    |                          |
| Operating pressure           | 2250 psia (15.51 MPa)                                                                    |                          |
| Core flow rate               | 59.738×10 <sup>6</sup> kg/hr (144.7 Mlbs/hr)                                             |                          |
| Inlet temperature            | 9% bypass<br>565 K (291.85°C)                                                            |                          |
| Number of fuel assemblies    | 193                                                                                      |                          |
| Region 1 (cycle 1)           | 2.11 wt % <sup>235</sup> U                                                               |                          |
| Region 2 (cycle 1)           | 2.619 wt % <sup>235</sup> U                                                              |                          |
| Region 3 (cycle 1)           | 3.10 wt % <sup>235</sup> U                                                               |                          |
| Region 4A (cycle 2)          | 3.709 wt % <sup>235</sup> U                                                              |                          |
| Region 5A (cycle 3)          | 3.80 wt % <sup>235</sup> U                                                               |                          |
| Region 5B (cycle 3)          | 4.40 wt % <sup>235</sup> U                                                               |                          |
| Pin lattice configuration    | 17 × 17                                                                                  |                          |
| Active fuel length           | 365.76 cm                                                                                | 367.6833 cm              |
| Number of fuel rods          | 264                                                                                      |                          |
| Number of grid spacers       | 8 (6 Zircaloy, 2 Inconel-718)                                                            |                          |
| Assembly pitch               | 21.50 cm                                                                                 | 21.60491 cm              |
| Pin pitch                    | 1.26 cm                                                                                  | 1.26213 cm               |
| Fuel pellet radius           | 0.4096 cm                                                                                | 0.41175 cm               |
| Cladding inner/outer radius  | 0.4180/0.4750 cm                                                                         | 0.41871/0.47580 cm       |
| Number of control banks      | 57                                                                                       |                          |
| Control rod material (upper) | B <sub>4</sub> C                                                                         |                          |
| Control rod material (lower) | AgInCd                                                                                   |                          |
| Burnable poison material     | Borosilicate glass, 12.5 w/o B <sub>2</sub> O <sub>3</sub><br>IFBA and WABA <sup>a</sup> | Cycle 1<br>Cycle 2 and 3 |

<sup>a</sup> Integral fuel burnable absorber and wet annular burnable absorber.

Table 2.3. Specification of the WBN1 fuel assemblies and reflectors.

| Item                          | Material                                                | Density<br>(g/cm <sup>3</sup> ) | Subitem                     | Thermal exp.<br>Coeff. (°C <sup>-1</sup> ) | Dimension (cm) |          |
|-------------------------------|---------------------------------------------------------|---------------------------------|-----------------------------|--------------------------------------------|----------------|----------|
|                               |                                                         |                                 |                             |                                            | Cold           | Hot      |
| Active height                 | UO <sub>2</sub>                                         |                                 |                             | 1.052E-05                                  | 365.76         | 367.6833 |
| Assembly pitch                | SS-304                                                  |                                 |                             | 1.730E-05                                  | 21.50          | 21.60491 |
| Pin pitch                     | Zr-4                                                    | 6.55                            |                             | 6.000E-06                                  | 1.26           | 1.26213  |
| Assembly gap                  |                                                         |                                 |                             |                                            | 0.25           | 0.1487   |
| Fuel                          | UO <sub>2</sub>                                         |                                 | Outer radius (cm)           | 1.052E-05                                  | 0.40960        | 0.41175  |
| IFBA                          | ZrB <sub>2</sub>                                        | 3.85                            | Outer radius (cm)           |                                            | 0.4106         |          |
| Pyrex                         | He                                                      |                                 | Outer radius (cm)           |                                            | 0.2140         |          |
|                               | SS-304                                                  | 8.03                            |                             |                                            | 0.2310         |          |
|                               | He                                                      |                                 |                             |                                            | 0.2410         |          |
|                               | B <sub>2</sub> O <sub>3</sub> -SiO <sub>2</sub>         | 2.23                            |                             |                                            | 0.4270         |          |
|                               | He                                                      |                                 |                             |                                            | 0.4370         |          |
|                               | SS-304                                                  | 8.03                            |                             |                                            | 0.4840         |          |
| WABA                          | H <sub>2</sub> O                                        |                                 | Outer radius (cm)           |                                            | 0.2860         |          |
|                               | Zr-4                                                    |                                 |                             |                                            | 0.3390         |          |
|                               | He                                                      |                                 |                             |                                            | 0.3530         |          |
|                               | B <sub>4</sub> C-Al <sub>2</sub> O <sub>3</sub>         | 3.65                            |                             |                                            | 0.4040         |          |
|                               | He                                                      |                                 |                             |                                            | 0.4180         |          |
|                               | Zr-4                                                    |                                 |                             |                                            | 0.4840         |          |
| TPBAR                         | He                                                      |                                 | Outer radius (cm)           |                                            | 0.2832         |          |
|                               | LiAlO <sub>2</sub>                                      | 1.57571                         |                             |                                            | 0.3835         |          |
|                               | SS-304                                                  |                                 |                             |                                            | 0.4840         |          |
| Clad                          | Zr-4                                                    | 6.55                            | Inner radius (cm)           | 6.000E-06                                  | 0.4180         | 0.41871  |
|                               |                                                         |                                 | Outer radius (cm)           | 6.000E-06                                  | 0.4570         | 0.45777  |
| Instrument tube               | Zr-4                                                    | 6.55                            | Inner radius(cm)            | 6.000E-06                                  | 0.5590         | 0.55995  |
|                               |                                                         |                                 | Outer radius(cm)            | 6.000E-06                                  | 0.6050         | 0.60602  |
| Guide tube                    | Zr-4                                                    | 6.55                            | Inner radius(cm)            | 6.000E-06                                  | 0.5610         | 0.56195  |
|                               |                                                         |                                 | Outer radius(cm)            | 6.000E-06                                  | 0.6020         | 0.60302  |
| Spacer grid<br>(top/bottom)   | Inc-718                                                 | 8.2                             | Inc-718 weight (g)          |                                            | 390.136        |          |
|                               | SS-304 (sleeve)                                         | 8.03                            | SS-304 weight (g)           |                                            | 91.0329        |          |
|                               |                                                         |                                 | Height (cm)                 |                                            | 3.358          |          |
|                               |                                                         |                                 | Volume (g/cm <sup>3</sup> ) |                                            | 58.9142        |          |
| Spacer grid<br>(intermediate) | Zr-4                                                    | 6.55                            | Zr-4 weight (g)             |                                            | 1169.23        |          |
|                               |                                                         |                                 | Height (cm)                 |                                            | 5.715          |          |
|                               |                                                         |                                 | Volume (cm <sup>3</sup> )   |                                            | 178.5084       |          |
| Radial reflector              | SS-304                                                  | 8.03                            | Water gap (cm)              |                                            | 0.1627         |          |
|                               |                                                         |                                 | Baffle thick. (cm)          |                                            | 2.2225         |          |
|                               |                                                         |                                 | Water                       |                                            | -              |          |
| Top reflector                 | Zr-4(29*), SS-304(55), H <sub>2</sub> O(16)             |                                 | Plenum                      |                                            | 8.556          |          |
|                               | Zr-4(23), SS-304(45), Inc-718(20), H <sub>2</sub> O(12) |                                 | Plenum + Grid               |                                            | 3.866          |          |
|                               | Zr-4(29), SS-304(55), H <sub>2</sub> O(16)              |                                 | Plenum                      |                                            | 3.578          |          |
|                               | Zr-4(88), H <sub>2</sub> O(12)                          |                                 | Plug                        |                                            | 1.670          |          |
|                               | Zr-4(7), H <sub>2</sub> O(93)                           |                                 | Nozzle + Gap                |                                            | 2.129          |          |
|                               | SS-304(76), H <sub>2</sub> O(24)                        |                                 | Top nozzle                  |                                            | 8.827          |          |
|                               | SS-304(93), H <sub>2</sub> O(7)                         |                                 | Support plate               |                                            | 7.600          |          |
| Bottom reflector              | Zr-4(44), Inc-718(21), H <sub>2</sub> O(35)             |                                 | Gap                         |                                            | 5.898          |          |
|                               | SS-304(80), H <sub>2</sub> O(20)                        |                                 | Bottom nozzle               |                                            | 6.053          |          |
|                               | SS-304(91), H <sub>2</sub> O(20)                        |                                 | Support plate               |                                            | 5.000          |          |
|                               | Zr-4(44), Inc-718(21), H <sub>2</sub> O(35)             |                                 | Gap                         |                                            | 20.000         |          |

**Table 2.4. Thermal expansion coefficients.**

| Material        | Thermal expansion coefficient (°C <sup>-1</sup> ) | Reference                                                                                                                                                 |
|-----------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zircaloy-4      | 6.0E-6                                            | <a href="https://www.azom.com/article.aspx?ArticleID=7644">https://www.azom.com/article.aspx?ArticleID=7644</a>                                           |
| SS-304          | 17.3E-6                                           | <a href="http://esmat.esa.int/Services/Preferred_Lists/Materials_Lists/a62.htm">http://esmat.esa.int/Services/Preferred_Lists/Materials_Lists/a62.htm</a> |
| UO <sub>2</sub> | 10.52E-6                                          | J. Chem. Phys. 30, 1524 (1959) <a href="https://doi.org/10.1063/1.1730230">https://doi.org/10.1063/1.1730230</a>                                          |
| Inc-718         | 13.0E-6                                           | <a href="https://www.espimetals.com/index.php/technical-data/91-inconel-718">https://www.espimetals.com/index.php/technical-data/91-inconel-718</a>       |

The following rules can be used in obtaining thermally expanded dimensions.

- Eq. (2.1) was used for x-, y-, and radial-expansions.
- Hot pin pitch was determined by expanding the Zircaloy-4 (Zr-4) spacer grid material.
- Hot assembly pitch was determined by expanding the SS-304 bottom plate material.
- Active height was determined by expanding UO<sub>2</sub>. When expanding UO<sub>2</sub>, the total UO<sub>2</sub> mass should be conserved.
- Thermal expansion was not applied to spacer grids, control rod materials, and burnable poison materials because they will not change moderator volume.

Even though the thermally expanded dimensions were not used in this study, the termally expanded data are included here for future use.

**Table 2.5. Atomic number densities for UO<sub>2</sub> fuels.**

| Fuel type | <sup>235</sup> U wt % |          | Density (g/cm <sup>3</sup> ) |         | Mass (kg U) | Nuclide          | ID    | Atomic number density (#/barn-cm) |             |
|-----------|-----------------------|----------|------------------------------|---------|-------------|------------------|-------|-----------------------------------|-------------|
|           | Nominal               | As-built | cold                         | hot     |             |                  |       | Cold                              | Hot         |
| 2.10_bp00 | 2.10                  | 2.11     | 10.257                       | 10.0966 | 460.1519    | <sup>234</sup> U | 92234 | 4.04814E-06                       | 3.9849E-06  |
|           |                       |          |                              |         |             | <sup>235</sup> U | 92235 | 4.88010E-04                       | 4.8039E-04  |
|           |                       |          |                              |         |             | <sup>236</sup> U | 92236 | 2.23756E-06                       | 2.2026E-06  |
|           |                       |          |                              |         |             | <sup>238</sup> U | 92238 | 2.23844E-02                       | 2.2035E-02  |
|           |                       |          |                              |         |             | <sup>16</sup> O  | 8016  | 4.57591E-02                       | 4.5045E-02  |
| 2.60_bp00 | 2.60                  | 2.619    | 10.257                       | 10.0969 | 460.1656    | <sup>234</sup> U | 92234 | 5.09503E-06                       | 5.0155E-06  |
|           |                       |          |                              |         |             | <sup>235</sup> U | 92235 | 6.06733E-04                       | 5.9726E-04  |
|           |                       |          |                              |         |             | <sup>236</sup> U | 92236 | 2.76809E-06                       | 2.7249E-06  |
|           |                       |          |                              |         |             | <sup>238</sup> U | 92238 | 2.22663E-02                       | 2.1919E-02  |
|           |                       |          |                              |         |             | <sup>16</sup> O  | 8016  | 4.57617E-02                       | 4.5047E-02  |
| 3.10_bp00 | 3.10                  | 3.10     | 10.257                       | 10.0969 | 460.1620    | <sup>234</sup> U | 92234 | 6.11864E-06                       | 6.02312E-06 |
|           |                       |          |                              |         |             | <sup>235</sup> U | 92235 | 7.18132E-04                       | 7.06921E-04 |
|           |                       |          |                              |         |             | <sup>236</sup> U | 92236 | 3.29861E-06                       | 3.24712E-06 |
|           |                       |          |                              |         |             | <sup>238</sup> U | 92238 | 2.21546E-02                       | 2.18087E-02 |
|           |                       |          |                              |         |             | <sup>16</sup> O  | 8016  | 4.57642E-02                       | 4.50498E-02 |
| 3.70_bp00 | 3.70                  | 3.709    | 10.257                       | 10.0969 | 460.1570    | <sup>234</sup> U | 92234 | 7.44448E-06                       | 7.32827E-06 |
|           |                       |          |                              |         |             | <sup>235</sup> U | 92235 | 8.59201E-04                       | 8.45788E-04 |
|           |                       |          |                              |         |             | <sup>236</sup> U | 92236 | 3.93555E-06                       | 3.87412E-06 |
|           |                       |          |                              |         |             | <sup>238</sup> U | 92238 | 2.20131E-02                       | 2.16695E-02 |
|           |                       |          |                              |         |             | <sup>16</sup> O  | 8016  | 4.57674E-02                       | 4.50529E-02 |
| 3.80_bp00 | 3.80                  | 3.807    | 10.257                       | 10.0969 | 460.1563    | <sup>234</sup> U | 92234 | 7.65787E-06                       | 7.53832E-06 |
|           |                       |          |                              |         |             | <sup>235</sup> U | 92235 | 8.81902E-04                       | 8.68135E-04 |
|           |                       |          |                              |         |             | <sup>236</sup> U | 92236 | 4.03953E-06                       | 3.97647E-06 |
|           |                       |          |                              |         |             | <sup>238</sup> U | 92238 | 2.19904E-02                       | 2.16471E-02 |
|           |                       |          |                              |         |             | <sup>16</sup> O  | 8016  | 4.57679E-02                       | 4.50535E-02 |
| 4.40_bp00 | 4.40                  | 4.401    | 10.257                       | 10.0969 | 460.1520    | <sup>234</sup> U | 92234 | 8.96072E-06                       | 8.82084E-06 |
|           |                       |          |                              |         |             | <sup>235</sup> U | 92235 | 1.01949E-03                       | 1.00358E-03 |
|           |                       |          |                              |         |             | <sup>236</sup> U | 92236 | 4.66977E-06                       | 4.59687E-06 |
|           |                       |          |                              |         |             | <sup>238</sup> U | 92238 | 2.18524E-02                       | 2.15113E-02 |
|           |                       |          |                              |         |             | <sup>16</sup> O  | 8016  | 4.57710E-02                       | 4.50565E-02 |

Table 2.6. Atomic number densities for structure materials.

| Material                   | Density (g/cm <sup>3</sup> ) |             | Nuclide           | ID    | Atomic number density (#/barn-cm) |             |
|----------------------------|------------------------------|-------------|-------------------|-------|-----------------------------------|-------------|
|                            | cold                         | hot         |                   |       | Cold                              | Hot         |
| Zr-4                       | 6.55                         | 6.51657     | <sup>16</sup> O   | 8016  | 3.07430E-04                       | 3.05866E-04 |
|                            |                              |             | <sup>17</sup> O   | 8017  | 1.17110E-07                       | 1.16514E-07 |
|                            |                              |             | <sup>18</sup> O   | 8018  | 6.31760E-07                       | 6.28547E-07 |
|                            |                              |             | <sup>50</sup> Cr  | 24050 | 3.29620E-06                       | 3.27943E-06 |
|                            |                              |             | <sup>52</sup> Cr  | 24052 | 6.35640E-05                       | 6.32407E-05 |
|                            |                              |             | <sup>53</sup> Cr  | 24053 | 7.20760E-06                       | 7.17094E-06 |
|                            |                              |             | <sup>54</sup> Cr  | 24054 | 1.79410E-06                       | 1.78497E-06 |
|                            |                              |             | <sup>54</sup> Fe  | 26054 | 8.66990E-06                       | 8.62580E-06 |
|                            |                              |             | <sup>56</sup> Fe  | 26056 | 1.36100E-04                       | 1.35408E-04 |
|                            |                              |             | <sup>57</sup> Fe  | 26057 | 3.14310E-06                       | 3.12711E-06 |
|                            |                              |             | <sup>58</sup> Fe  | 26058 | 4.18290E-07                       | 4.16162E-07 |
|                            |                              |             | <sup>90</sup> Zr  | 40090 | 2.18270E-02                       | 2.17160E-02 |
|                            |                              |             | <sup>91</sup> Zr  | 40091 | 4.76000E-03                       | 4.73579E-03 |
|                            |                              |             | <sup>92</sup> Zr  | 40092 | 7.27580E-03                       | 7.23879E-03 |
|                            |                              |             | <sup>94</sup> Zr  | 40094 | 7.37340E-03                       | 7.33590E-03 |
|                            |                              |             | <sup>96</sup> Zr  | 40096 | 1.18790E-03                       | 1.18186E-03 |
|                            |                              |             | <sup>112</sup> Sn | 50112 | 4.67350E-06                       | 4.64973E-06 |
|                            |                              |             | <sup>114</sup> Sn | 50114 | 3.17990E-06                       | 3.16373E-06 |
|                            |                              |             | <sup>115</sup> Sn | 50115 | 1.63810E-06                       | 1.62977E-06 |
|                            |                              |             | <sup>116</sup> Sn | 50116 | 7.00550E-05                       | 6.96987E-05 |
|                            |                              |             | <sup>117</sup> Sn | 50117 | 3.70030E-05                       | 3.68148E-05 |
|                            |                              |             | <sup>118</sup> Sn | 50118 | 1.16690E-04                       | 1.16096E-04 |
|                            |                              |             | <sup>119</sup> Sn | 50119 | 4.13870E-05                       | 4.11765E-05 |
|                            |                              |             | <sup>120</sup> Sn | 50120 | 1.56970E-04                       | 1.56172E-04 |
|                            |                              |             | <sup>122</sup> Sn | 50122 | 2.23080E-05                       | 2.21945E-05 |
|                            |                              |             | <sup>124</sup> Sn | 50124 | 2.78970E-05                       | 2.77551E-05 |
| Inconel 718                | 8.2                          | Inconel 718 | <sup>50</sup> Cr  | 24050 | 7.8239e-04                        |             |
|                            |                              |             | <sup>52</sup> Cr  | 24052 | 1.5088e-02                        |             |
|                            |                              |             | <sup>53</sup> Cr  | 24053 | 1.7108e-03                        |             |
|                            |                              |             | <sup>54</sup> Cr  | 24054 | 4.2586e-04                        |             |
|                            |                              |             | <sup>54</sup> Fe  | 26054 | 1.4797e-03                        |             |
|                            |                              |             | <sup>56</sup> Fe  | 26056 | 2.3229e-02                        |             |
|                            |                              |             | <sup>57</sup> Fe  | 26057 | 5.3645e-04                        |             |
|                            |                              |             | <sup>58</sup> Fe  | 26058 | 7.1392e-05                        |             |
|                            |                              |             | <sup>55</sup> Mn  | 25055 | 7.8201e-04                        |             |
|                            |                              |             | <sup>58</sup> Ni  | 28058 | 2.9320e-02                        |             |
|                            |                              |             | <sup>60</sup> Ni  | 28060 | 1.1294e-02                        |             |
|                            |                              |             | <sup>61</sup> Ni  | 28061 | 4.9094e-04                        |             |
|                            |                              |             | <sup>62</sup> Ni  | 28062 | 1.5653e-03                        |             |
|                            |                              |             | <sup>64</sup> Ni  | 28064 | 3.9864e-04                        |             |
|                            |                              |             | <sup>28</sup> Si  | 14028 | 5.6757e-04                        |             |
| Borosilicate glass (Pyrex) | 2.26                         |             | <sup>29</sup> Si  | 14029 | 2.8820e-05                        |             |
|                            |                              |             | <sup>30</sup> Si  | 14030 | 1.8998e-05                        |             |
|                            |                              |             | <sup>27</sup> Al  | 13027 | 1.7352e-03                        |             |
|                            |                              |             | <sup>10</sup> B   | 5010  | 9.6506e-04                        |             |
|                            |                              |             | <sup>11</sup> B   | 5011  | 3.9189e-03                        |             |
|                            |                              |             | <sup>16</sup> O   | 8016  | 4.6514e-02                        |             |
|                            |                              |             | <sup>17</sup> O   | 8017  | 1.7671e-05                        |             |
| IFBA                       | 3.85                         |             | <sup>18</sup> O   | 8018  | 9.3268e-05                        |             |
|                            |                              |             | <sup>28</sup> Si  | 14028 | 1.6926e-02                        |             |
|                            |                              |             | <sup>29</sup> Si  | 14029 | 8.5944e-04                        |             |
|                            |                              |             | <sup>30</sup> Si  | 14030 | 5.6654e-04                        |             |
|                            |                              |             | <sup>10</sup> B   | 5010  | 2.16410e-02                       |             |
|                            |                              |             | <sup>11</sup> B   | 5011  | 1.96824e-02                       |             |
|                            |                              |             | Zr                | 40000 | 2.06617e-02                       |             |
|                            |                              |             |                   |       |                                   |             |
|                            |                              |             |                   |       |                                   |             |

**Table 2.6. Atomic number densities for structure materials (continued).**

| Material         | Density (g/cm <sup>3</sup> ) |     | Nuclide           | ID    | Atomic number density (#/barn-cm) |     |
|------------------|------------------------------|-----|-------------------|-------|-----------------------------------|-----|
|                  | cold                         | hot |                   |       | Cold                              | Hot |
| WABA             | 3.65                         |     | <sup>10</sup> B   | 5010  | 2.98553e-03                       |     |
|                  |                              |     | <sup>11</sup> B   | 5011  | 1.21192e-02                       |     |
|                  |                              |     | C                 | 6000  | 3.77001e-03                       |     |
|                  |                              |     | <sup>27</sup> Al  | 13027 | 3.90223e-02                       |     |
|                  |                              |     | <sup>16</sup> O   | 8016  | 5.85563e-02                       |     |
| TPBAR            | 1.57571                      |     | <sup>6</sup> Li   | 3006  | 4.63439E-03                       |     |
|                  |                              |     | <sup>7</sup> Li   | 3007  | 9.87705E-03                       |     |
|                  |                              |     | <sup>27</sup> Al  | 13027 | 1.44413E-02                       |     |
|                  |                              |     | <sup>16</sup> O   | 8016  | 2.88900E-02                       |     |
| AgInCd           | 10.16                        |     | <sup>107</sup> Ag | 47107 | 2.3523e-02                        |     |
|                  |                              |     | <sup>109</sup> Ag | 47109 | 2.1854e-02                        |     |
|                  |                              |     | <sup>106</sup> Cd | 48106 | 3.3882e-05                        |     |
|                  |                              |     | <sup>108</sup> Cd | 48108 | 2.4166e-05                        |     |
|                  |                              |     | <sup>110</sup> Cd | 48110 | 3.3936e-04                        |     |
|                  |                              |     | <sup>111</sup> Cd | 48111 | 3.4821e-04                        |     |
|                  |                              |     | <sup>112</sup> Cd | 48112 | 6.5611e-04                        |     |
|                  |                              |     | <sup>113</sup> Cd | 48113 | 3.3275e-04                        |     |
|                  |                              |     | <sup>114</sup> Cd | 48114 | 7.8252e-04                        |     |
|                  |                              |     | <sup>116</sup> Cd | 48116 | 2.0443e-04                        |     |
|                  |                              |     | <sup>113</sup> In | 49113 | 3.4219e-04                        |     |
| B <sub>4</sub> C | 1.76                         |     | <sup>10</sup> B   | 5010  | 1.5206e-02                        |     |
|                  |                              |     | <sup>11</sup> B   | 5011  | 6.1514e-02                        |     |
|                  |                              |     | <sup>12</sup> C   | 6012  | 1.8972e-02                        |     |
|                  |                              |     | <sup>13</sup> C   | 6013  | 2.1252e-04                        |     |
| SS 304           | 8.03                         |     | <sup>50</sup> Cr  | 24050 | 7.6778e-04                        |     |
|                  |                              |     | <sup>52</sup> Cr  | 24052 | 1.4806e-02                        |     |
|                  |                              |     | <sup>53</sup> Cr  | 24053 | 1.6789e-03                        |     |
|                  |                              |     | <sup>54</sup> Cr  | 24054 | 4.1791e-04                        |     |
|                  |                              |     | <sup>54</sup> Fe  | 26054 | 3.4620e-03                        |     |
|                  |                              |     | <sup>56</sup> Fe  | 26056 | 5.4345e-02                        |     |
|                  |                              |     | <sup>57</sup> Fe  | 26057 | 1.2551e-03                        |     |
|                  |                              |     | <sup>58</sup> Fe  | 26058 | 1.6703e-04                        |     |
|                  |                              |     | <sup>55</sup> Mn  | 25055 | 1.7604e-03                        |     |
|                  |                              |     | <sup>58</sup> Ni  | 28058 | 5.6089e-03                        |     |
|                  |                              |     | <sup>60</sup> Ni  | 28060 | 2.1605e-03                        |     |
|                  |                              |     | <sup>61</sup> Ni  | 28061 | 9.3917e-05                        |     |
|                  |                              |     | <sup>62</sup> Ni  | 28062 | 2.9945e-04                        |     |
|                  |                              |     | <sup>64</sup> Ni  | 28064 | 7.6261e-05                        |     |
|                  |                              |     | <sup>28</sup> Si  | 14028 | 9.5281e-04                        |     |
|                  |                              |     | <sup>29</sup> Si  | 14029 | 4.8381e-05                        |     |
|                  |                              |     | <sup>30</sup> Si  | 14030 | 3.1893e-05                        |     |

Table 2.5 provides the cold and hot atomic number densities for UO<sub>2</sub> fuel by conserving total uranium mass per each fuel assembly. The cold atomic number densities were used in the Polaris calculations to generate assembly homogenized 2-group cross sections for nodal diffusion calculations. Table 2.6 provides atomic number densities for structure materials such as Zr-4, Inc-718, and SS-304 and burnable absorber (BA) and control rod materials. The WBN1 cycles 1–3 include various BAs including Pyrex, Wet Annular Burnable Absorber (WABA), Integral Fuel Burnable Absorber (IFBA), and Tritium-Producing Burnable Absorber Rod (TPBAR). Figures 2.2 and 2.3 illustrate fuel assembly configurations with various loadings of BAs including Pyrex, WABA, and TPBAR.

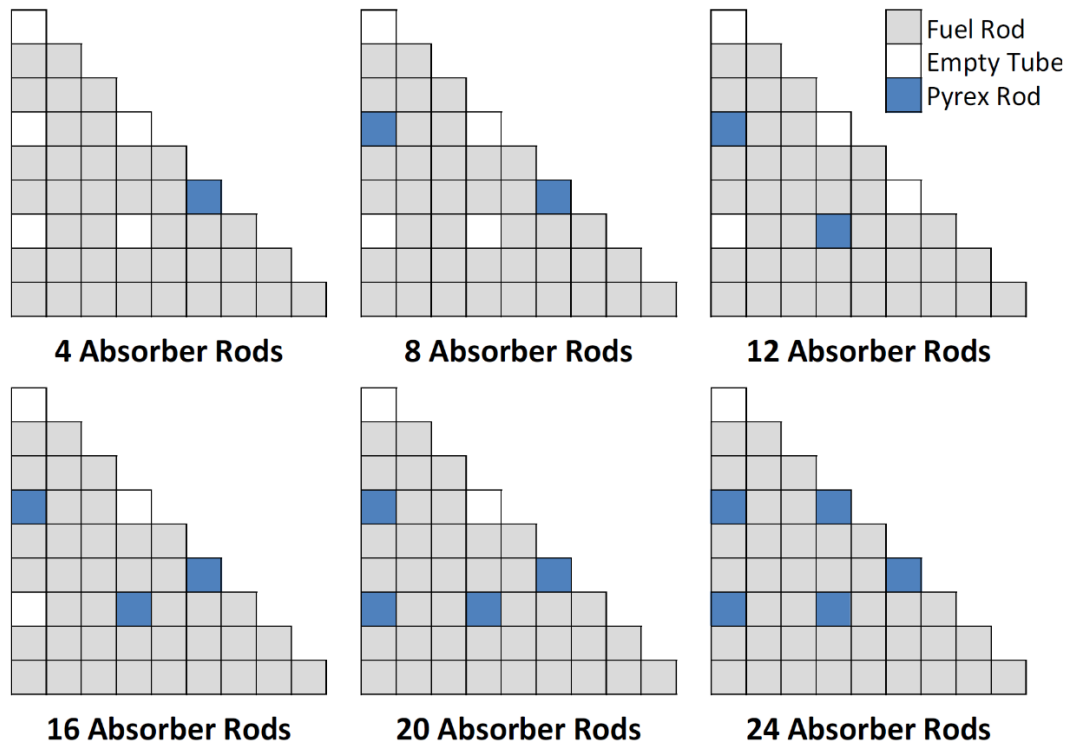


Figure 2.2. Fuel assembly configurations with burnable absorbers (Pyrex, WABA, and TPBAR).

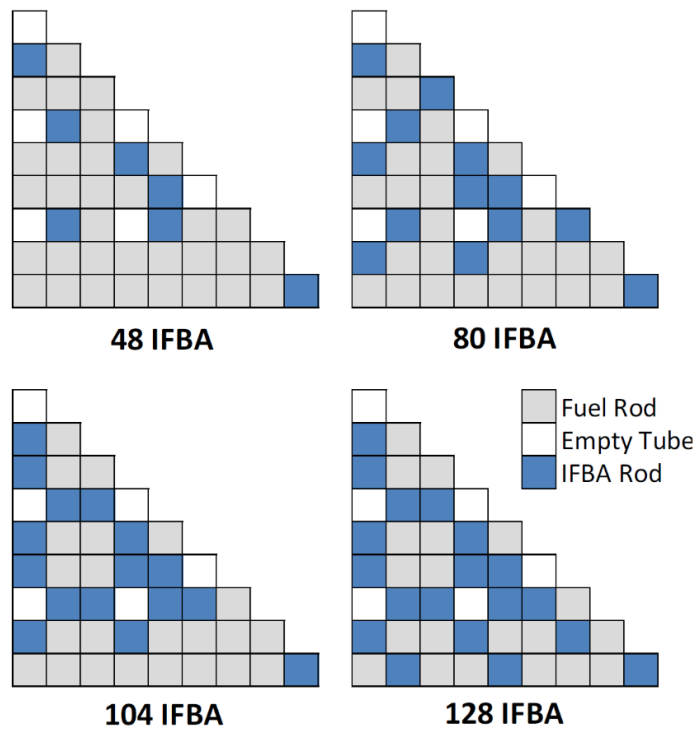


Figure 2.3. Fuel assembly configurations with IFBA.



Figure 2.4 provides the cycle 1 fuel assembly loading pattern for which 3  $^{235}\text{U}$  enrichment  $\text{UO}_2$  fuels and 5 types of BA patterns were used with various combinations to have 10 types of fuel assemblies. Figure 2.5 illustrates the cycle 2 loading pattern with fresh assembly loadings with 3.709 wt%  $^{235}\text{U}$  fuels and shuffling of once-burned fuel assemblies. Figure 2.6 illustrates the cycle 3 loading pattern with fresh assembly loadings with 3.8 and 4.4 wt%  $^{235}\text{U}$  fuels and shuffling of once- and twice-burned fuel assemblies.

Control and shutdown bank positions are provided in Figure 2.7. WBN1 includes the  $^{235}\text{U}$  fission chambers as in-core detectors of which locations are provided in Figure 2.8. Table 2.7 provides the fuel assembly types and numbers loaded in cycles 1–3. The WBN1 axial configuration for fuel, spacer grids, and reflectors is shown in Table 2.8.

|    | H         | G         | F         | E         | D                                  | C         | B         | A         |
|----|-----------|-----------|-----------|-----------|------------------------------------|-----------|-----------|-----------|
| 8  | 2.1<br>20 | 2.6<br>20 | 2.1<br>20 | 2.6<br>20 | 2.1<br>20                          | 2.6<br>20 | 2.1<br>20 | 3.1<br>12 |
| 9  | 2.6<br>20 | 2.1<br>24 | 2.6<br>24 | 2.1<br>20 | 2.6<br>20                          | 2.1<br>24 | 3.1<br>24 | 3.1       |
| 10 | 2.1<br>24 | 2.6<br>24 | 2.1<br>20 | 2.6<br>20 | 2.1<br>16                          | 2.6<br>16 | 2.1<br>8  | 3.1       |
| 11 | 2.6<br>20 | 2.1<br>20 | 2.6<br>20 | 2.1<br>20 | 2.6<br>16                          | 2.1<br>16 | 3.1       | 3.1       |
| 12 | 2.1<br>20 | 2.6<br>20 | 2.1<br>20 | 2.6<br>24 | 2.6<br>24                          | 2.1<br>16 | 3.1       |           |
| 13 | 2.6<br>20 | 2.1<br>16 | 2.6<br>24 | 2.1<br>12 | 2.6<br>12                          | 3.1       | 3.1       |           |
| 14 | 2.1<br>24 | 3.1<br>24 | 2.1<br>16 | 3.1<br>16 | 3.1                                | 3.1       |           |           |
| 15 | 3.1<br>12 | 3.1<br>8  | 3.1<br>8  | 3.1       | Enrichment<br>Number of Pyrex Rods |           |           |           |

**Figure 2.4. Layout of fuel assemblies and burnable poisons in the WBN1 cycle 1.**

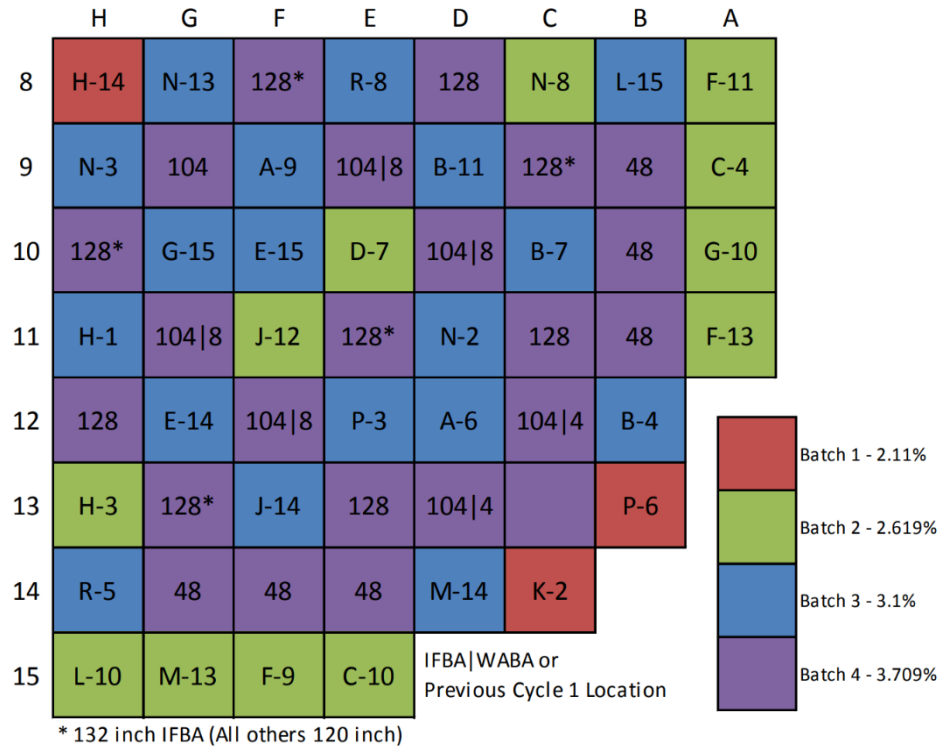


Figure 2.5. Fresh assembly loading and shuffling patterns in the WBN1 cycle 2.

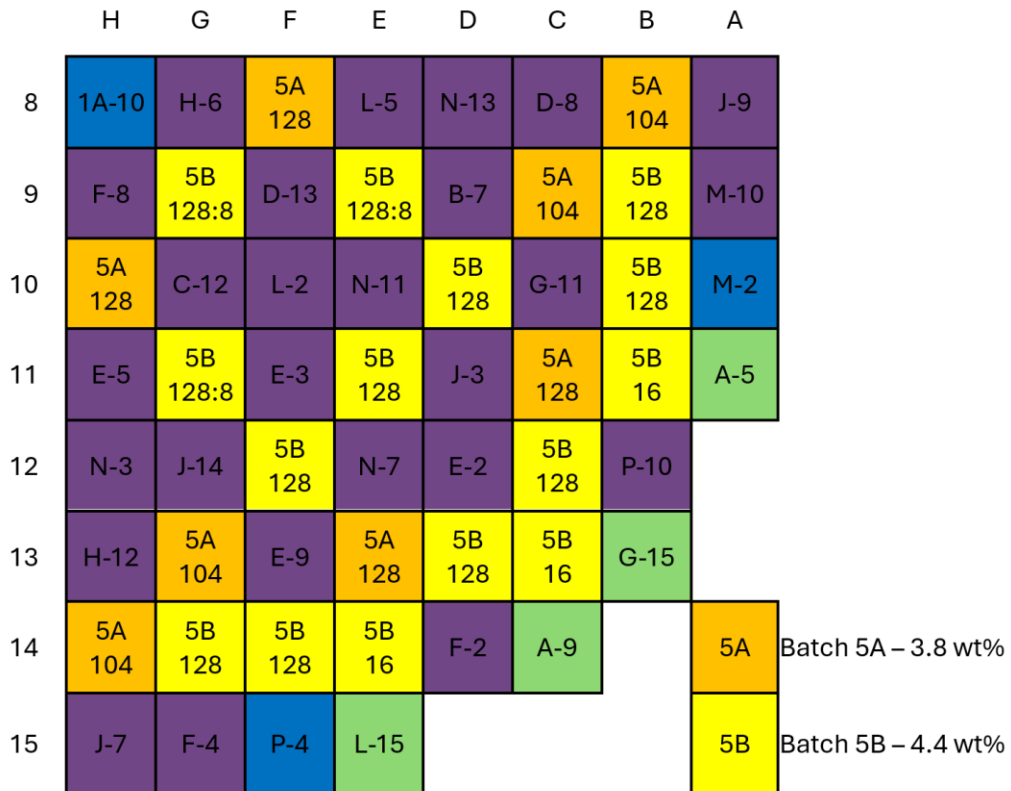


Figure 2.6. Fresh assembly loading and shuffling patterns in the WBN1 cycle 3.

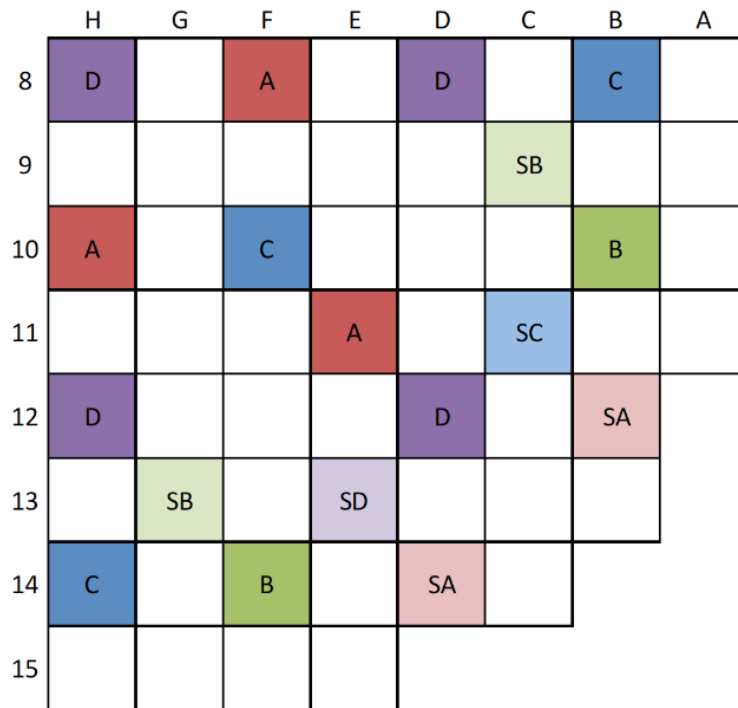


Figure 2.7. WBN1 control rod and shutdown bank positions.

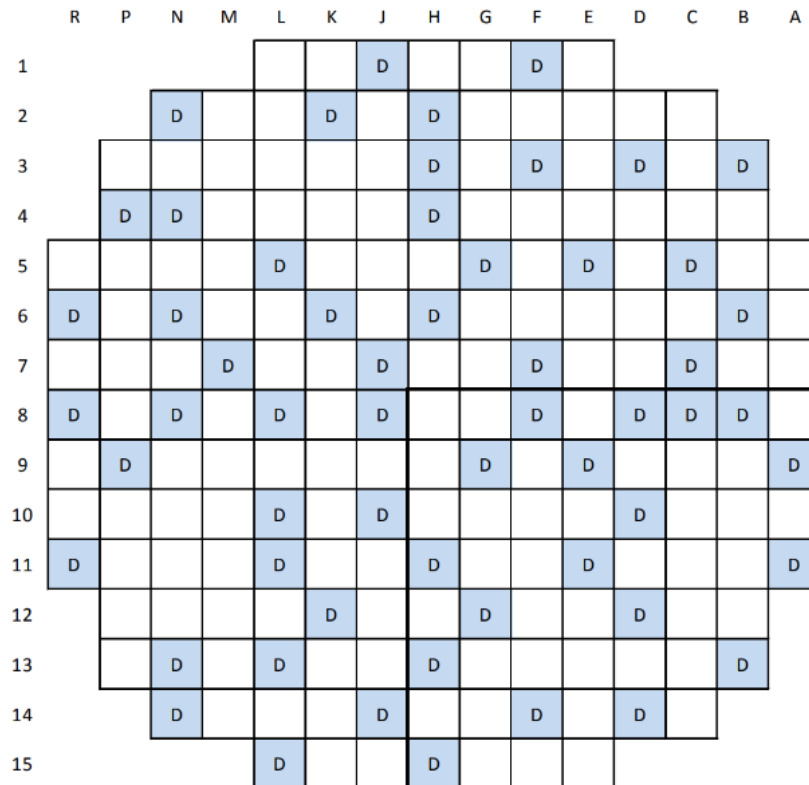


Figure 2.8. WBN1 core instrumentation locations.

Table 2.7. Assembly types for the WBN1 cycles 1–3.

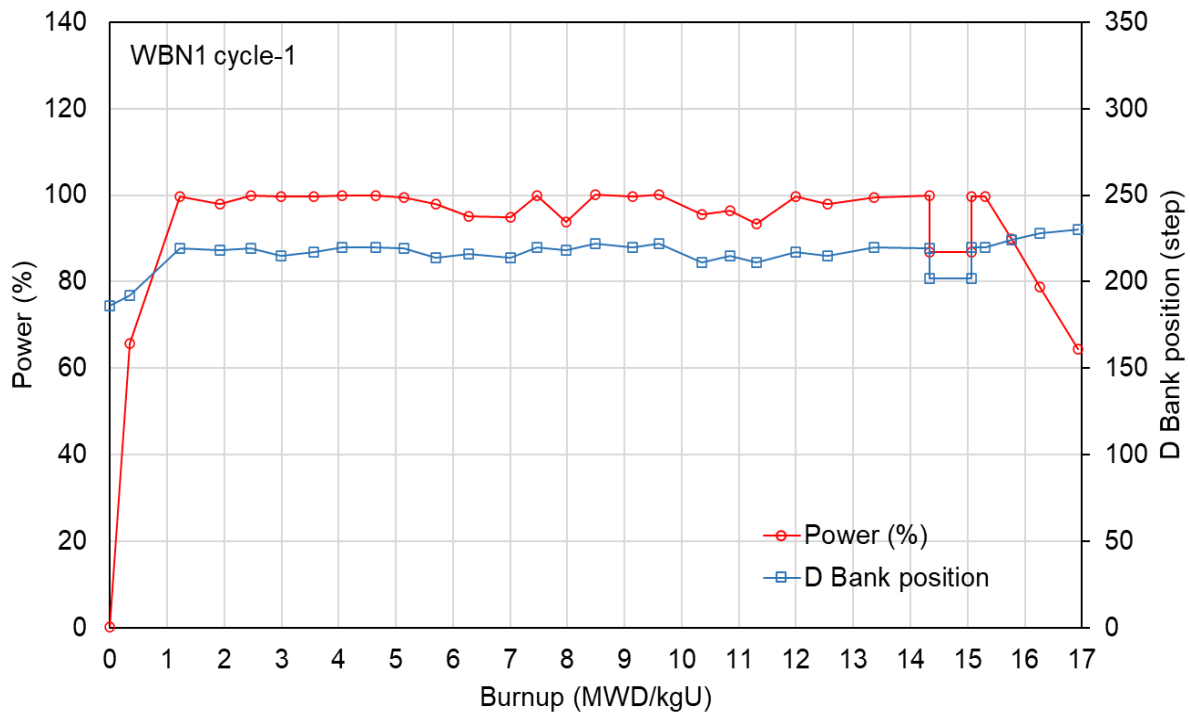
| Cycle | <sup>235</sup> U wt% | Power density (W/g) | BA | IFBA | PARCS assembly no. | Description      | Remark            |
|-------|----------------------|---------------------|----|------|--------------------|------------------|-------------------|
| 1     | 2.10                 | 38.4                | 0  | 0    | 10                 | 2.1wt%           |                   |
|       |                      |                     | 0  | 0    | 20                 | 2.6wt%           |                   |
|       | 2.60                 | 38.4                | 16 | 0    | 21                 | 2.6wt%-bp16      | Pyrex             |
|       |                      |                     | 20 | 0    | 22                 | 2.6wt%-bp20      | Pyrex             |
|       |                      |                     | 24 | 0    | 23                 | 2.6wt%-bp24      | Pyrex             |
|       |                      |                     | 0  | 0    | 30                 | 3.1wt%           |                   |
|       | 3.10                 | 38.4                | 8  | 0    | 31                 | 3.1wt%-bp08      | Pyrex             |
|       |                      |                     | 12 | 0    | 32                 | 3.1wt%-bp12      | Pyrex             |
|       |                      |                     | 16 | 0    | 33                 | 3.1wt%-bp16      | Pyrex             |
|       |                      |                     | 24 | 0    | 34                 | 3.1wt%-bp24      | Pyrex             |
| 2     | 3.709                | 38.4                | 0  | 0    | 40                 | 3.7wt%           |                   |
|       |                      |                     | 0  | 48   | 41                 | 3.7wt%-i048      |                   |
|       |                      |                     | 8  | 104  | 42                 | 3.7wt%-i104-tpb  | TPBAR             |
|       |                      |                     | 4  | 104  | 43                 | 3.7wt%-i104-w4   | WABA              |
|       |                      |                     | 8  | 104  | 44                 | 3.7wt%-i104-w8   | WABA              |
|       |                      |                     | 0  | 128  | 45                 | 3.7wt%-i128      |                   |
|       |                      |                     | 0  | 128  | 46                 | 3.7wt%-i128-LONG | long              |
| 3     | 3.80                 | 38.4                | 0  | 104  | 51                 | 3.8wt%-i104      |                   |
|       |                      |                     | 0  | 128  | 52                 | 3.8wt%-i128      |                   |
|       | 4.40                 | 38.4                | 0  | 0    | 61                 | 4.4wt%-i016      | 16 hollow pellets |
|       |                      |                     | 0  | 128  | 62                 | 4.4wt%-i128      |                   |
|       |                      |                     | 8  | 128  | 63                 | 4.4wt%-i128-w8   | WABA              |

Table 2.8. WBN1 core axial configurations.

| No. | Axial materials  | Cold height (cm) |         |          | Hot height (cm) |         |          | No. of meshes |
|-----|------------------|------------------|---------|----------|-----------------|---------|----------|---------------|
|     |                  | Core             | Active  | Distance | Core            | Active  | Distance |               |
| 1   | Top reflector    | 418.937          | -       | 36.226   | 420.884         | -       | 36.226   | 1             |
| 2   | fuel rod         | 382.711          | 365.760 | 1.270    | 384.658         | 367.710 | 1.277    | 1             |
| 3   | fuel rod         | 381.441          | 364.490 | 13.970   | 383.381         | 366.433 | 14.044   | 1             |
| 4   | fuel rod         | 367.471          | 350.520 | 15.240   | 369.337         | 352.389 | 15.321   | 1             |
| 5   | fuel rod         | 352.231          | 335.280 | 9.126    | 354.016         | 337.068 | 9.175    | 1             |
| 6   | Spacer grid 7    | 343.105          | 326.154 | 3.810    | 344.841         | 327.893 | 3.830    | 1             |
| 7   | fuel rod         | 339.295          | 322.344 | 48.390   | 341.011         | 324.063 | 48.648   | 3             |
| 8   | Spacer grid 6    | 290.905          | 273.954 | 3.810    | 292.363         | 275.415 | 3.830    | 1             |
| 9   | fuel rod         | 287.095          | 270.144 | 48.390   | 288.533         | 271.584 | 48.648   | 3             |
| 10  | Spacer grid 5    | 238.705          | 221.754 | 3.810    | 239.885         | 222.936 | 3.830    | 1             |
| 11  | fuel rod         | 234.895          | 217.944 | 48.390   | 236.055         | 219.106 | 48.648   | 3             |
| 12  | Spacer grid 4    | 186.505          | 169.554 | 3.810    | 187.407         | 170.458 | 3.830    | 1             |
| 13  | fuel rod         | 182.695          | 165.744 | 48.390   | 183.577         | 166.628 | 48.648   | 3             |
| 14  | Spacer grid 3    | 134.305          | 117.354 | 3.810    | 134.929         | 117.980 | 3.830    | 1             |
| 15  | fuel rod         | 130.495          | 113.544 | 48.390   | 131.099         | 114.149 | 48.648   | 3             |
| 16  | Spacer grid 2    | 82.105           | 65.154  | 3.810    | 82.451          | 65.501  | 3.830    | 1             |
| 17  | fuel rod         | 78.295           | 61.344  | 30.864   | 78.621          | 61.671  | 31.028   | 2             |
| 18  | fuel rod         | 47.431           | 30.480  | 15.240   | 47.593          | 30.643  | 15.321   | 1             |
| 19  | fuel rod         | 32.191           | 15.240  | 11.430   | 32.272          | 15.321  | 11.491   | 1             |
| 20  | Spacer grid 1    | 20.761           | 3.810   | 3.810    | 20.781          | 3.830   | 3.830    | 1             |
| 21  | Bottom reflector | 16.951           | 0       | 16.951   | 16.951          | 0       | 16.951   | 1             |
|     |                  | 0                |         |          |                 |         |          | 32            |

## 2.2.2 Measured data

The measured data used in the current study were obtained from the publicly available WBN1 cycles 1–3 HZP physics test data and operating history, including a critical boron history and in-core detector responses. Figures 2.9–2.11 and Table 2.9 provide the operational histories for powers, control bank D positions, and inlet temperatures at burnups.



**Figure 2.9. Power and D bank position history for the WBN1 cycle 1.**

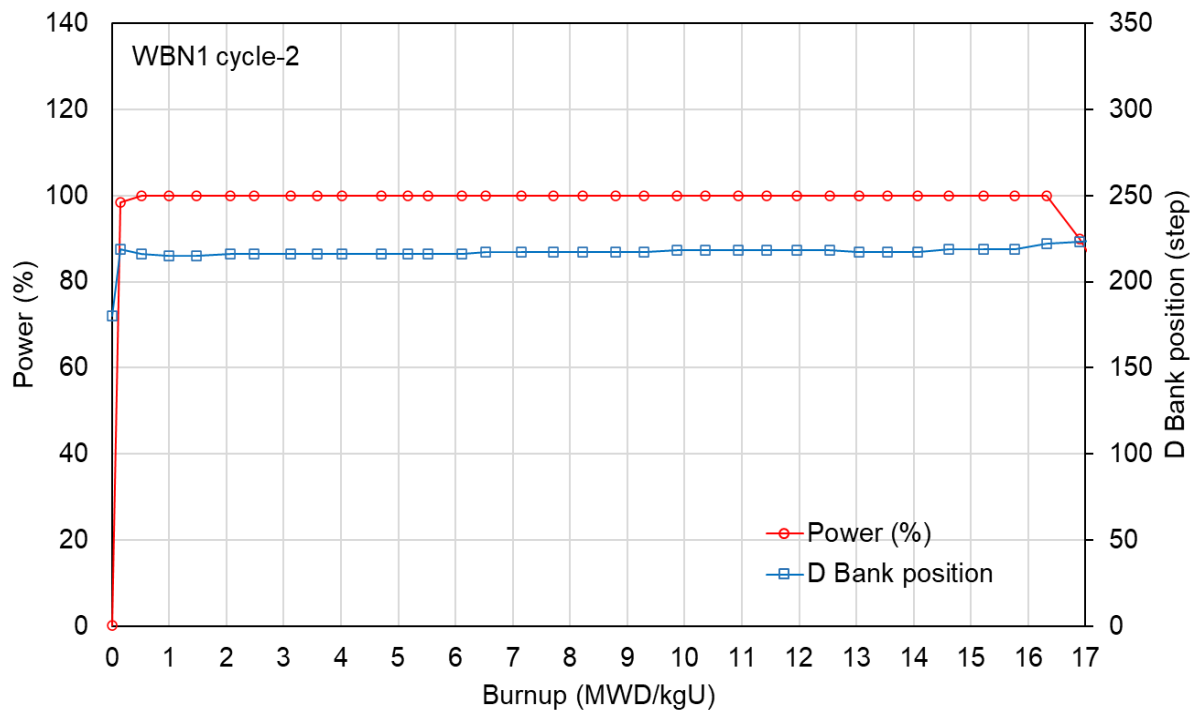


Figure 2.10. Power and D bank position history for the WBN1 cycle 2.

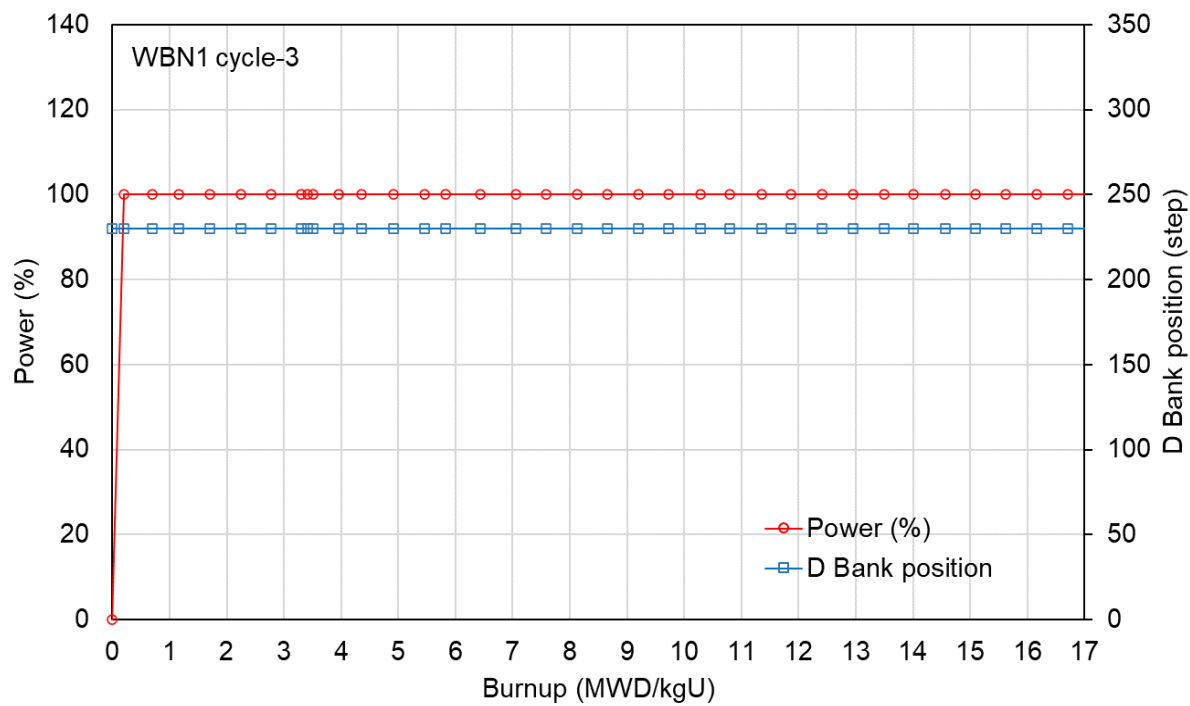


Figure 2.11. Power and D bank position history for the WBN1 cycle 3.

**Table 2.9. WBN1 operational history.**

| No. | Cycle 1             |              |                |                        | Cycle 2             |              |                |                        | Cycle 3             |              |                |                        |
|-----|---------------------|--------------|----------------|------------------------|---------------------|--------------|----------------|------------------------|---------------------|--------------|----------------|------------------------|
|     | Burnup<br>(MWD/kgU) | Power<br>(%) | D bank<br>step | T <sub>in</sub><br>(K) | Burnup<br>(MWD/kgU) | Power<br>(%) | D bank<br>step | T <sub>in</sub><br>(K) | Burnup<br>(MWD/kgU) | Power<br>(%) | D bank<br>step | T <sub>in</sub><br>(K) |
| 1   | 0                   | 0.1          | 186            | 564.9                  | 0.000               | 0.1          | 180            | 564.81                 | 0.000               | 0.1          | 230            | 564.81                 |
| 2   | 0.346               | 65.7         | 192            | 565.2                  | 0.142               | 98.4         | 219            | 564.81                 | 0.195               | 100.0        | 230            | 564.81                 |
| 3   | 1.229               | 99.7         | 219            | 565.5                  | 0.521               | 100.0        | 216            | 564.81                 | 0.692               | 100.0        | 230            | 564.81                 |
| 4   | 1.92                | 98.0         | 218            | 565.5                  | 1.000               | 100.0        | 215            | 564.81                 | 1.166               | 100.0        | 230            | 564.81                 |
| 5   | 2.458               | 100          | 219            | 565.8                  | 1.479               | 100.0        | 215            | 564.81                 | 1.698               | 100.0        | 230            | 564.81                 |
| 6   | 2.996               | 99.7         | 215            | 565.8                  | 2.065               | 100.0        | 216            | 564.81                 | 2.245               | 100.0        | 230            | 564.81                 |
| 7   | 3.561               | 99.7         | 217            | 565.8                  | 2.475               | 100.0        | 216            | 564.81                 | 2.769               | 100.0        | 230            | 564.81                 |
| 8   | 4.064               | 99.8         | 220            | 565.9                  | 3.130               | 100.0        | 216            | 564.81                 | 3.304               | 100.0        | 230            | 564.81                 |
| 9   | 4.644               | 99.8         | 220            | 565.6                  | 3.579               | 100.0        | 216            | 564.81                 | 3.419               | 100.0        | 230            | 564.81                 |
| 10  | 5.139               | 99.5         | 219            | 565.4                  | 4.019               | 100.0        | 216            | 564.81                 | 3.511               | 100.0        | 230            | 564.81                 |
| 11  | 5.7                 | 98.0         | 214            | 565.4                  | 4.698               | 100.0        | 216            | 564.81                 | 3.958               | 100.0        | 230            | 564.81                 |
| 12  | 6.272               | 95.1         | 216            | 565.4                  | 5.161               | 100.0        | 216            | 564.81                 | 4.352               | 100.0        | 230            | 564.81                 |
| 13  | 6.998               | 94.8         | 214            | 565.4                  | 5.525               | 100.0        | 216            | 564.81                 | 4.910               | 100.0        | 230            | 564.81                 |
| 14  | 7.463               | 99.8         | 220            | 565.3                  | 6.111               | 100.0        | 216            | 564.81                 | 5.457               | 100.0        | 230            | 564.81                 |
| 15  | 7.978               | 93.9         | 218            | 565.1                  | 6.525               | 100.0        | 217            | 564.81                 | 5.824               | 100.0        | 230            | 564.81                 |
| 16  | 8.492               | 100.1        | 222            | 565.4                  | 7.146               | 100.0        | 217            | 564.81                 | 6.436               | 100.0        | 230            | 564.81                 |
| 17  | 9.141               | 99.7         | 220            | 565.3                  | 7.705               | 100.0        | 217            | 564.81                 | 7.059               | 100.0        | 230            | 564.81                 |
| 18  | 9.602               | 100.2        | 222            | 565.2                  | 8.219               | 100.0        | 217            | 564.81                 | 7.583               | 100.0        | 230            | 564.81                 |
| 19  | 10.344              | 95.6         | 211            | 565.4                  | 8.794               | 100.0        | 217            | 564.81                 | 8.134               | 100.0        | 230            | 564.81                 |
| 20  | 10.843              | 96.4         | 215            | 565.5                  | 9.288               | 100.0        | 217            | 564.81                 | 8.654               | 100.0        | 230            | 564.81                 |
| 21  | 11.315              | 93.4         | 211            | 565.1                  | 9.859               | 100.0        | 218            | 564.81                 | 9.201               | 100.0        | 230            | 564.81                 |
| 22  | 11.987              | 99.7         | 217            | 565.1                  | 10.357              | 100.0        | 218            | 564.81                 | 9.725               | 100.0        | 230            | 564.81                 |
| 23  | 12.552              | 98.0         | 215            | 565.2                  | 10.932              | 100.0        | 218            | 564.81                 | 10.276              | 100.0        | 230            | 564.81                 |
| 24  | 13.359              | 99.4         | 220            | 565.3                  | 11.430              | 100.0        | 218            | 564.81                 | 10.796              | 100.0        | 230            | 564.81                 |
| 25  | 14.334              | 99.9         | 219            | 565.3                  | 11.962              | 100.0        | 218            | 564.81                 | 11.346              | 100.0        | 230            | 564.81                 |
| 26  | 14.33401            | 86.9         | 202            | 564.7                  | 12.541              | 100.0        | 218            | 564.81                 | 11.870              | 100.0        | 230            | 564.81                 |
| 27  | 15.06801            | 86.9         | 202            | 564.7                  | 13.039              | 100.0        | 217            | 564.81                 | 12.409              | 100.0        | 230            | 564.81                 |
| 28  | 15.06802            | 99.6         | 220            | 565.4                  | 13.537              | 100.0        | 217            | 564.81                 | 12.945              | 100.0        | 230            | 564.81                 |
| 29  | 15.31002            | 99.6         | 220            | 565.4                  | 14.066              | 100.0        | 217            | 564.81                 | 13.488              | 100.0        | 230            | 564.81                 |
| 30  | 15.77502            | 89.9         | 224            | 564.9                  | 14.618              | 100.0        | 219            | 564.81                 | 14.012              | 100.0        | 230            | 564.81                 |
| 31  | 16.27002            | 78.8         | 228            | 564.5                  | 15.215              | 100.0        | 219            | 564.81                 | 14.559              | 100.0        | 230            | 564.81                 |
| 32  | 16.93902            | 64.5         | 230            | 563.7                  | 15.763              | 100.0        | 219            | 564.81                 | 15.086              | 100.0        | 230            | 564.81                 |
| 33  |                     |              |                |                        | 16.319              | 100.0        | 222            | 564.81                 | 15.622              | 100.0        | 230            | 564.81                 |
| 34  |                     |              |                |                        | 16.901              | 89.9         | 223            | 564.81                 | 16.161              | 100.0        | 230            | 564.81                 |
| 35  |                     |              |                |                        | 17.476              | 77.3         | 225            | 564.81                 | 16.700              | 100.0        | 230            | 564.81                 |
| 36  |                     |              |                |                        | 18.062              | 63.2         | 225            | 564.81                 | 17.232              | 100.0        | 230            | 564.81                 |
| 37  |                     |              |                |                        |                     |              |                |                        | 17.748              | 100.0        | 230            | 564.81                 |
| 38  |                     |              |                |                        |                     |              |                |                        | 18.142              | 100.0        | 230            | 564.81                 |
| 39  |                     |              |                |                        |                     |              |                |                        | 18.788              | 85.9         | 230            | 564.81                 |
| 40  |                     |              |                |                        |                     |              |                |                        | 19.324              | 75.8         | 230            | 564.81                 |

## 2.3 FUTURE IMPROVEMENT IN POLARIS AND PARCS

The benchmark calculations for WBN1 using the SCALE 6.3/Polaris–GenPMAXS–PARCS v3.4.2 code package was completed which will be used in the validation of the code package for LWR reactor physics by obtaining statistical uncertainties for key nuclear parameters. However, some further improvement needs to be considered in SCALE 6.3.0/Polaris and PARCS v3.4.2 to have a better performance as follows:

- Gamma smeared power distribution is considered in pin power form factors. Gamma energy deposition is very flat over fuel assembly, which would decrease pin peaking factor. If gamma smeared power distribution is not considered, bigger  $F_{\Delta H}^N$  is expected which may result in violating the  $F_{\Delta H}^N$  limit.

- b. Different PMAXS file assignment and burnups for assembly quadrants, and
- c. Limitation on assembly rotation.

PARCS requires the following state point calculations for each type of fuel assembly which results in very expensive computing time in Polaris.

- no BP assembly:  $8(\text{reference case}) \times 2(\text{spacer grid}) \times 21(\text{burnup point}) \times 45(\text{branch}) = 15120$ , and
- BP assembly:  $8(\text{reference case}) \times 2(\text{spacer grid}) \times 21(\text{burnup point}) \times 36(\text{branch}) = 12096$ .

Advanced 2-step analysis code package based on micro depletion capability requires about 300 state point calculations for each assembly.

PARCS does not support 2×2 nodes for each fuel assembly which would limit to treat fuel assembly with asymmetric BPs and quadrant-dependent burnup due to different surrounding fuel assemblies. The 2×2 node capability with different cross sections and burnups would enhance overall accuracy of the Polaris-PARCS code package.



### 3. BENCHMARK CALCULATIONS AND RESULTS

#### 3.1 BENCHMARK CALCULATION

##### 3.1.1 Polaris Calculations

The SCALE 6.3/Polaris inputs were prepared using the data provided in Section 2 to process assembly-homogenized 2-group cross sections for whole-core nodal diffusion calculations. To cover all the possible reactor states, the 2-group cross sections were tabulated as a function of burnup, moderator, and fuel temperatures, moderator density, soluble boron concentration, and control rod. Table 3.1 provides state information for eight history cases. Another eight history cases are for assembly with spacer grids. There are two types of spacer grids in WBN1, including Zr-4 (six intermediates) and Inc-718 (two top and bottom), which should be considered in the Polaris lattice physics calculation. Six intermediate Zr-4 and one bottom Inc-718 spacer grids are included in the active fuel zone. Therefore three sets of cross section table-sets should be provided for each assembly. Since there is only one Inc-718 spacer grid, it was treated as the Zr-4 spacer grid for simplicity. This should be updated in later calculations.

To complete the functionalization of the cross section table-sets, various branch calculations should be performed to cover all the possible reactor states at each burnup point for each history case. Tables 3.2 and 3.3 provide the information for branch states for assembly without and with BA, respectively. Since the Pyrex burnable absorbers are located at the guide tubes, the branch cases 28–36 for the assembly without BA are not included in the assembly with BA. Table 3.4 provides the branch cases for reflectors for which no depletion calculation was performed. Figure 3.1 illustrates the Polaris models for radial and top and bottom axial reflectors.

Appendices A.1–A.4 provide the Polaris inputs for the fuel assembly 2.60\_bp16 (A.1) and radial (A.2), axial top (A.3), and bottom (A.4) reflectors. Table 3.5 provides the Polaris input and output files for the WBN1 cycles 1–3. All the files produced for this benchmark calculation will be available on <https://code.ornl.gov/scale/analysis/> with ‘2023 SCALE 6.3/Polaris–PARCS v3.4.2 Validation for LWR’.

**Table 3.1. History cases for each assembly type.**

|    | File name       | Rod-in | Moderator temp. (K) | Density (g/cm <sup>3</sup> ) | Soluble boron (ppm) | Fuel temp. (K) | Spacer grid |
|----|-----------------|--------|---------------------|------------------------------|---------------------|----------------|-------------|
| A1 | CR0-HDC-MPC-MTF | no     | 550                 | 0.76971                      | 500                 | 800            | no          |
| A2 | CR0-LDC-MPC-MTF | no     | 615                 | 0.60811                      | 500                 | 800            | no          |
| A3 | CR0-MDC-HPC-MTF | no     | 585                 | 0.70045                      | 1,500               | 800            | no          |
| A4 | CR0-MDC-LPC-MTF | no     | 585                 | 0.70045                      | 0                   | 800            | no          |
| A5 | CR0-MDC-MPC-HTF | no     | 585                 | 0.70045                      | 500                 | 1,600          | no          |
| A6 | CR0-MDC-MPC-LTF | no     | 585                 | 0.70045                      | 500                 | 560            | no          |
| A7 | CR0-MDC-MPC-MTF | no     | 585                 | 0.70045                      | 500                 | 800            | no          |
| A8 | CR1-MDC-MPC-MTF | yes    | 585                 | 0.70045                      | 500                 | 800            | no          |
| B1 | CR0-HDC-MPC-MTF | no     | 550                 | 0.76971                      | 500                 | 800            | yes         |
| B2 | CR0-LDC-MPC-MTF | no     | 615                 | 0.60811                      | 500                 | 800            | yes         |
| B3 | CR0-MDC-HPC-MTF | no     | 585                 | 0.70045                      | 1,500               | 800            | yes         |
| B4 | CR0-MDC-LPC-MTF | no     | 585                 | 0.70045                      | 0                   | 800            | yes         |
| B5 | CR0-MDC-MPC-HTF | no     | 585                 | 0.70045                      | 500                 | 1,600          | yes         |
| B6 | CR0-MDC-MPC-LTF | no     | 585                 | 0.70045                      | 500                 | 560            | yes         |
| B7 | CR0-MDC-MPC-MTF | no     | 585                 | 0.70045                      | 500                 | 800            | yes         |
| B8 | CR1-MDC-MPC-MTF | yes    | 585                 | 0.70045                      | 500                 | 800            | yes         |

**Table 3.2. Branch states for assembly type without BA.**

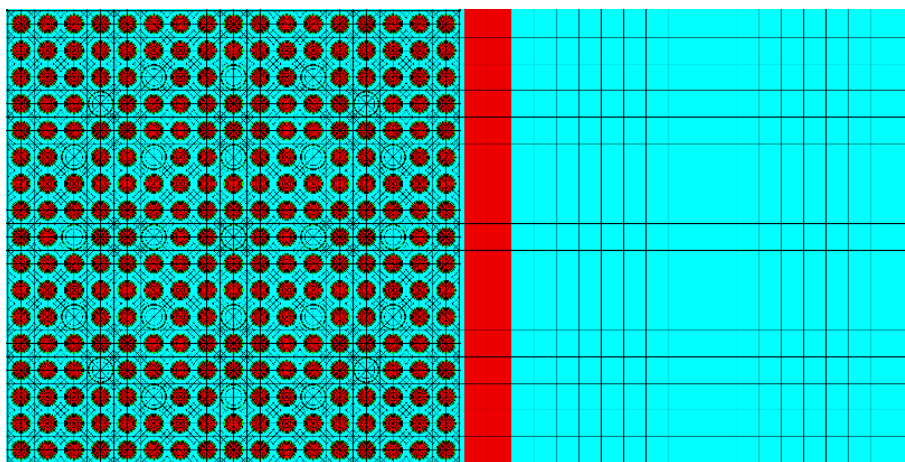
| Case | Bank  |       | All       | Moderator                    |                     | Fuel      | Reference |
|------|-------|-------|-----------|------------------------------|---------------------|-----------|-----------|
|      | B     | A     | Temp. (K) | Density (g/cm <sup>3</sup> ) | Soluble boron (ppm) | Temp. (K) |           |
| 1    | false | false | 615.0     | 0.60811                      | 0.1                 | 560.0     |           |
| 2    | false | false | 615.0     | 0.60811                      | 500                 | 560.0     |           |
| 3    | false | false | 615.0     | 0.60811                      | 1,500               | 560.0     |           |
| 4    | false | false | 585.0     | 0.70045                      | 0.1                 | 560.0     |           |
| 5    | false | false | 585.0     | 0.70045                      | 500                 | 560.0     | A6        |
| 6    | false | false | 585.0     | 0.70045                      | 1,500               | 560.0     |           |
| 7    | false | false | 550.0     | 0.76971                      | 0.1                 | 560.0     |           |
| 8    | false | false | 550.0     | 0.76971                      | 500                 | 560.0     |           |
| 9    | false | false | 550.0     | 0.76971                      | 1,500               | 560.0     |           |
| 10   | false | false | 615.0     | 0.60811                      | 0.1                 | 800.0     |           |
| 11   | false | false | 615.0     | 0.60811                      | 500                 | 800.0     | A2, B2    |
| 12   | false | false | 615.0     | 0.60811                      | 1,500               | 800.0     |           |
| 13   | false | false | 585.0     | 0.70045                      | 0.1                 | 800.0     | A4, B4    |
| 14   | false | false | 585.0     | 0.70045                      | 500                 | 800.0     | A7, B7    |
| 15   | false | false | 585.0     | 0.70045                      | 1,500               | 800.0     | A3, B3    |
| 16   | false | false | 550.0     | 0.76971                      | 0.1                 | 800.0     |           |
| 17   | false | false | 550.0     | 0.76971                      | 500                 | 800.0     | A1, B1    |
| 18   | false | false | 550.0     | 0.76971                      | 1,500               | 800.0     |           |
| 19   | false | false | 615.0     | 0.60811                      | 0.1                 | 1,600.0   |           |
| 20   | false | false | 615.0     | 0.60811                      | 500                 | 1,600.0   |           |
| 21   | false | false | 615.0     | 0.60811                      | 1,500               | 1,600.0   |           |
| 22   | false | false | 585.0     | 0.70045                      | 0.1                 | 1,600.0   |           |
| 23   | false | false | 585.0     | 0.70045                      | 500                 | 1,600.0   | A5, B5    |
| 24   | false | false | 585.0     | 0.70045                      | 1,500               | 1,600.0   |           |
| 25   | false | false | 550.0     | 0.76971                      | 0.1                 | 1,600.0   |           |
| 26   | false | false | 550.0     | 0.76971                      | 500                 | 1,600.0   |           |
| 27   | false | false | 550.0     | 0.76971                      | 1,500               | 1,600.0   |           |
| 28   | true  | false | 615.0     | 0.60811                      | 0.1                 | 800.0     |           |
| 29   | true  | false | 615.0     | 0.60811                      | 500                 | 800.0     |           |
| 30   | true  | false | 615.0     | 0.60811                      | 1,500               | 800.0     |           |
| 31   | true  | false | 585.0     | 0.70045                      | 0.1                 | 800.0     |           |
| 32   | true  | false | 585.0     | 0.70045                      | 500                 | 800.0     | A8, B8    |
| 33   | true  | false | 585.0     | 0.70045                      | 1,500               | 800.0     |           |
| 34   | true  | false | 550.0     | 0.76971                      | 0.1                 | 800.0     |           |
| 35   | true  | false | 550.0     | 0.76971                      | 500                 | 800.0     |           |
| 36   | true  | false | 550.0     | 0.76971                      | 1,500               | 800.0     |           |
| 37   | false | true  | 615.0     | 0.60811                      | 0.1                 | 800.0     |           |
| 38   | false | true  | 615.0     | 0.60811                      | 500                 | 800.0     |           |
| 39   | false | true  | 615.0     | 0.60811                      | 1,500               | 800.0     |           |
| 40   | false | true  | 585.0     | 0.70045                      | 0.1                 | 800.0     |           |
| 41   | false | true  | 585.0     | 0.70045                      | 500                 | 800.0     |           |
| 42   | false | true  | 585.0     | 0.70045                      | 1,500               | 800.0     |           |
| 43   | false | true  | 550.0     | 0.76971                      | 0.1                 | 800.0     |           |
| 44   | false | true  | 550.0     | 0.76971                      | 500                 | 800.0     |           |
| 45   | false | true  | 550.0     | 0.76971                      | 1,500               | 800.0     |           |

**Table 3.3. Branch states for assembly type with BA.**

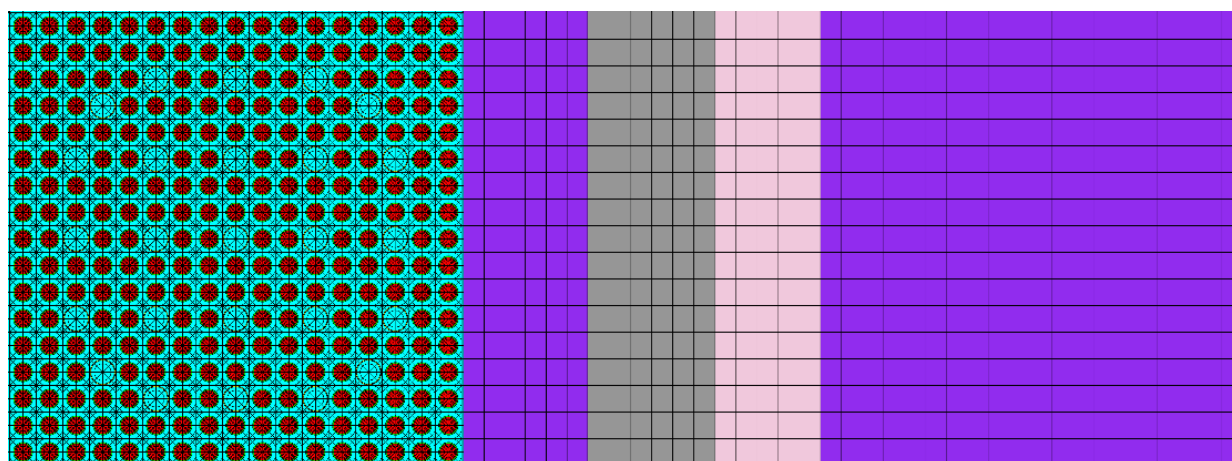
| Case | Bank  |             | All       | Moderator                    |                     | Fuel      | Remark |
|------|-------|-------------|-----------|------------------------------|---------------------|-----------|--------|
|      | BP    | Control rod | Temp. (K) | Density (g/cm <sup>3</sup> ) | Soluble boron (ppm) | Temp. (K) |        |
| 1    | false | –           | 615.0     | 0.60811                      | 0.1                 | 560.0     |        |
| 2    | false | –           | 615.0     | 0.60811                      | 500                 | 560.0     |        |
| 3    | false | –           | 615.0     | 0.60811                      | 1,500               | 560.0     |        |
| 4    | false | –           | 585.0     | 0.70045                      | 0.1                 | 560.0     |        |
| 5    | false | –           | 585.0     | 0.70045                      | 500                 | 560.0     | A6     |
| 6    | false | –           | 585.0     | 0.70045                      | 1,500               | 560.0     |        |
| 7    | false | –           | 550.0     | 0.76971                      | 0.1                 | 560.0     |        |
| 8    | false | –           | 550.0     | 0.76971                      | 500                 | 560.0     |        |
| 9    | false | –           | 550.0     | 0.76971                      | 1,500               | 560.0     |        |
| 10   | false | –           | 615.0     | 0.60811                      | 0.1                 | 800.0     |        |
| 11   | false | –           | 615.0     | 0.60811                      | 500                 | 800.0     | A2, B2 |
| 12   | false | –           | 615.0     | 0.60811                      | 1,500               | 800.0     |        |
| 13   | false | –           | 585.0     | 0.70045                      | 0.1                 | 800.0     | A4, B4 |
| 14   | false | –           | 585.0     | 0.70045                      | 500                 | 800.0     | A7, B7 |
| 15   | false | –           | 585.0     | 0.70045                      | 1,500               | 800.0     | A3, B3 |
| 16   | false | –           | 550.0     | 0.76971                      | 0.1                 | 800.0     |        |
| 17   | false | –           | 550.0     | 0.76971                      | 500                 | 800.0     | A1, B1 |
| 18   | false | –           | 550.0     | 0.76971                      | 1,500               | 800.0     |        |
| 19   | false | –           | 615.0     | 0.60811                      | 0.1                 | 1,600.0   |        |
| 20   | false | –           | 615.0     | 0.60811                      | 500                 | 1,600.0   |        |
| 21   | false | –           | 615.0     | 0.60811                      | 1,500               | 1,600.0   |        |
| 22   | false | –           | 585.0     | 0.70045                      | 0.1                 | 1,600.0   |        |
| 23   | false | –           | 585.0     | 0.70045                      | 500                 | 1,600.0   | A5, B5 |
| 24   | false | –           | 585.0     | 0.70045                      | 1,500               | 1,600.0   |        |
| 25   | false | –           | 550.0     | 0.76971                      | 0.1                 | 1,600.0   |        |
| 26   | false | –           | 550.0     | 0.76971                      | 500                 | 1,600.0   |        |
| 27   | false | –           | 550.0     | 0.76971                      | 1,500               | 1,600.0   |        |
| 28   | true  | –           | 615.0     | 0.60811                      | 0.1                 | 800.0     |        |
| 29   | true  | –           | 615.0     | 0.60811                      | 500                 | 800.0     |        |
| 30   | true  | –           | 615.0     | 0.60811                      | 1,500               | 800.0     |        |
| 31   | true  | –           | 585.0     | 0.70045                      | 0.1                 | 800.0     |        |
| 32   | true  | –           | 585.0     | 0.70045                      | 500                 | 800.0     | A8, B8 |
| 33   | true  | –           | 585.0     | 0.70045                      | 1,500               | 800.0     |        |
| 34   | true  | –           | 550.0     | 0.76971                      | 0.1                 | 800.0     |        |
| 35   | true  | –           | 550.0     | 0.76971                      | 500                 | 800.0     |        |
| 36   | true  | –           | 550.0     | 0.76971                      | 1,500               | 800.0     |        |

**Table 3.4. Branch states for reflectors.**

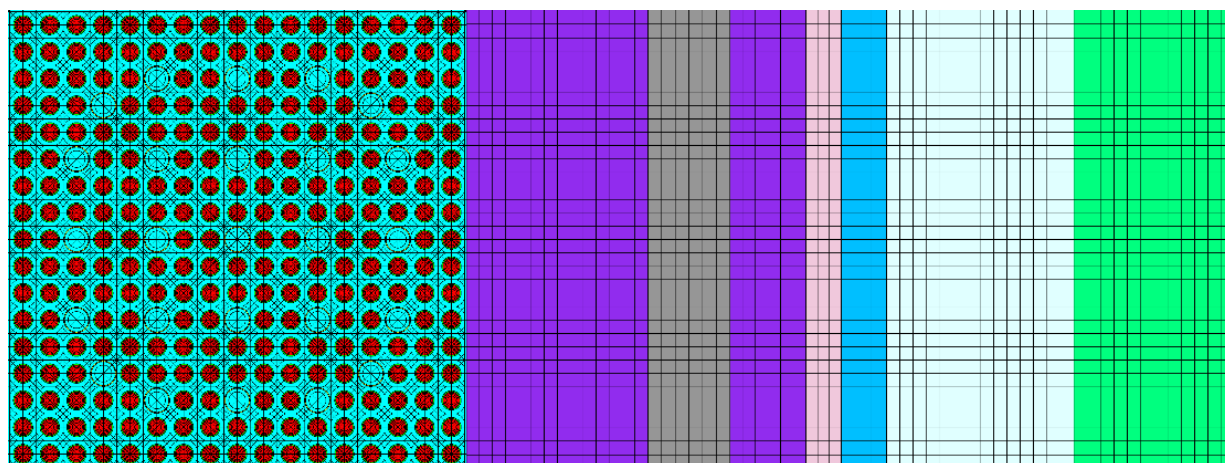
| Case             | Bank  |             | All       | Moderator                    |                     | Fuel      | Remark |
|------------------|-------|-------------|-----------|------------------------------|---------------------|-----------|--------|
|                  | BP    | Control rod | Temp. (K) | Density (g/cm <sup>3</sup> ) | Soluble boron (ppm) | Temp. (K) |        |
| <b>Reference</b> | false | false       | 550.0     | 0.76971                      | 600                 | 800.0     |        |
| 1                | false | false       | 550.0     | 0.76971                      | 0                   | 800.0     |        |
| 1                | false | false       | 550.0     | 0.76971                      | 1,800               | 800.0     |        |
| 2                | false | false       | 580.0     | 0.70045                      | 600                 | 800.0     |        |



(A) Radial reflector.



(B) Bottom axial reflector.



(C) Top axial reflector.

**Figure 3.1. Polaris models for reflectors.**

**Table 3.5. Polaris input and output files for WBN1.**

| Assembly type    | Dir          | File name                      | Description                                                                                                                                                                                                                      |
|------------------|--------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.10_bp00        | ./spacer_000 | 2.1-CR?-?DC-?PC-?TF.*          | Dir<br>./spacer_000: no spacer grid<br>./spacer_230: Zr-4 spacer grid<br>./reflector: reflector                                                                                                                                  |
| 2.60_bp00        | ./spacer_000 | 2.6-CR?-?DC-?PC-?TF.*          |                                                                                                                                                                                                                                  |
| 2.60_bp16        | ./spacer_000 | 2.6_bp16-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 2.60_bp20        | ./spacer_000 | 2.6_bp20-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 2.60_bp24        | ./spacer_000 | 2.6_bp24-CR?-?DC-?PC-?TF.*     | Input, output, T16<br>CR? Control rod<br>CR0: out<br>CR1: in<br>?DC moderator density<br>LDC: 0.60811<br>MDC: 0.70045<br>HDC: 0.76971                                                                                            |
| 3.10_bp00        | ./spacer_000 | 3.1-CR?-?DC-?PC-?TF.*          |                                                                                                                                                                                                                                  |
| 3.10_bp08        | ./spacer_000 | 3.1_bp08-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 3.10_bp12        | ./spacer_000 | 3.1_bp12-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 3.10_bp16        | ./spacer_000 | 3.1_bp16-CR?-?DC-?PC-?TF.*     | ?PC boron concentration<br>LPC: 0 ppm<br>MPC: 500 ppm<br>HPC: 1500 ppm                                                                                                                                                           |
| 3.10_bp24        | ./spacer_000 | 3.1_bp24-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 2.60_blank       | ./spacer_000 | BLKT-CR?-?DC-?PC-?TF.*         |                                                                                                                                                                                                                                  |
| 2.60_t08_blank   | ./spacer_000 | BLKT_t08-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 3.70_bp00        | ./spacer_000 | 3.7-CR?-?DC-?PC-?TF.*          | ?TF Fuel temperature<br>LTF: 560 K<br>MTF: 800 K<br>HTF: 1600 K                                                                                                                                                                  |
| 3.70_i048        | ./spacer_000 | 3.7_i048-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 3.70_i104_t08    | ./spacer_000 | 3.7_i104_t08-CR?-?DC-?PC-?TF.* |                                                                                                                                                                                                                                  |
| 3.70_i104_w04    | ./spacer_000 | 3.7_i104_w04-CR?-?DC-?PC-?TF.* |                                                                                                                                                                                                                                  |
| 3.70_i104_w08    | ./spacer_000 | 3.7_i104_w08-CR?-?DC-?PC-?TF.* | * File names<br>inp: input file<br>idc: IDC format input file<br>msg: message file<br>out: output file<br>png: geometry image<br>t16: few-group cross sections<br>f71: ORIGEN concentration<br>f33: ORIGEN cross section library |
| 3.70_i128        | ./spacer_000 | 3.7_i128-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 3.70_t08         | ./spacer_000 | 3.7_t08-CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| 3.70_w04         | ./spacer_000 | 3.7_w04-CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| 3.70_w08         | ./spacer_000 | 3.7_w08-CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| 3.70_t08         | ./spacer_000 | 3.7_t08-CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| 3.80_i000        | ./spacer_000 | B5A0I-CR?-?DC-?PC-?TF.*        |                                                                                                                                                                                                                                  |
| 3.80_i104        | ./spacer_000 | B5A104I-CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| 3.80_i128        | ./spacer_000 | B5A128I-CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| 4.40_i000        | ./spacer_000 | B5B0I-CR?-?DC-?PC-?TF.*        |                                                                                                                                                                                                                                  |
| 4.40_w08         | ./spacer_000 | B5B0I_w08-CR?-?DC-?PC-?TF.*    |                                                                                                                                                                                                                                  |
| 4.40_i128        | ./spacer_000 | B5B128I -CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 4.40_i128_w08    | ./spacer_000 | B5B128I_w08 -CR?-?DC-?PC-?TF.* |                                                                                                                                                                                                                                  |
| 4.40_i16         | ./spacer_000 | B5B16I -CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| 4.40_i16_blank   | ./spacer_000 | BL16I-CR?-?DC-?PC-?TF.*        |                                                                                                                                                                                                                                  |
| 4.40_i104_blank  | ./spacer_000 | BL104I-CR?-?DC-?PC-?TF.*       |                                                                                                                                                                                                                                  |
| 4.40_i128_blank  | ./spacer_000 | BL128I-CR?-?DC-?PC-?TF.*       |                                                                                                                                                                                                                                  |
| 2.10_bp00        | ./spacer_230 | 2.1-CR?-?DC-?PC-?TF.*          |                                                                                                                                                                                                                                  |
| 2.60_bp00        | ./spacer_230 | 2.6-CR?-?DC-?PC-?TF.*          |                                                                                                                                                                                                                                  |
| 2.60_bp16        | ./spacer_230 | 2.6_bp16-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 2.60_bp20        | ./spacer_230 | 2.6_bp20-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 2.60_bp24        | ./spacer_230 | 2.6_bp24-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 3.10_bp00        | ./spacer_230 | 3.1-CR?-?DC-?PC-?TF.*          |                                                                                                                                                                                                                                  |
| 3.10_bp08        | ./spacer_230 | 3.1_bp08-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 3.10_bp12        | ./spacer_230 | 3.1_bp12-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 3.10_bp16        | ./spacer_230 | 3.1_bp16-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 3.10_bp24        | ./spacer_230 | 3.1_bp24-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 3.70_bp00        | ./spacer_230 | 3.7-CR?-?DC-?PC-?TF.*          |                                                                                                                                                                                                                                  |
| 3.70_i048        | ./spacer_230 | 3.7_i048-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 3.70_i104_t08    | ./spacer_230 | 3.7_i104_t08-CR?-?DC-?PC-?TF.* |                                                                                                                                                                                                                                  |
| 3.70_i104_w04    | ./spacer_230 | 3.7_i104_w04-CR?-?DC-?PC-?TF.* |                                                                                                                                                                                                                                  |
| 3.70_i104_w08    | ./spacer_230 | 3.7_i104_w08-CR?-?DC-?PC-?TF.* |                                                                                                                                                                                                                                  |
| 3.70_i128        | ./spacer_230 | 3.7_i128-CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 3.70_t08         | ./spacer_230 | 3.7_t08-CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| 3.70_w04         | ./spacer_230 | 3.7_w04-CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| 3.70_w08         | ./spacer_230 | 3.7_w08-CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| 3.70_t08         | ./spacer_230 | 3.7_t08-CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| 3.80_i000        | ./spacer_230 | B5A0I-CR?-?DC-?PC-?TF.*        |                                                                                                                                                                                                                                  |
| 3.80_i104        | ./spacer_230 | B5A104I-CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| 3.80_i128        | ./spacer_230 | B5A128I-CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| 4.40_i000        | ./spacer_230 | B5B0I-CR?-?DC-?PC-?TF.*        |                                                                                                                                                                                                                                  |
| 4.40_w08         | ./spacer_230 | B5B0I_w08-CR?-?DC-?PC-?TF.*    |                                                                                                                                                                                                                                  |
| 4.40_i128        | ./spacer_230 | B5B128I -CR?-?DC-?PC-?TF.*     |                                                                                                                                                                                                                                  |
| 4.40_i128_w08    | ./spacer_230 | B5B128I_w08 -CR?-?DC-?PC-?TF.* |                                                                                                                                                                                                                                  |
| 4.40_i16         | ./spacer_230 | B5B16I -CR?-?DC-?PC-?TF.*      |                                                                                                                                                                                                                                  |
| Radial reflector | ./reflector  | polaris_rad.*                  |                                                                                                                                                                                                                                  |
| Top reflector    | ./reflector  | polaris_top.*                  |                                                                                                                                                                                                                                  |
| Bottom reflector | ./reflector  | polaris_bot.*                  |                                                                                                                                                                                                                                  |

### 3.1.2 Cross-Section Table-Set Generation Using GenPMAXS

Appendices B.1–B.4 provide the GenPMAXS inputs for the fuel assembly 2.60\_bp16 and radial edge, axial top, and bottom reflectors.

- B.1 GenPMAXS input for the fuel assembly 2.60\_bp16
- B.2 GenPMAXS input for radial edge reflector
- B.3 GenPMAXS input for axial top reflector
- B.4 GenPMAXS input for axial bottom reflector

Table 3.6 provides the GenPMAXS input and output files. There can be various GenPMAXS input files for various radial reflectors including edge, east, south-east, and south reflectors.

**Table 3.6. GenPMAXS input and output files for WBN1.**

| Assembly type   | Dir          | Filename       | Description                                                                                                                                                   |
|-----------------|--------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.10_bp00       | ./spacer_000 | 2.1.*          | Dir<br>./bare: no spacer grid<br>./zirc: Zr-4 spacer grid<br>*<br>inp: input files<br>kinf: kinf summary<br>out: output file<br>PMAX: cross section table-set |
| 2.60_bp00       | ./spacer_000 | 2.6.*          |                                                                                                                                                               |
| 2.60_bp16       | ./spacer_000 | 2.6_bp16.*     |                                                                                                                                                               |
| 2.60_bp20       | ./spacer_000 | 2.6_bp20.*     |                                                                                                                                                               |
| 2.60_bp24       | ./spacer_000 | 2.6_bp24.*     |                                                                                                                                                               |
| 3.10_bp00       | ./spacer_000 | 3.1.*          |                                                                                                                                                               |
| 3.10_bp08       | ./spacer_000 | 3.1_bp08.*     |                                                                                                                                                               |
| 3.10_bp12       | ./spacer_000 | 3.1_bp12.*     |                                                                                                                                                               |
| 3.10_bp16       | ./spacer_000 | 3.1_bp16.*     |                                                                                                                                                               |
| 3.10_bp24       | ./spacer_000 | 3.1_bp24.*     |                                                                                                                                                               |
| 2.60_blank      | ./spacer_000 | BLKT.*         |                                                                                                                                                               |
| 2.60_t08_blank  | ./spacer_000 | BLKT_t08.*     |                                                                                                                                                               |
| 3.70_bp00       | ./spacer_000 | 3.7.*          |                                                                                                                                                               |
| 3.70_i048       | ./spacer_000 | 3.7_i048.*     |                                                                                                                                                               |
| 3.70_i104_t08   | ./spacer_000 | 3.7_i104.*     |                                                                                                                                                               |
| 3.70_i104_w04   | ./spacer_000 | 3.7_i104_t08.* |                                                                                                                                                               |
| 3.70_i104_w08   | ./spacer_000 | 3.7_i104_w04.* |                                                                                                                                                               |
| 3.70_i128       | ./spacer_000 | 3.7_i104_w08.* |                                                                                                                                                               |
| 3.70_t08        | ./spacer_000 | 3.7_i128.*     |                                                                                                                                                               |
| 3.70_w04        | ./spacer_000 | 3.7_t08.*      |                                                                                                                                                               |
| 3.70_w08        | ./spacer_000 | 3.7_w04.*      |                                                                                                                                                               |
| 3.70_t08        | ./spacer_000 | 3.7_w08.*      |                                                                                                                                                               |
| 3.80_i000       | ./spacer_000 | B5A0I.*        |                                                                                                                                                               |
| 3.80_i104       | ./spacer_000 | B5A104I.*      |                                                                                                                                                               |
| 3.80_i128       | ./spacer_000 | B5A128I.*      |                                                                                                                                                               |
| 4.40_i000       | ./spacer_000 | B5B0I.*        |                                                                                                                                                               |
| 4.40_w08        | ./spacer_000 | B5B0I_w08.*    |                                                                                                                                                               |
| 4.40_i128       | ./spacer_000 | B5B128I.*      |                                                                                                                                                               |
| 4.40_i128_w08   | ./spacer_000 | B5B128I_w08.*  |                                                                                                                                                               |
| 4.40_i16        | ./spacer_000 | B5B16I.*       |                                                                                                                                                               |
| 4.40_i16_blank  | ./spacer_000 | BL16I.*        |                                                                                                                                                               |
| 4.40_i104_blank | ./spacer_000 | BL104I.*       |                                                                                                                                                               |
| 4.40_i128_blank | ./spacer_000 | BL128I.*       |                                                                                                                                                               |
| 2.10_bp00       | ./spacer_230 | 2.1.*          |                                                                                                                                                               |
| 2.60_bp00       | ./spacer_230 | 2.6.*          |                                                                                                                                                               |
| 2.60_bp16       | ./spacer_230 | 2.6_bp16.*     |                                                                                                                                                               |
| 2.60_bp20       | ./spacer_230 | 2.6_bp20.*     |                                                                                                                                                               |
| 2.60_bp24       | ./spacer_230 | 2.6_bp24.*     |                                                                                                                                                               |
| 3.10_bp00       | ./spacer_230 | 3.1.*          |                                                                                                                                                               |
| 3.10_bp08       | ./spacer_230 | 3.1_bp08.*     |                                                                                                                                                               |
| 3.10_bp12       | ./spacer_230 | 3.1_bp12.*     |                                                                                                                                                               |
| 3.10_bp16       | ./spacer_230 | 3.1_bp16.*     |                                                                                                                                                               |
| 3.10_bp24       | ./spacer_230 | 3.1_bp24.*     |                                                                                                                                                               |
| 3.70_bp00       | ./spacer_230 | 3.7.*          |                                                                                                                                                               |

**Table 3.6. GenPMAXS input and output files for WBN1. (continued)**

| Assembly type               | Dir          | Filename        | Description |
|-----------------------------|--------------|-----------------|-------------|
| 3.70_i048                   | ./spacer_230 | 3.7_i048.*      |             |
| 3.70_i104_t08               | ./spacer_230 | 3.7_i104.*      |             |
| 3.70_i104_w04               | ./spacer_230 | 3.7_i104_t08.*  |             |
| 3.70_i104_w08               | ./spacer_230 | 3.7_i104_w04.*  |             |
| 3.70_i128                   | ./spacer_230 | 3.7_i104_w08.*  |             |
| 3.70_t08                    | ./spacer_230 | 3.7_i128.*      |             |
| 3.70_w04                    | ./spacer_230 | 3.7_t08.*       |             |
| 3.70_w08                    | ./spacer_230 | 3.7_w04.*       |             |
| 3.70_t08                    | ./spacer_230 | 3.7_w08.*       |             |
| 3.80_i000                   | ./spacer_230 | B5A0I.*         |             |
| 3.80_i104                   | ./spacer_230 | B5A104I.*       |             |
| 3.80_i128                   | ./spacer_230 | B5A128I.*       |             |
| 4.40_i000                   | ./spacer_230 | B5B0I.*         |             |
| 4.40_w08                    | ./spacer_230 | B5B0I_w08.*     |             |
| 4.40_i128                   | ./spacer_230 | B5B128I.*       |             |
| 4.40_i128_w08               | ./spacer_230 | B5B128I_w08.*   |             |
| 4.40_i16                    | ./spacer_230 | B5B16I.*        |             |
| Radial reflector diagonal   | ./bare       | refl_rad_diag.* |             |
| Radial reflector edge       | ./bare       | refl_rad_edge.* |             |
| Radial reflector east       | ./bare       | refl_rad_E.*    |             |
| Radial reflector south east | ./bare       | refl_rad_SE.*   |             |
| Radial reflector south      | ./bare       | refl_rad_S.*    |             |
| Top reflector               | ./bare       | refl_top.*      |             |
| Bottom reflector            | ./bare       | refl_bot.*      |             |

### 3.1.3 Whole-Core Nodal Diffusion Calculations Using PARCS

Appendices C.1–C.6 provide the PARCS inputs for the WBN1 cycles-1 and 2 HZP and HFP calculations.

- C.1 PARCS input for the cross section set “pmax.dir” file
- C.2 PARCS input for the assembly geometry “depl\_assm.geom” file
- C.3 PARCS input for the in-core detector map “depl\_detect.geom” file
- C.4 PARCS input for the HFP cycle 1
- C.5 PARCS input for the HZP cycle 2 fuel shuffling
- C.6 PARCS input for the HFP cycle 2
- C.7 PARCS input for the HZP cycle 3 fuel shuffling
- C.8 PARCS input for the HFP cycle 3

Table 3.7 provides the PARCS input and output files.

**Table 3.7. PARCS input and output files for WBN1.**

| Cycle | Condition | File name                                    | Description              |
|-------|-----------|----------------------------------------------|--------------------------|
| All   |           | depl_assm.geom                               | Assembly geometry        |
|       |           | depl_detect.geom                             | Detector                 |
|       |           | pmax.dir                                     | Cross section table-sets |
| 1     | HZIP      | wbn1_c1_zppt_01-100.*                        |                          |
|       |           | wbn1_c1_zppt_01.*                            |                          |
|       |           | wbn1_c1_zppt_02.*                            |                          |
|       |           | wbn1_c1_zppt_03.*                            |                          |
|       |           | wbn1_c1_zppt_04.*                            |                          |
|       |           | wbn1_c1_zppt_05.*                            |                          |
|       |           | wbn1_c1_zppt_06.*                            |                          |
|       |           | wbn1_c1_zppt_07.*                            |                          |
|       |           | wbn1_c1_zppt_08.*                            |                          |
|       |           | wbn1_c1_zppt_09.*                            |                          |
|       |           | wbn1_c1_zppt_10.*                            |                          |
|       |           | wbn1_c1_zppt_11.*                            |                          |
|       |           | wbn1_c1_zppt_12.*                            |                          |
|       |           | wbn1_c1_zppt_13.*                            |                          |
|       |           | wbn1_c1_zppt_14.*                            |                          |
|       |           | wbn1_c1_zppt_15.*                            |                          |
|       |           | wbn1_c1_zppt_16.*                            |                          |
|       |           | wbn1_c1_zppt_17.*                            |                          |
|       |           | wbn1_c1_zppt_18.*                            |                          |
|       |           | wbn1_c1_zppt_19.*                            |                          |
|       |           | wbn1_c1_zppt_20.*                            |                          |
|       |           | wbn1_c1_zppt_21.*                            |                          |
|       | HFP       | wbn1_c1.inp                                  | PARCS input              |
|       |           | wbn1_cycle1_hfp.parc_out                     | output                   |
|       |           | wbn1_cycle1_hfp.parc_cyc-01                  | Restart file             |
|       |           | wbn1_cycle1_hfp.parc_det                     | Detector response        |
|       |           | wbn1_cycle1_hfp.parc_dpl                     | Summary                  |
|       |           | ./cy1_pinpower/wbn1_cycle1_hfp.parc_pin*     | Pin powers               |
| 2     | HZIP      | wbn1_c1_c2_shuf.inp                          | PARCS input              |
|       |           | wbn1_cycle2_hzp_BOC.parc_out                 | output                   |
|       |           | wbn1_cycle2_hzp_BOC.parc_cyc-02              | Restart file             |
|       |           | wbn1_cycle2_hzp_BOC.parc_det                 | Detector response        |
|       |           | wbn1_cycle2_hzp_BOC.parc_dpl                 | Summary                  |
|       |           | ./cy2_hzp_pinpower/wbn1_cycle2_hzp.parc_pin* | Pin powers               |
|       | HFP       | wbn1_c2.inp                                  | PARCS input              |
|       |           | wbn1_cycle2_hfp.parc_out                     | output                   |
|       |           | wbn1_cycle2_hfp.parc_cyc-01                  | Restart file             |
|       |           | wbn1_cycle2_hfp.parc_det                     | Detector response        |
|       |           | wbn1_cycle2_hfp.parc_dpl                     | Summary                  |
|       |           | ./cy2_pinpower/wbn1_cycle2_hfp.parc_pin*     | Pin powers               |
| 3     | HZIP      | wbn1_c2_c3_shuf.inp                          | PARCS input              |
|       |           | wbn1_cycle3_hzp_BOC.parc_out                 | output                   |
|       |           | wbn1_cycle3_hzp_BOC.parc_cyc-02              | Restart file             |
|       |           | wbn1_cycle3_hzp_BOC.parc_det                 | Detector response        |
|       |           | wbn1_cycle3_hzp_BOC.parc_dpl                 | Summary                  |
|       |           | ./cy3_hzp_pinpower/wbn1_cycle2_hzp.parc_pin* | Pin powers               |
|       | HFP       | wbn1_c3.inp                                  | PARCS input              |
|       |           | wbn1_cycle3_hfp.parc_out                     | output                   |
|       |           | wbn1_cycle3_hfp.parc_cyc-01                  | Restart file             |
|       |           | wbn1_cycle3_hfp.parc_det                     | Detector response        |
|       |           | wbn1_cycle3_hfp.parc_dpl                     | Summary                  |
|       |           | ./cy3_pinpower/wbn1_cycle3_hfp.parc_pin*     | Pin powers               |



## 3.2 BENCHMARK RESULTS

### 3.2.1 HZP Physics Tests

Table 3.8 compares the control bank worths, critical boron concentrations, isothermal thermal temperature coefficients (ITCs), and boron worth between the measured data and the calculated results in the WBN1 cycle 1 HZP physics tests.

**Table 3.8. Comparison of the control bank worth and ITC for WBN1.**

| Cycle | Case | Critical boron (ppm) |       | Control bank worth (pcm) |       |       | ITC (pcm/°F) |       |       | Boron worth (pcm/ppm) |        |       |
|-------|------|----------------------|-------|--------------------------|-------|-------|--------------|-------|-------|-----------------------|--------|-------|
|       |      | Meas.                | Calc. | Meas.                    | Calc. | Diff. | Meas.        | Calc. | Diff. | Meas.                 | Calc.  | Diff. |
| 1     | ARO  | 1,291                | 1281  |                          |       |       | -2.17        | -3.39 | -1.22 | -10.77                | -10.18 | 0.59  |
|       | A    |                      |       | 843                      | 977   | 134   |              |       |       |                       |        |       |
|       | B    |                      |       | 879                      | 840   | -39   |              |       |       |                       |        |       |
|       | C    |                      |       | 951                      | 1,031 | 80    |              |       |       |                       |        |       |
|       | D    |                      |       | 1,342                    | 1,450 | 108   |              |       |       |                       |        |       |
|       | SA   |                      |       | 435                      | 421   | -14   |              |       |       |                       |        |       |
|       | SB   |                      |       | 1,056                    | 1,077 | 21    |              |       |       |                       |        |       |
|       | SC   |                      |       | 480                      | 458   | -22   |              |       |       |                       |        |       |
|       | SD   |                      |       | 480                      | 458   | -22   |              |       |       |                       |        |       |
|       | All  |                      |       | 6,466                    | 6,713 | 247   |              |       |       |                       |        |       |

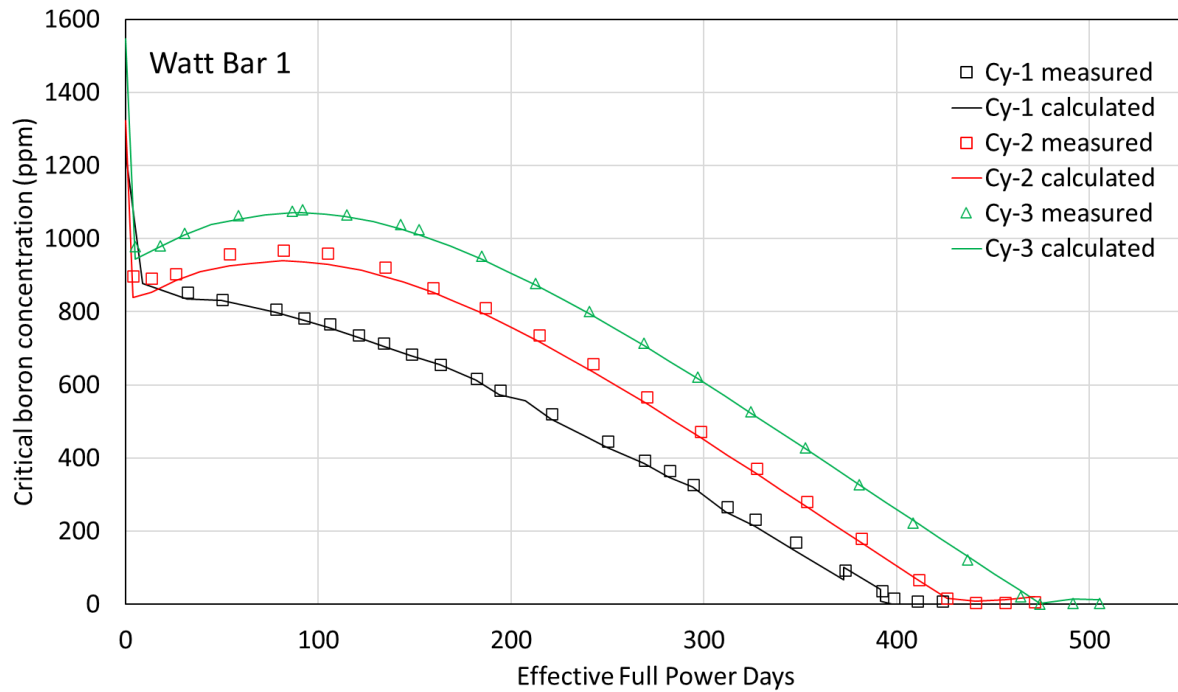
### 3.2.2 HFP Results

Comparison of the HFP critical boron concentrations is illustrated in Figure 3.2 and Table 3.9. Figure 3.3 illustrates the boron worths as a function burnup which are used in converting the differences of critical boron concentrations between measurement and simulation into reactivity differences. Since boron worths are changing smoothly as burnup increases, boron worths were obtained only for the selected burnup points. The enthalpy-rise hot channel factors ( $F_{\Delta H}^N$ ) for WBN1 cycles 1–3 are provided in Figure 3.4, which does not satisfy the limit of 1.435 considering with 8% uncertainty. The  $F_{\Delta H}^N$  limit comes from departure from nucleate boiling. This violation might come from no gamma smeared pin power form factors, which should be addressed by performing neutron-gamma coupled calculation and considering gamma energy deposition.

Figure 3.5 and Table 3.10 provide comparisons of the calculated flux maps with the measured flux maps for the 3D axially integrated 2D radial and radially integrated 1D axial distributions. Appendix B provides the utility program used to compare the calculated results with the measured data. The total 3D, 2D, and 1D RMS errors for WBN1 are 4.55% 1.66%, and 3.35%, respectively. Appendix D provides the utility program to compare the calculated flux map results with the measured data.

- D.1 Program source
- D.2 Cycle 1 input
- D.3 Cycle 2 input
- D.4 Cycle 3 input
- D.5 Cycle 1 flux map measured data
- D.6 Cycle 2 flux map measured data
- D.7 Cycle 3 flux map measured data

Table 3.11 compares the total 3D, 2D, and 1D flux map RMS errors for WBN1 between the VERA and the Polaris–PARCS results compared with the measured data. Although the 2D RMS errors of the Polaris–PARCS results are very comparable to the RMS errors of the VERA results, there are about 1.5% differences in the 1D and 3D RMS errors between the VERA and Polaris–PARCS results. This indicates that the axial reflector models for Polaris–PARCS need to be improved. Figure 3.6 compares the 1D axial flux maps at burnups 1.92 and 13.36 MWd/kgU for WBN1 cycle 1, which indicates some axial power tilt—probably resulting from poor axial reflector cross sections.



**Figure 3.2. WBN1 cycles 1–3 HFP boron letdown comparison.**

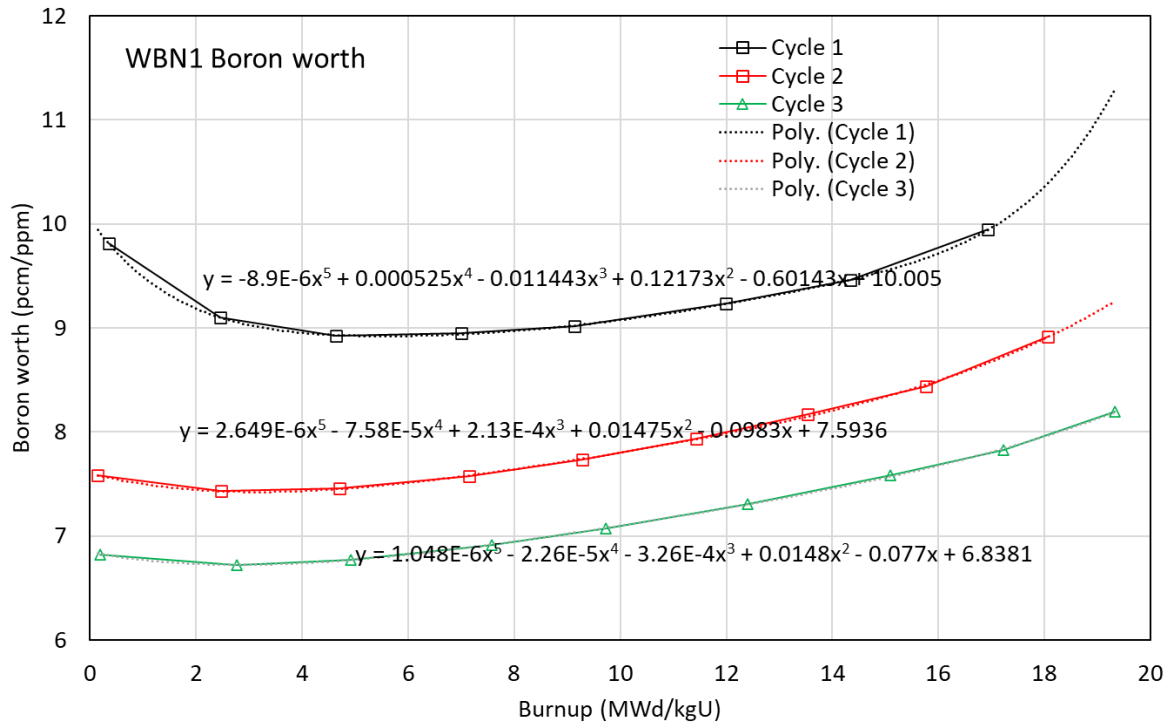


Figure 3.3. Boron worths as a function of burnup for WBN1 cycles 1–3.

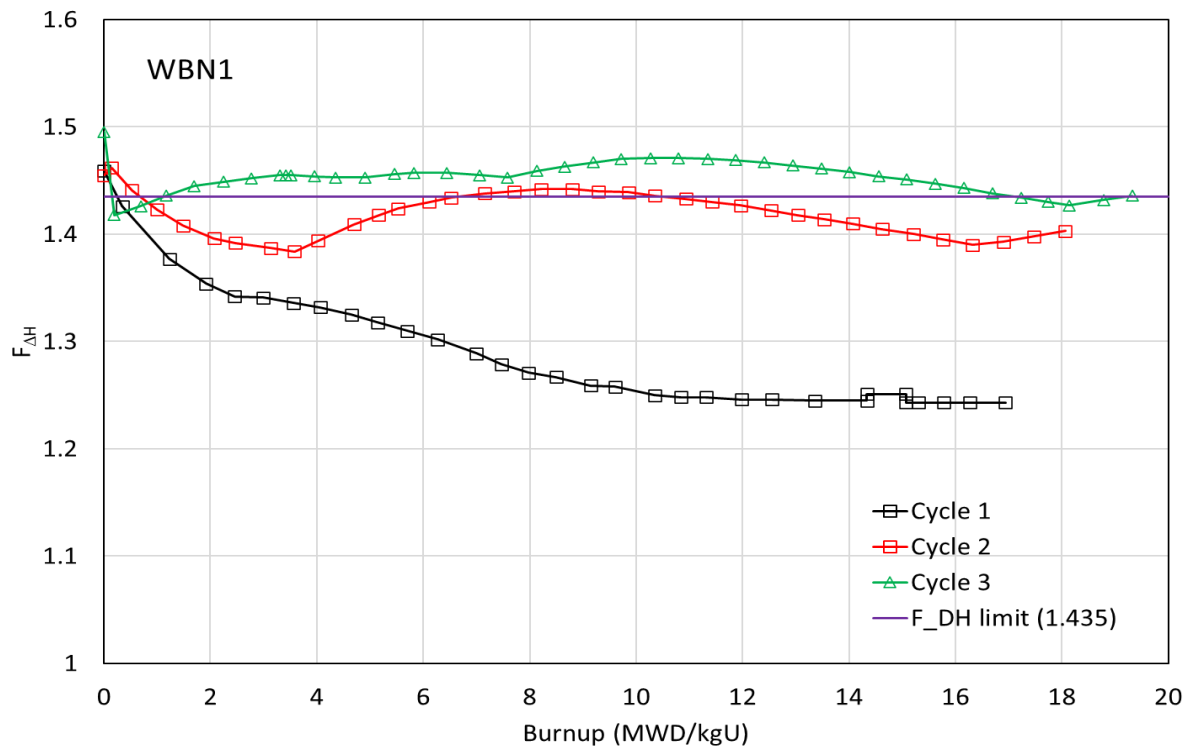


Figure 3.4. Enthalpy rise hot channel factors for WBN1 cycles 1–3.

**Table 3.9. Comparison of the critical boron concentrations.**

| Cycle 1             |                   |                      |              |       |                   | Cycle 2 |       |                      |              |       |                   | Cycle 3 |       |                      |              |       |                   |
|---------------------|-------------------|----------------------|--------------|-------|-------------------|---------|-------|----------------------|--------------|-------|-------------------|---------|-------|----------------------|--------------|-------|-------------------|
| Burnup <sup>a</sup> | EFPD <sup>b</sup> | Critical boron (ppm) |              |       | $\Delta p$<br>pcm | Burnup  | EFPD  | Critical boron (ppm) |              |       | $\Delta p$<br>pcm | Burnup  | EFPD  | Critical boron (ppm) |              |       | $\Delta p$<br>pcm |
|                     |                   | Meas.<br>[a]         | Calc.<br>[b] | [a-b] |                   |         |       | Meas.<br>[c]         | Calc.<br>[d] | [c-d] |                   |         |       | Meas.<br>[e]         | Calc.<br>[f] | [e-f] |                   |
| 1.232               | 32                | 853                  | 834.2        | 18.8  | -177              | 8.937   | 3.7   | 898                  | 840          | 58    | -440              | 13.008  | 5.1   | 979                  | 944.8        | 34.2  | -233              |
| 1.925               | 50                | 834                  | 831.2        | 2.8   | -26               | 9.316   | 13.6  | 892                  | 853.4        | 38.6  | -291              | 13.505  | 18.1  | 980                  | 978.8        | 1.2   | -8                |
| 3.002               | 78                | 807                  | 797.9        | 9.1   | -82               | 9.795   | 26.1  | 904                  | 885.9        | 18.1  | -136              | 13.979  | 30.5  | 1015                 | 1011.5       | 3.5   | -24               |
| 3.568               | 92.7              | 783                  | 776.4        | 6.6   | -59               | 10.86   | 53.9  | 958                  | 925.7        | 32.3  | -241              | 15.058  | 58.7  | 1062                 | 1052.7       | 9.3   | -63               |
| 4.073               | 105.8             | 767                  | 755.1        | 11.9  | -106              | 11.926  | 81.7  | 968                  | 940.5        | 27.5  | -204              | 16.117  | 86.4  | 1076                 | 1071.4       | 4.6   | -31               |
| 4.654               | 120.9             | 737                  | 728.9        | 8.1   | -72               | 12.815  | 104.9 | 960                  | 929.9        | 30.1  | -224              | 16.324  | 91.8  | 1079                 | 1070.8       | 8.2   | -55               |
| 5.15                | 133.8             | 714                  | 705.8        | 8.2   | -73               | 13.956  | 134.7 | 921                  | 896          | 25    | -187              | 17.203  | 114.8 | 1064                 | 1059.7       | 4.3   | -29               |
| 5.712               | 148.4             | 684                  | 678.8        | 5.2   | -46               | 14.907  | 159.5 | 865                  | 852.2        | 12.8  | -96               | 18.27   | 142.7 | 1039                 | 1026.2       | 12.8  | -87               |
| 6.286               | 163.3             | 655                  | 654.6        | 0.4   | -4                | 15.941  | 186.5 | 810                  | 791.9        | 18.1  | -137              | 18.638  | 152.3 | 1024                 | 1010.6       | 13.4  | -91               |
| 7.013               | 182.2             | 618                  | 611.8        | 6.2   | -55               | 17.014  | 214.5 | 737                  | 717.3        | 19.7  | -151              | 19.873  | 184.6 | 953                  | 944          | 9     | -62               |
| 7.479               | 194.3             | 585                  | 572.6        | 12.4  | -111              | 18.083  | 242.4 | 658                  | 635.6        | 22.4  | -173              | 20.947  | 212.7 | 878                  | 872.7        | 5.3   | -37               |
| 8.511               | 221.1             | 521                  | 504.3        | 16.7  | -150              | 19.152  | 270.3 | 567                  | 548.3        | 18.7  | -147              | 22.015  | 240.6 | 801                  | 793.7        | 7.3   | -51               |
| 9.623               | 250               | 446                  | 428.4        | 17.6  | -159              | 20.225  | 298.3 | 473                  | 455.5        | 17.5  | -139              | 23.089  | 268.7 | 714                  | 707.3        | 6.7   | -48               |
| 10.366              | 269.3             | 393                  | 382.9        | 10.1  | -92               | 21.337  | 327.3 | 372                  | 356.5        | 15.5  | -125              | 24.16   | 296.7 | 622                  | 616.4        | 5.6   | -40               |
| 10.867              | 282.3             | 365                  | 345.7        | 19.3  | -176              | 22.333  | 353.3 | 281                  | 265.7        | 15.3  | -125              | 25.223  | 324.5 | 527                  | 522.8        | 4.2   | -31               |
| 11.34               | 294.6             | 326                  | 317.8        | 8.2   | -75               | 23.413  | 381.5 | 179                  | 168.7        | 10.3  | -85               | 26.302  | 352.7 | 427                  | 425.4        | 1.6   | -12               |
| 12.014              | 312.1             | 267                  | 251.2        | 15.8  | -146              | 24.559  | 411.4 | 67                   | 64.6         | 2.4   | -20               | 27.372  | 380.7 | 327                  | 327.5        | -0.5  | 4                 |
| 12.579              | 326.8             | 233                  | 210.9        | 22.1  | -205              | 25.115  | 425.9 | 16                   | 16.3         | -0.3  | 3                 | 28.436  | 408.5 | 223                  | 229.6        | -6.6  | 50                |
| 13.388              | 347.8             | 169                  | 144.9        | 24.1  | -225              | 25.697  | 441.1 | 5                    | 8.9          | -3.9  | 34                | 29.514  | 436.7 | 122                  | 131.5        | -9.5  | 74                |
| 14.366              | 373.2             | 93                   | 97.8         | -4.8  | 45                | 26.272  | 456.1 | 5                    | 12.4         | -7.4  | 65                | 30.562  | 464.1 | 21                   | 37.6         | -16.6 | 131               |
| 15.101              | 392.3             | 37                   | 7.8          | 29.2  | -280              | 26.857  | 471.4 | 7                    | 19.9         | -12.9 | 115               | 30.956  | 474.4 | 0                    | 2.8          | -2.8  | 22                |
| 15.343              | 398.6             | 16                   | 0.1          | 15.9  | -153              |         |       |                      |              |       |                   | 31.602  | 491.3 | 2                    | 13.7         | -11.7 | 95                |
| 15.809              | 410.7             | 9                    | 0.1          | 8.9   | -86               |         |       |                      |              |       |                   | 32.137  | 505.3 | 2                    | 13.3         | -11.3 | 93                |
| 16.306              | 423.6             | 9                    | 0.1          | 8.9   | -87               |         |       |                      |              |       |                   |         |       |                      |              |       |                   |

<sup>a</sup> Cycle burnup (MWD/kgU). <sup>b</sup> Effective full power day

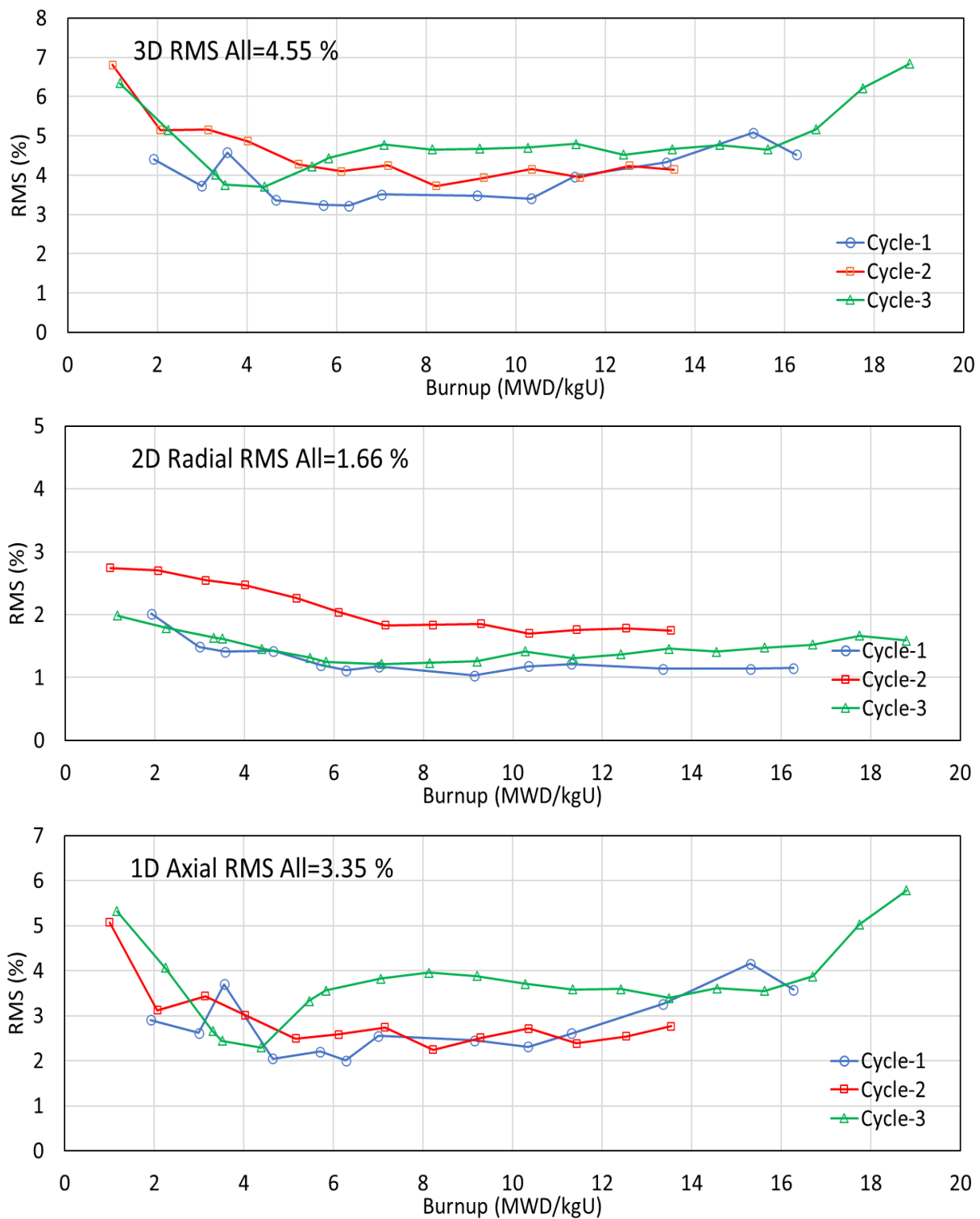


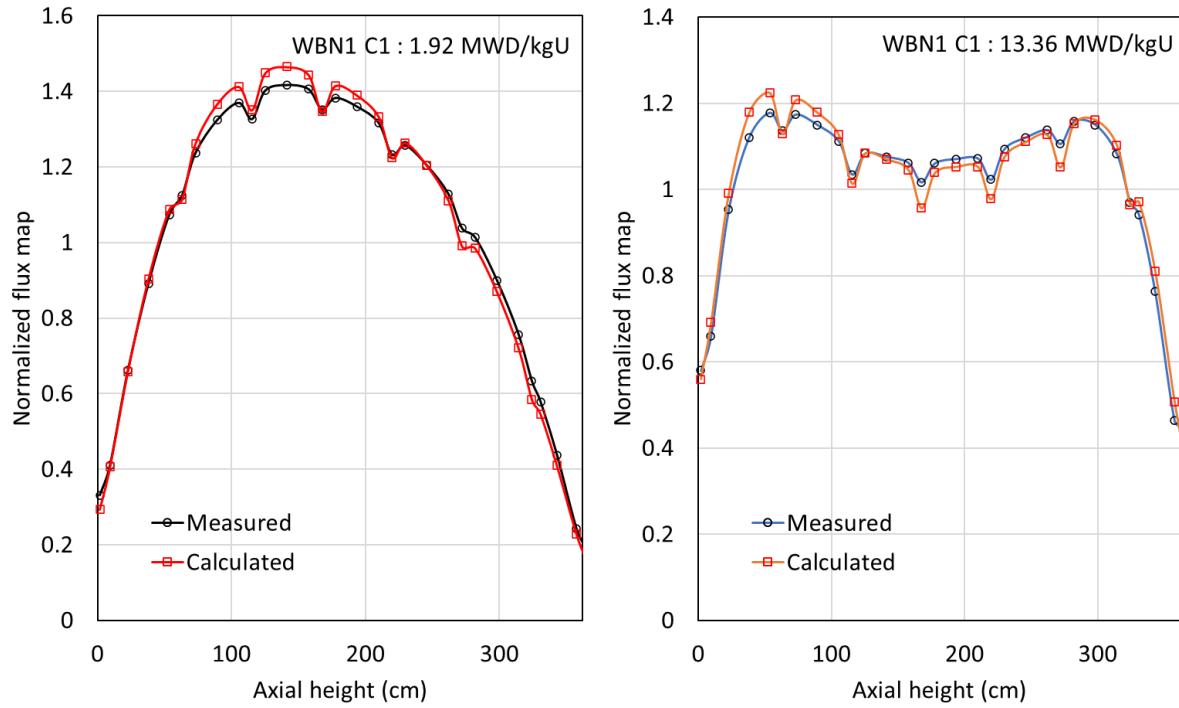
Figure 3.5. Comparison of the HFP flux maps for WBN1.

**Table 3.10. Comparison of the flux maps for WBN1.**

| Cycle | Data point | Burnup (MWD/kgU) | PARCS point | 3D     |        | Radial |       | Axial  |       |
|-------|------------|------------------|-------------|--------|--------|--------|-------|--------|-------|
|       |            |                  |             | RMS(%) | cases  | RMS(%) | cases | RMS(%) | cases |
| 1     | 9          | 1.919            | 4           | 4.408  | 1,740  | 2.014  | 58    | 2.908  | 30    |
|       | 10         | 2.994            | 6           | 3.724  | 1,710  | 1.487  | 57    | 2.619  | 30    |
|       | 11         | 3.562            | 7           | 4.583  | 1,710  | 1.411  | 57    | 3.703  | 30    |
|       | 12         | 4.642            | 9           | 3.371  | 1,710  | 1.421  | 57    | 2.05   | 30    |
|       | 13         | 5.7              | 11          | 3.244  | 1,710  | 1.199  | 57    | 2.2    | 30    |
|       | 14         | 6.273            | 12          | 3.224  | 1,710  | 1.113  | 57    | 2.004  | 30    |
|       | 15         | 7                | 13          | 3.514  | 1,650  | 1.173  | 55    | 2.548  | 30    |
|       | 17         | 9.14             | 17          | 3.483  | 1,650  | 1.034  | 55    | 2.451  | 30    |
|       | 19         | 10.344           | 19          | 3.406  | 1,740  | 1.182  | 58    | 2.311  | 30    |
|       | 20         | 11.314           | 21          | 3.96   | 1,740  | 1.215  | 58    | 2.611  | 30    |
|       | 22         | 13.36            | 24          | 4.323  | 1,740  | 1.141  | 58    | 3.26   | 30    |
|       | 25         | 15.308           | 29          | 5.086  | 1,740  | 1.141  | 58    | 4.151  | 30    |
|       | 26         | 16.27            | 31          | 4.527  | 1,740  | 1.15   | 58    | 3.577  | 30    |
| 2     | 3          | 0.999            | 4           | 6.812  | 1,680  | 2.743  | 56    | 5.073  | 30    |
|       | 4          | 2.065            | 6           | 5.149  | 1,650  | 2.702  | 55    | 3.122  | 30    |
|       | 5          | 3.131            | 8           | 5.16   | 1,650  | 2.55   | 55    | 3.433  | 30    |
|       | 6          | 4.02             | 10          | 4.863  | 1,680  | 2.469  | 56    | 3.017  | 30    |
|       | 7          | 5.16             | 12          | 4.281  | 1,650  | 2.263  | 55    | 2.495  | 30    |
|       | 8          | 6.112            | 14          | 4.095  | 1,650  | 2.04   | 55    | 2.582  | 30    |
|       | 9          | 7.148            | 16          | 4.252  | 1,650  | 1.833  | 55    | 2.742  | 30    |
|       | 10         | 8.22             | 18          | 3.724  | 1,650  | 1.837  | 55    | 2.245  | 30    |
|       | 11         | 9.287            | 20          | 3.939  | 1,650  | 1.854  | 55    | 2.512  | 30    |
|       | 12         | 10.359           | 22          | 4.151  | 1,650  | 1.698  | 55    | 2.719  | 30    |
|       | 13         | 11.429           | 24          | 3.947  | 1,650  | 1.757  | 55    | 2.389  | 30    |
|       | 14         | 12.54            | 26          | 4.243  | 1,650  | 1.782  | 55    | 2.539  | 30    |
|       | 15         | 13.539           | 28          | 4.149  | 1,650  | 1.747  | 55    | 2.763  | 30    |
| 3     | 5          | 1.165            | 4           | 6.345  | 1,560  | 1.986  | 52    | 5.323  | 30    |
|       | 6          | 2.246            | 6           | 5.148  | 1,560  | 1.791  | 52    | 4.063  | 30    |
|       | 7          | 3.305            | 8           | 4.014  | 1,560  | 1.638  | 52    | 2.655  | 30    |
|       | 8          | 3.511            | 10          | 3.756  | 1,560  | 1.616  | 52    | 2.444  | 30    |
|       | 9          | 4.39             | 12          | 3.706  | 1,560  | 1.458  | 52    | 2.291  | 30    |
|       | 10         | 5.457            | 14          | 4.23   | 1,560  | 1.317  | 52    | 3.331  | 30    |
|       | 11         | 5.825            | 15          | 4.437  | 1,560  | 1.248  | 52    | 3.556  | 30    |
|       | 12         | 7.061            | 17          | 4.78   | 1,560  | 1.216  | 52    | 3.824  | 30    |
|       | 13         | 8.135            | 19          | 4.658  | 1,560  | 1.232  | 52    | 3.958  | 30    |
|       | 14         | 9.2              | 21          | 4.671  | 1,500  | 1.254  | 50    | 3.88   | 30    |
|       | 15         | 10.274           | 23          | 4.705  | 1,560  | 1.418  | 52    | 3.712  | 30    |
|       | 16         | 11.347           | 25          | 4.798  | 1,560  | 1.303  | 52    | 3.58   | 30    |
|       | 17         | 12.409           | 27          | 4.52   | 1,560  | 1.367  | 52    | 3.594  | 30    |
|       | 18         | 13.49            | 29          | 4.663  | 1,560  | 1.457  | 52    | 3.397  | 30    |
|       | 19         | 14.558           | 31          | 4.769  | 1,560  | 1.409  | 52    | 3.61   | 30    |
|       | 20         | 15.622           | 33          | 4.661  | 1,560  | 1.478  | 52    | 3.552  | 30    |
|       | 21         | 16.7             | 35          | 5.172  | 1,560  | 1.521  | 52    | 3.87   | 30    |
|       | 22         | 17.75            | 37          | 6.217  | 1,560  | 1.667  | 52    | 5.034  | 30    |
|       | 23         | 18.788           | 39          | 6.833  | 1,560  | 1.586  | 52    | 5.781  | 30    |
| All   |            |                  |             | 4.548  | 73,380 | 1.656  | 2,446 | 3.351  | 1,350 |

**Table 3.11. Comparison of the flux maps for WBN1 (VERA vs. Polaris-PARCS).**

| Cycle | VERA RMS (%) |     |     | Polaris-PARCS RMS (%) |     |     |
|-------|--------------|-----|-----|-----------------------|-----|-----|
|       | 3D           | 2D  | 1D  | 3D                    | 2D  | 1D  |
| 1     | 3.0          | 1.3 | 1.8 | 4.0                   | 1.3 | 2.9 |
| 2     | 3.2          | 1.9 | 1.4 | 4.6                   | 2.1 | 3.0 |
| 3     | 3.3          | 1.5 | 2.4 | 4.9                   | 1.5 | 3.9 |



**Figure 3.6. Axial 1D flux map comparison for WBN1 cycle 1.**

#### 4. CONCLUSION

The benchmark calculations were performed for WBN1 cycles 1–3 using the SCALE 6.3/Polaris–PARCS v3.4.2 code package with the ENDF/B-VII.1 AMPX 56-group library which has been used in confirmatory analysis by US Nuclear Regulatory Commission. The benchmark results were compared with the measured data for the HZP physics test including control rod worths and ITC and the HFP critical boron concentrations and in-core detector flux maps. The final goal of this benchmark calculation is to validate the SCALE/Polaris–GenPMAX–PARCS code package by assessing uncertainties through statistical analysis for key nuclear parameters such as reactivity, temperature reactivity coefficients, control rod worth, and pin and assembly power peaking factors. [10]

The SCALE 6.3/Polaris–PARCS v3.4.2 code package with the ENDF/B-VII.1 AMPX 56-group library shows reasonable reactivity biases compared to other 2-step code packages. However, the flux map comparison between measurement and simulation shows relatively large RMS errors for axial and radial flux distributions which must be improved for better agreement by enhancing effective reflector cross sections. The PARCS code also needs to be improved to have better user convenience options such as automatic temperature coefficients and boron worth, and flexible in-core fuel management capability. The developed models for SCALE/Polaris, GenPMAXS and PARCS can be utilized in validating new version of the SCALE/Polaris–PARCS code package with new ENDF/B evaluated nuclear data.

The WBN1 cycles 1–3 design data and measured data were released to the public through the CASL project and the OECD/NEA uncertainty analysis in modeling (UAM) benchmarks for PWR. However, the plant measured data were not well documented for benchmark calculation. This technical report would help benchmark calculations because most of the required design and measured data are included in this report.



## 5. REFERENCES

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## APPENDIX A. SAMPLE OF POLARIS INPUT FILES

### A.1 Polaris Input for the Fuel Assembly 2.60\_bp16

```
=polaris
% -----
%   general options
% -----
title "WEC 17x17 spacer_000/2.6_bp16-CR0-HDC-MPC-MTF lattice"
lib   "broad_lwr"
opt   PRINT XFile16=true
system PWR
% -----
%   geometry
% -----
geom W17 : ASSM 17 1.26 SE
hgap 0.04
% -----
%   compositions and materials
% -----
comp c_uox : UOX 2.619
comp c_enrB : WT B10=50 B11=50
comp c_ifba : FORM Zr=1 c_enrB=2
comp c_waba : CONC
              5010=2.98553E-03
              5011=1.21192E-02
              6000=3.77001E-03
              8016=5.85563E-02
              13027=3.90223E-02
comp c_tpbar : WT
              3006=2.93769
              3007=7.30281
              13027=41.0625
              8016=-100
mat FUEL.1 : c_uox 10.257
mat BP.1   : PYREX
mat BP.2   : c_waba
mat BP.3   : c_ifba 3.85
mat BP.4   : c_tpbar 1.57571
mat GRID.1 : ZIRC4
% -----
%   pins definitions
% -----
pin F : 0.4096 0.418 0.475 0.63
       : FUEL.1 GAP CLAD COOL.1
       : CIR CIR CIR SQR
pin Y : 0.1969 0.4096 0.418 0.475 0.63
       : GAP FUEL.1 GAP CLAD COOL.1
       : CIR CIR CIR CIR SQR
pin Z : 0.4096 0.4106 0.418 0.475 0.63
       : FUEL.1 BP.3 GAP CLAD COOL.1
       : CIR CIR CIR CIR SQR
pin I : 0.559 0.605 0.63
       : COOL.1 TUBE COOL.1
       : CIR CIR SQR
pin H : 0.2832 0.3835 0.484 0.561 0.602 0.63
       : GAP BP.4 STRUCT COOL.1 TUBE COOL.1
       : CIR CIR CIR CIR CIR SQR
pin G : 0.561 0.602 0.63
       : COOL.1 TUBE COOL.1
       : CIR CIR SQR
pin B : 0.373 0.386 0.484 0.561 0.602 0.63
       : CNTL.2 GAP STRUCT COOL.1 TUBE COOL.1
       : CIR CIR CIR CIR CIR SQR
pin A : 0.382 0.386 0.484 0.561 0.602 0.63
       : CNTL.1 GAP STRUCT COOL.1 TUBE COOL.1
       : CIR CIR CIR CIR CIR SQR
pin W : 0.286 0.339 0.353 0.404 0.418 0.484 0.561 0.602 0.63
       : COOL.1 TUBE GAP BP.2 GAP TUBE COOL.1 TUBE COOL.1
       : CIR CIR CIR CIR CIR CIR CIR CIR SQR
pin P : 0.214 0.231 0.241 0.427 0.437 0.484 0.561 0.602 0.63
       : GAP STRUCT GAP BP.1 GAP STRUCT COOL.1 TUBE COOL.1
       : CIR CIR CIR CIR CIR CIR CIR CIR SQR
mesh GAP : ns=1
mesh FUEL : nr=4
mesh COOL : nx=4 ny=4
mesh GRID : nx=4 ny=4
shield ALL=N FUEL=P CNTL=P BP=P
opt KEFF RaySpacing=0.04
```

```

% -----
%      detector
% -----
comp c_u235 : CONC 92235=1.0e-15
mat DET.1 : c_u235
pin D : 0.4 : COOL.2
det d_u235 : pin.D at=pin.I : R(n,FIS) COOL.2 DET.1
opt FG DetectorEdit=d_u235
% -----
%      maps
% -----
pinmap
I
F F
F F F
G F F G
F F F F F
F F F F F G
G F F G F F F
F F F F F F F
F F F F F F F F

control BankB : RODLET

-
- -
B - - B
- - - - - B
B - - B - - -
- - - - - -
- - - - - -

control BankA : RODLET

-
- -
A - - A
- - - - - A
A - - A - - -
- - - - - -
- - - - - -

insert py16

-
- -
P - - -
- - - - - P
- - - P - - -
- - - - - -
- - - - - -

insert py20

-
- -
P - - -
- - - - - P
P - - P - - -
- - - - - -
- - - - - -

insert py24

-
- -
P - - P
- - - - - P
P - - P - - -
- - - - - -
- - - - - -

insert py08

-
- -
- - -

```

```

P _ _ _ _
_ _ _ _ _
_ _ _ _ _ P
_ _ _ _ _
_ _ _ _ _
_ _ _ _ _
_ _ _ _ _

insert py12
_
_ _
_ _ _
P _ _ _
_ _ _ _ _
_ _ _ _ _
_ _ _ _ _ P _ _
_ _ _ _ _
_ _ _ _ _
_ _ _ _ _

insert waba04
_
_ _
_ _ _
_ _ _ _ _
_ _ _ _ _
_ _ _ _ _ W
_ _ _ _ _
_ _ _ _ _
_ _ _ _ _
_ _ _ _ _

insert waba08
_
_ _
_ _ _
W _ _ _
_ _ _ _ _
_ _ _ _ _ W
_ _ _ _ _
_ _ _ _ _
_ _ _ _ _
_ _ _ _ _

insert tpbar08
_
_ _
_ _ _
H _ _ _
_ _ _ _ _
_ _ _ _ _
_ _ _ _ _ H
_ _ _ _ _
_ _ _ _ _
_ _ _ _ _

%
% -----
% states
% -----
power 38.4
bu 0 0.1 0.5 1 2 4 6 8 10 12.5 15 17.5
    20 25 30 35 40 45 50 55 60

state py16 : in= no ALL : temp= 550 FUEL : temp= 800 COOL : dens=0.76971 boron= 600

read branch
add py16 : in= no ALL : temp= 615 FUEL : temp= 560 COOL : dens=0.60811 boron= 0
add py16 : in= no ALL : temp= 615 FUEL : temp= 560 COOL : dens=0.60811 boron= 600
add py16 : in= no ALL : temp= 615 FUEL : temp= 560 COOL : dens=0.60811 boron=1800
add py16 : in= no ALL : temp= 585 FUEL : temp= 560 COOL : dens=0.70045 boron= 0
add py16 : in= no ALL : temp= 585 FUEL : temp= 560 COOL : dens=0.70045 boron= 600
add py16 : in= no ALL : temp= 585 FUEL : temp= 560 COOL : dens=0.70045 boron=1800
add py16 : in= no ALL : temp= 550 FUEL : temp= 560 COOL : dens=0.76971 boron= 0
add py16 : in= no ALL : temp= 550 FUEL : temp= 560 COOL : dens=0.76971 boron= 600
add py16 : in= no ALL : temp= 550 FUEL : temp= 560 COOL : dens=0.76971 boron=1800
add py16 : in= no ALL : temp= 615 FUEL : temp= 800 COOL : dens=0.60811 boron= 0
add py16 : in= no ALL : temp= 615 FUEL : temp= 800 COOL : dens=0.60811 boron= 600
add py16 : in= no ALL : temp= 615 FUEL : temp= 800 COOL : dens=0.60811 boron=1800
add py16 : in= no ALL : temp= 585 FUEL : temp= 800 COOL : dens=0.70045 boron= 0
add py16 : in= no ALL : temp= 585 FUEL : temp= 800 COOL : dens=0.70045 boron= 600
add py16 : in= no ALL : temp= 585 FUEL : temp= 800 COOL : dens=0.70045 boron=1800
add py16 : in= no ALL : temp= 550 FUEL : temp= 800 COOL : dens=0.76971 boron= 0
add py16 : in= no ALL : temp= 550 FUEL : temp= 800 COOL : dens=0.76971 boron= 600
add py16 : in= no ALL : temp= 550 FUEL : temp= 800 COOL : dens=0.76971 boron=1800
add py16 : in= no ALL : temp= 615 FUEL : temp=1600 COOL : dens=0.60811 boron= 0
add py16 : in= no ALL : temp= 615 FUEL : temp=1600 COOL : dens=0.60811 boron= 600
add py16 : in= no ALL : temp= 615 FUEL : temp=1600 COOL : dens=0.60811 boron=1800

```

```

add pyl6 : in= no ALL : temp= 585 FUEL : temp=1600 COOL : dens=0.70045 boron= 0
add pyl6 : in= no ALL : temp= 585 FUEL : temp=1600 COOL : dens=0.70045 boron= 600
add pyl6 : in= no ALL : temp= 585 FUEL : temp=1600 COOL : dens=0.70045 boron=1800
add pyl6 : in= no ALL : temp= 550 FUEL : temp=1600 COOL : dens=0.76971 boron= 0
add pyl6 : in= no ALL : temp= 550 FUEL : temp=1600 COOL : dens=0.76971 boron= 600
add pyl6 : in= no ALL : temp= 550 FUEL : temp=1600 COOL : dens=0.76971 boron=1800
add pyl6 : in=yes ALL : temp= 615 FUEL : temp= 800 COOL : dens=0.60811 boron= 0
add pyl6 : in=yes ALL : temp= 615 FUEL : temp= 800 COOL : dens=0.60811 boron= 600
add pyl6 : in=yes ALL : temp= 615 FUEL : temp= 800 COOL : dens=0.60811 boron=1800
add pyl6 : in=yes ALL : temp= 585 FUEL : temp= 800 COOL : dens=0.70045 boron= 0
add pyl6 : in=yes ALL : temp= 585 FUEL : temp= 800 COOL : dens=0.70045 boron= 600
add pyl6 : in=yes ALL : temp= 585 FUEL : temp= 800 COOL : dens=0.70045 boron=1800
add pyl6 : in=yes ALL : temp= 550 FUEL : temp= 800 COOL : dens=0.76971 boron= 0
add pyl6 : in=yes ALL : temp= 550 FUEL : temp= 800 COOL : dens=0.76971 boron= 600
add pyl6 : in=yes ALL : temp= 550 FUEL : temp= 800 COOL : dens=0.76971 boron=1800
end branch

end

```

## A.2 Polaris Input for Radial Reflector

```

=polaris
% -----
%   general options
% -----
title "WEC 17x17 lattice"
lib   "broad_lwr"
opt   PRINT XFile16=true
system PWR
% -----
%   compositions and materials
% -----
comp c_uox : UOX 3.1
mat FUEL.1 : c_uox 10.257
% -----
%   geometry
% -----
geom W17 : ASSM 17 1.26
hgap 0.04
pin F : 0.4096 0.418 0.475 : FUEL.1 GAP CLAD
pin I : 0.559 0.605 : COOL.1 TUBE
pin G : 0.561 0.602 : COOL.1 TUBE
mesh GAP : ns=1
mesh FUEL : nr=4
shield ALL=N FUEL=P
pinmap
  I
  F F
  F F F
  G F F G
  F F F F F
  F F F F F G
  G F F G F F F
  F F F F F F F
  F F F F F F F F
geom rad_ref : REFL 21.5
slab rad_ref : 0.19 2.85 18.46
               : COOL.1 STRUCT COOL.1
               : 1 4 25
               : 34 34 34
% -----
%   states
% -----
state ALL : temp=550 FUEL : temp=800 COOL : dens=0.76971 boron=600 rad_ref : in=yes
read branch
  add COOL : boron=0
  add COOL : boron=1800
  add ALL : temp=585 FUEL : temp=800 COOL : boron=600 dens=0.70045
end branch

end

```

### A.3 Polaris Input for Axial Top Reflector

```
=polaris
% -----
%   general options
% -----
title "WEC 17x17 spacer_000/2.1-CR0-MDC-MPC-MTF lattice"
lib   "broad_lwr"
opt   PRINT XFile16=true
system PWR
% -----
%   geometry
% -----
geom W17 : ASSM 17 1.26 FULL
hgap 0.04
% -----
%   compositions and materials
% -----
comp c_uox : UOX 2.11
mat FUEL.1 : c_uox 10.257
% -----
%   pins definitions
% -----
pin F : 0.4096 0.418 0.475
       : FUEL.1 GAP CLAD
pin I : 0.559 0.605
       : COOL.1 TUBE COOL.1
pin G : 0.561 0.602
       : COOL.1 TUBE COOL.1
mesh GAP : ns=1
mesh FUEL : nr=4
shield ALL=N FUEL=P
% -----
%   maps
% -----
pinmap
  I
  F F
  F F F
  G F F G
  F F F F F
  F F F F F G
  G F F G F F F
  F F F F F F F
  F F F F F F F F
% -----
%   reflector
% -----
geom top_refl_lpc : REFL 36.226
geom top_refl_mpc : REFL 36.226
geom top_refl_hpc : REFL 36.226
geom top_refl_mdc : REFL 36.226
slab top_refl_lpc : 8.556 3.866 3.578 1.67 2.129 8.827 7.6
                  : PLE.1 PLE_GRID.1 PLE.1 PLUG.1 NOZ_GAP.1 TOP_NOZ.1 TOP_CP.1
                  : 14 6 6 3 3 14 12
                  : 34 34 34 34 34 34 34
slab top_refl_mpc : 8.556 3.866 3.578 1.67 2.129 8.827 7.6
                  : PLE.2 PLE_GRID.2 PLE.2 PLUG.2 NOZ_GAP.2 TOP_NOZ.2 TOP_CP.2
                  : 14 6 6 3 3 14 12
                  : 34 34 34 34 34 34 34
slab top_refl_hpc : 8.556 3.866 3.578 1.67 2.129 8.827 7.6
                  : PLE.3 PLE_GRID.3 PLE.3 PLUG.3 NOZ_GAP.3 TOP_NOZ.3 TOP_CP.3
                  : 14 6 6 3 3 14 12
                  : 34 34 34 34 34 34 34
slab top_refl_mdc : 8.556 3.866 3.578 1.67 2.129 8.827 7.6
                  : PLE.4 PLE_GRID.4 PLE.4 PLUG.4 NOZ_GAP.4 TOP_NOZ.4 TOP_CP.4
                  : 14 6 6 3 3 14 12
                  : 34 34 34 34 34 34 34
mat PLE.1 : c_ple_lpc 2.25
mat PLE_GRID.1 : c_ple_grid_lpc 2.78
mat PLUG.1 : c_plug_lpc 3.07
mat NOZ_GAP.1 : c_noz_gap_lpc 0.66
mat TOP_NOZ.1 : c_top_noz_lpc 2.02
mat TOP_CP.1 : c_top_cp_lpc 4.27

mat PLE.2 : c_ple_mpc 2.25
mat PLE_GRID.2 : c_ple_grid_mpc 2.78
mat PLUG.2 : c_plug_mpc 3.07
mat NOZ_GAP.2 : c_noz_gap_mpc 0.66
mat TOP_NOZ.2 : c_top_noz_mpc 2.02
mat TOP_CP.2 : c_top_cp_mpc 4.27
```

```

mat PLE.3 : c_ple_hpc 2.25
mat PLE_GRID.3 : c_ple_grid_hpc 2.78
mat PLUG.3 : c_plug_hpc 3.07
mat NOZ_GAP.3 : c_noz_gap_hpc 0.66
mat TOP_NOZ.3 : c_top_noz_hpc 2.02
mat TOP_CP.3 : c_top_cp_hpc 4.27

mat PLE.4 : c_ple_mdc 2.31
mat PLE_GRID.4 : c_ple_grid_mdc 2.83
mat PLUG.4 : c_plug_mdc 3.12
mat NOZ_GAP.4 : c_noz_gap_mdc 0.75
mat TOP_NOZ.4 : c_top_noz_mdc 2.10
mat TOP_CP.4 : c_top_cp_mdc 4.32

comp c_cool_lpc : LW 0
comp c_ple_lpc : WT ZIRC4=29 SS304=55 c_cool_lpc=-100
comp c_ple_grid_lpc : WT ZIRC4=23 SS304=45 INC718=20 c_cool_lpc=-100
comp c_plug_lpc : WT ZIRC4=88 c_cool_lpc=-100
comp c_noz_gap_lpc : WT ZIRC4=8 c_cool_lpc=-100
comp c_top_noz_lpc : WT SS304=76 c_cool_lpc=-100
comp c_top_cp_lpc : WT SS304=93 c_cool_lpc=-100

comp c_cool_mpc : LW 600
comp c_ple_mpc : WT ZIRC4=29 SS304=55 c_cool_mpc=-100
comp c_ple_grid_mpc : WT ZIRC4=23 SS304=45 INC718=20 c_cool_mpc=-100
comp c_plug_mpc : WT ZIRC4=88 c_cool_mpc=-100
comp c_noz_gap_mpc : WT ZIRC4=8 c_cool_mpc=-100
comp c_top_noz_mpc : WT SS304=76 c_cool_mpc=-100
comp c_top_cp_mpc : WT SS304=93 c_cool_mpc=-100

comp c_cool_hpc : LW 1800
comp c_ple_hpc : WT ZIRC4=29 SS304=55 c_cool_hpc=-100
comp c_ple_grid_hpc : WT ZIRC4=23 SS304=45 INC718=20 c_cool_hpc=-100
comp c_plug_hpc : WT ZIRC4=88 c_cool_hpc=-100
comp c_noz_gap_hpc : WT ZIRC4=8 c_cool_hpc=-100
comp c_top_noz_hpc : WT SS304=76 c_cool_hpc=-100
comp c_top_cp_hpc : WT SS304=93 c_cool_hpc=-100

comp c_cool_mdc : LW 600
comp c_ple_mdc : WT ZIRC4=28 SS304=54 c_cool_mdc=-100
comp c_ple_grid_mdc : WT ZIRC4=23 SS304=44 INC718=20 c_cool_mdc=-100
comp c_plug_mdc : WT ZIRC4=87 c_cool_mdc=-100
comp c_noz_gap_mdc : WT ZIRC4=7 c_cool_mdc=-100
comp c_top_noz_mdc : WT SS304=73 c_cool_mdc=-100
comp c_top_cp_mdc : WT SS304=92 c_cool_mdc=-100

opt ESSM RaySpacing=0.0425
opt KEFF RaySpacing=0.046
% -----
% states
% -----
state ALL : temp=615
FUEL : temp=800
COOL : dens=0.60811 boron=600
top_refl_mpc : in=yes
top_refl_lpc : in=no
top_refl_hpc : in=no
top_refl_mdc : in=no
read branch
add COOL : boron=0
top_refl_mpc : in=no
top_refl_lpc : in=yes
top_refl_hpc : in=no
top_refl_mdc : in=no

add COOL : boron=1800
top_refl_mpc : in=no
top_refl_lpc : in=no
top_refl_hpc : in=yes
top_refl_mdc : in=no

add ALL : temp=585
FUEL : temp=800
COOL : boron=600 dens=0.70045
top_refl_mpc : in=no
top_refl_lpc : in=no
top_refl_hpc : in=no
top_refl_mdc : in=yes
end branch
end

```

## A.4 Polaris Input for Axial Bottom Reflector

```
=polaris
% -----
%   general options
% -----
title "WEC 17x17 spacer_000/2.1-CR0-MDC-MPC-MTF lattice"
lib   "broad_lwr"
opt   PRINT XFile16=true
system PWR
% -----
%   geometry
% -----
geom W17 : ASSM 17 1.26 FULL
hgap 0.04
% -----
%   compositions and materials
% -----
comp c_uox : UOX 2.11
mat FUEL.1 : c_uox 10.257
% -----
%   pins definitions
% -----
pin F : 0.4096 0.418 0.475
       : FUEL.1 GAP CLAD
pin I : 0.559 0.605
       : COOL.1 TUBE COOL.1
pin G : 0.561 0.602
       : COOL.1 TUBE COOL.1
mesh GAP : ns=1
mesh FUEL : nr=4
shield ALL=N FUEL=P
% -----
%   maps
% -----
pinmap
  I
  F F
  F F F
  G F F G
  F F F F F
  F F F F F G
  G F F G F F F
  F F F F F F F
  F F F F F F F F
% -----
%   reflector
% -----
geom bot_refl_lpc : REFL 16.951
geom bot_refl_mpc : REFL 16.951
geom bot_refl_hpc : REFL 16.951
geom bot_refl_mdc : REFL 16.951
slab bot_refl_lpc : 5.898 6.053 5.000
                  : BOT_GAP.1 BOT_NOZ.1 BOT_CP.1
                  : 9 9 8
                  : 34 34 34
slab bot_refl_mpc : 5.898 6.053 5.000
                  : BOT_GAP.2 BOT_NOZ.2 BOT_CP.2
                  : 9 9 8
                  : 34 34 34
slab bot_refl_hpc : 5.898 6.053 5.000
                  : BOT_GAP.3 BOT_NOZ.3 BOT_CP.3
                  : 9 9 8
                  : 34 34 34
slab bot_refl_mdc : 5.898 6.053 5.000
                  : BOT_GAP.4 BOT_NOZ.4 BOT_CP.4
                  : 9 9 8
                  : 34 34 34
mat BOT_GAP.1 : c_bot_gap_lpc 1.82
mat BOT_NOZ.1 : c_bot_noz_lpc 2.79
mat BOT_CP.1 : c_bot_cp_lpc 4.35

mat BOT_GAP.2 : c_bot_gap_mpc 1.82
mat BOT_NOZ.2 : c_bot_noz_mpc 2.79
mat BOT_CP.2 : c_bot_cp_mpc 4.35

mat BOT_GAP.3 : c_bot_gap_hpc 1.82
mat BOT_NOZ.3 : c_bot_noz_hpc 2.79
mat BOT_CP.3 : c_bot_cp_hpc 4.35

mat BOT_GAP.4 : c_bot_gap_mdc 1.76
```



```

mat BOT_NOZ.4 : c_bot_noz_mdc 2.74
mat BOT_CP.4 : c_bot_cp_mdc 4.32

comp c_cool_lpc : LW 0
comp c_bot_gap_lpc : WT ZIRC4=44 INC718=21 c_cool_lpc=-100
comp c_bot_noz_lpc : WT SS304=80 c_cool_lpc=-100
comp c_bot_cp_lpc : WT SS304=91 c_cool_lpc=-100

comp c_cool_mpc : LW 600
comp c_bot_gap_mpc : WT ZIRC4=44 INC718=21 c_cool_mpc=-100
comp c_bot_noz_mpc : WT SS304=80 c_cool_mpc=-100
comp c_bot_cp_mpc : WT SS304=91 c_cool_mpc=-100

comp c_cool_hpc : LW 1800
comp c_bot_gap_hpc : WT ZIRC4=44 INC718=21 c_cool_hpc=-100
comp c_bot_noz_hpc : WT SS304=80 c_cool_hpc=-100
comp c_bot_cp_hpc : WT SS304=91 c_cool_hpc=-100

comp c_cool_mdc : LW 600
comp c_bot_gap_mdc : WT ZIRC4=46 INC718=21 c_cool_mdc=-100
comp c_bot_noz_mdc : WT SS304=82 c_cool_mdc=-100
comp c_bot_cp_mdc : WT SS304=92 c_cool_mdc=-100

opt ESSM RaySpacing=0.045
opt KEFF RaySpacing=.066
% -----
% states
% -----
state ALL : temp=550
    FUEL : temp=800
    COOL : dens=0.76971 boron=600
    bot_refl_mpc : in=yes
    bot_refl_lpc : in=no
    bot_refl_hpc : in=no
    bot_refl_mdc : in=no
read branch
    add      COOL : boron=0
    bot_refl_mpc : in=no
    bot_refl_lpc : in=yes
    bot_refl_hpc : in=no
    bot_refl_mdc : in=no

    add      COOL : boron=1800
    bot_refl_mpc : in=no
    bot_refl_lpc : in=no
    bot_refl_hpc : in=yes
    bot_refl_mdc : in=no

    add      ALL : temp=585
    FUEL : temp=800
    COOL : boron=600 dens=0.70045
    bot_refl_mpc : in=no
    bot_refl_lpc : in=no
    bot_refl_hpc : in=no
    bot_refl_mdc : in=yes
end branch
end

```

## APPENDIX B. SAMPLE OF GENPMAXS INPUT FILES

### B.1 GenPMAXS Input for the Fuel Assembly 2.60\_bp16

```
%JOB_TIT
'2.60_bp16.PMAX' T 3.0
%JOB_OPT
T T F F T F T T T T T T F F
!adf,xes,ded,jlf,chi, chd,inv,det,yld,cdf,gff,bet,lam,dec,zdf
%DAT_SRC
8 8 1
%STA_VAR
4
CR DC PC TF
%HISTORY
8 4
'CR0-MDC-MPC-LTF' 0 0.70045 600 560
'CR0-LDC-MPC-MTF' 0 0.60811 600 800
'CR0-MDC-LPC-MTF' 0 0.70045 0 800
'CR0-MDC-MPC-MTF' 0 0.70045 600 800
'CR0-MDC-HPC-MTF' 0 0.70045 1800 800
'CR0-HDC-MPC-MTF' 0 0.76971 600 800
'CR0-MDC-MPC-HTF' 0 0.70045 600 1600
'CR1-MDC-MPC-MTF' 1 0.70045 600 800
%BRANCH
36 14
1 0 0.60811 0 560
2 0 0.60811 600 560
3 0 0.60811 1800 560
4 0 0.70045 0 560
5 0 0.70045 600 560
6 0 0.70045 1800 560
7 0 0.76971 0 560
8 0 0.76971 600 560
9 0 0.76971 1800 560
10 0 0.60811 0 800
11 0 0.60811 600 800
12 0 0.60811 1800 800
13 0 0.70045 0 800
14 0 0.70045 600 800
15 0 0.70045 1800 800
16 0 0.76971 0 800
17 0 0.76971 600 800
18 0 0.76971 1800 800
19 0 0.60811 0 1600
20 0 0.60811 600 1600
21 0 0.60811 1800 1600
22 0 0.70045 0 1600
23 0 0.70045 600 1600
24 0 0.70045 1800 1600
25 0 0.76971 0 1600
26 0 0.76971 600 1600
27 0 0.76971 1800 1600
28 1 0.60811 0 800
29 1 0.60811 600 800
30 1 0.60811 1800 800
31 1 0.70045 0 800
32 1 0.70045 600 800
33 1 0.70045 1800 800
34 1 0.76971 0 800
35 1 0.76971 600 800
36 1 0.76971 1800 800
%BURNUP
1
'BP' 21
0 0.1 0.5 1 2 4 6 8 10 12.5 15 17.5
20 25 30 35 40 45 50 55 60
'CR0-MDC-MPC-LTF' 36*1
'CR0-LDC-MPC-MTF' 36*1
'CR0-MDC-LPC-MTF' 36*1
'CR0-MDC-MPC-MTF' 36*1
'CR0-MDC-HPC-MTF' 36*1
'CR0-HDC-MPC-MTF' 36*1
'CR0-MDC-MPC-HTF' 36*1
'CR1-MDC-MPC-MTF' 36*1
%USERINP
1
PIN 1.26 0.04 0.04 0
%FIL_CNT
```

```

1 './../Polaris_630/T16/spacer_000/2.6_bp16-CR0-MDC-MPC-LTF.t16' 36 0
  1 1 5 1 21
  2 1 1 1 21
  3 1 2 1 21
  4 1 3 1 21
  5 1 4 1 21
  6 1 6 1 21
  7 1 7 1 21
  8 1 8 1 21
  9 1 9 1 21
 10 1 10 1 21
 11 1 11 1 21
 12 1 12 1 21
 13 1 13 1 21
 14 1 14 1 21
 15 1 15 1 21
 16 1 16 1 21
 17 1 17 1 21
 18 1 18 1 21
 19 1 19 1 21
 20 1 20 1 21
 21 1 21 1 21
 22 1 22 1 21
 23 1 23 1 21
 24 1 24 1 21
 25 1 25 1 21
 26 1 26 1 21
 27 1 27 1 21
 28 1 28 1 21
 29 1 29 1 21
 30 1 30 1 21
 31 1 31 1 21
 32 1 32 1 21
 33 1 33 1 21
 34 1 34 1 21
 35 1 35 1 21
 36 1 36 1 21
2 './../Polaris_630/T16/spacer_000/2.6_bp16-CR0-LDC-MPC-MTF.t16' 36 0
  1 2 11 1 21
  2 2 1 1 21
  3 2 2 1 21
  4 2 3 1 21
  5 2 4 1 21
  6 2 5 1 21
  7 2 6 1 21
  8 2 7 1 21
  9 2 8 1 21
 10 2 9 1 21
 11 2 10 1 21
 12 2 12 1 21
 13 2 13 1 21
 14 2 14 1 21
 15 2 15 1 21
 16 2 16 1 21
 17 2 17 1 21
 18 2 18 1 21
 19 2 19 1 21
 20 2 20 1 21
 21 2 21 1 21
 22 2 22 1 21
 23 2 23 1 21
 24 2 24 1 21
 25 2 25 1 21
 26 2 26 1 21
 27 2 27 1 21
 28 2 28 1 21
 29 2 29 1 21
 30 2 30 1 21
 31 2 31 1 21
 32 2 32 1 21
 33 2 33 1 21
 34 2 34 1 21
 35 2 35 1 21
 36 2 36 1 21
3 './../Polaris_630/T16/spacer_000/2.6_bp16-CR0-MDC-LPC-MTF.t16' 36 0
  1 3 13 1 21
  2 3 1 1 21
  3 3 2 1 21
  4 3 3 1 21
  5 3 4 1 21
  6 3 5 1 21

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7 3 6 1 21
8 3 7 1 21
9 3 8 1 21
10 3 9 1 21
11 3 10 1 21
12 3 11 1 21
13 3 12 1 21
14 3 14 1 21
15 3 15 1 21
16 3 16 1 21
17 3 17 1 21
18 3 18 1 21
19 3 19 1 21
20 3 20 1 21
21 3 21 1 21
22 3 22 1 21
23 3 23 1 21
24 3 24 1 21
25 3 25 1 21
26 3 26 1 21
27 3 27 1 21
28 3 28 1 21
29 3 29 1 21
30 3 30 1 21
31 3 31 1 21
32 3 32 1 21
33 3 33 1 21
34 3 34 1 21
35 3 35 1 21
36 3 36 1 21
4 '..../Polaris_630/T16/spacer_000/2.6_bp16-CR0-MDC-MPC-MTF.t16' 36 0
1 4 14 1 21
2 4 1 1 21
3 4 2 1 21
4 4 3 1 21
5 4 4 1 21
6 4 5 1 21
7 4 6 1 21
8 4 7 1 21
9 4 8 1 21
10 4 9 1 21
11 4 10 1 21
12 4 11 1 21
13 4 12 1 21
14 4 13 1 21
15 4 15 1 21
16 4 16 1 21
17 4 17 1 21
18 4 18 1 21
19 4 19 1 21
20 4 20 1 21
21 4 21 1 21
22 4 22 1 21
23 4 23 1 21
24 4 24 1 21
25 4 25 1 21
26 4 26 1 21
27 4 27 1 21
28 4 28 1 21
29 4 29 1 21
30 4 30 1 21
31 4 31 1 21
32 4 32 1 21
33 4 33 1 21
34 4 34 1 21
35 4 35 1 21
36 4 36 1 21
5 '..../Polaris_630/T16/spacer_000/2.6_bp16-CR0-MDC-HPC-MTF.t16' 36 0
1 5 15 1 21
2 5 1 1 21
3 5 2 1 21
4 5 3 1 21
5 5 4 1 21
6 5 5 1 21
7 5 6 1 21
8 5 7 1 21
9 5 8 1 21
10 5 9 1 21
11 5 10 1 21
12 5 11 1 21
13 5 12 1 21

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14 5 13 1 21
15 5 14 1 21
16 5 16 1 21
17 5 17 1 21
18 5 18 1 21
19 5 19 1 21
20 5 20 1 21
21 5 21 1 21
22 5 22 1 21
23 5 23 1 21
24 5 24 1 21
25 5 25 1 21
26 5 26 1 21
27 5 27 1 21
28 5 28 1 21
29 5 29 1 21
30 5 30 1 21
31 5 31 1 21
32 5 32 1 21
33 5 33 1 21
34 5 34 1 21
35 5 35 1 21
36 5 36 1 21
6 '.../Polaris_630/T16/spacer_000/2.6_bp16-CR0-HDC-MPC-MTF.t16' 36 0
1 6 17 1 21
2 6 1 1 21
3 6 2 1 21
4 6 3 1 21
5 6 4 1 21
6 6 5 1 21
7 6 6 1 21
8 6 7 1 21
9 6 8 1 21
10 6 9 1 21
11 6 10 1 21
12 6 11 1 21
13 6 12 1 21
14 6 13 1 21
15 6 14 1 21
16 6 15 1 21
17 6 16 1 21
18 6 18 1 21
19 6 19 1 21
20 6 20 1 21
21 6 21 1 21
22 6 22 1 21
23 6 23 1 21
24 6 24 1 21
25 6 25 1 21
26 6 26 1 21
27 6 27 1 21
28 6 28 1 21
29 6 29 1 21
30 6 30 1 21
31 6 31 1 21
32 6 32 1 21
33 6 33 1 21
34 6 34 1 21
35 6 35 1 21
36 6 36 1 21
7 '.../Polaris_630/T16/spacer_000/2.6_bp16-CR0-MDC-MPC-HTF.t16' 36 0
1 7 23 1 21
2 7 1 1 21
3 7 2 1 21
4 7 3 1 21
5 7 4 1 21
6 7 5 1 21
7 7 6 1 21
8 7 7 1 21
9 7 8 1 21
10 7 9 1 21
11 7 10 1 21
12 7 11 1 21
13 7 12 1 21
14 7 13 1 21
15 7 14 1 21
16 7 15 1 21
17 7 16 1 21
18 7 17 1 21
19 7 18 1 21
20 7 19 1 21

```

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21 7 20 1 21
22 7 21 1 21
23 7 22 1 21
24 7 24 1 21
25 7 25 1 21
26 7 26 1 21
27 7 27 1 21
28 7 28 1 21
29 7 29 1 21
30 7 30 1 21
31 7 31 1 21
32 7 32 1 21
33 7 33 1 21
34 7 34 1 21
35 7 35 1 21
36 7 36 1 21
8 ' ../../Polaris_630/T16/spacer_000/2.6_bp16-CR1-MDC-MPC-MTF.t16' 36 0
1 8 32 1 21
2 8 1 1 21
3 8 2 1 21
4 8 3 1 21
5 8 4 1 21
6 8 5 1 21
7 8 6 1 21
8 8 7 1 21
9 8 8 1 21
10 8 9 1 21
11 8 10 1 21
12 8 11 1 21
13 8 12 1 21
14 8 13 1 21
15 8 14 1 21
16 8 15 1 21
17 8 16 1 21
18 8 17 1 21
19 8 18 1 21
20 8 19 1 21
21 8 20 1 21
22 8 21 1 21
23 8 22 1 21
24 8 23 1 21
25 8 24 1 21
26 8 25 1 21
27 8 26 1 21
28 8 27 1 21
29 8 28 1 21
30 8 29 1 21
31 8 30 1 21
32 8 31 1 21
33 8 33 1 21
34 8 34 1 21
35 8 35 1 21
36 8 36 1 21
%JOB_END

```

## B.2 GenPMAXS Input for Radial Edge Reflector

```
%JOB_TIT
'refl_rad_edge.PMAX' T 3.0
%JOB_OPT
F F F F F F F F F F F F F F
!adf,xes,ded,jlf,chi, chd,inv,det,yld,cdf,gff,bet,lam,dec,zdf
%DAT_SRC
8 1 0
%STA_VAR
3
CR DC PC
%HISTORY
1 1
'CR0-HDC-MPC' 0 0.76971 600
%BRANCH
4 1
'CR0-HDC-MPC' 0 0.76971 600
'CR0-HDC-LPC' 0 0.76971 0
'CR0-HDC-HPC' 0 0.76971 1800
'CR0-MDC-MPC' 0 0.70045 600
%BURNUP
1
'NOBP' 1
0
'CR0-MDC-MPC' 4*1
%FIL_CNT
1 '..//reflector/WBN1_refl_rad.t16' 4 0
1 1 1 1 1
2 1 2 1 1
3 1 3 1 1
4 1 4 1 1
%JOB_END
```

## B.3 GenPMAXS Input for Axial Top Reflector

```
%JOB_TIT
'refl_top.PMAX' T 3.0
%JOB_OPT
F F F F F F T F F F F F F F
!adf,xes,ded,jlf,chi, chd,inv,det,yld,cdf,gff,bet,lam,dec,zdf
%DAT_SRC
8 1 0
%STA_VAR
3
CR DC PC
%HISTORY
1 1
'CR0-LDC-MPC' 0 0.60811 600
%BRANCH
4 1
'CR0-LDC-MPC' 0 0.60811 600
'CR0-LDC-LPC' 0 0.60811 0
'CR0-LDC-HPC' 0 0.60811 1800
'CR0-MDC-MPC' 0 0.70045 600
%BURNUP
1
'NOBP' 1
0
'CR0-LDC-MPC' 4*1
%FIL_CNT
1 '..//reflector/WBN1_refl_top.t16' 4 0
1 1 1 1 1
2 1 2 1 1
3 1 3 1 1
4 1 4 1 1
%JOB_END
```

## B.4 GenPMAXS Input for Axial Bottom Reflector

```
%JOB_TIT
'refl_bot.PMAX' T 3.0
%JOB_OPT
F F F F F F F F F F F F F F
!adf,xes,ded,jlf,chi, chd,inv,det,yld,cdf,gff,bet,lam,dec,zdf
%DAT_SRC
8 1 0
%STA_VAR
3
CR DC PC
%HISTORY
1 1
'CR0-HDC-MPC' 0 0.76971 600
%BRANCH
4 1
'CR0-HDC-MPC' 0 0.76971 600
'CR0-HDC-LPC' 0 0.76971 0
'CR0-HDC-HPC' 0 0.76971 1800
'CR0-MDC-MPC' 0 0.70045 600
%BURNUP
1
'NOBP' 1
0
'CR0-MDC-MPC' 4*1
%FIL_CNT
1 ' ../reflector/WBN1_refl_bot.t16' 4 0
1 1 1 1 1
2 1 2 1 1
3 1 3 1 1
4 1 4 1 1
%JOB_END
```





# APPENDIX C. PARCS INPUT FILES

## C.1 PARCS Input for the Cross-Section Set “pmax.dir” File

| ! | Index      | PMAXS File Name                               | Br_struct                     |
|---|------------|-----------------------------------------------|-------------------------------|
| ! | *****      | *****                                         | *****                         |
|   | PMAXS_F 1  | '../GenPMAX_630/reflector/refl_bot.PMAX'      | 1                             |
|   | PMAXS_F 2  | '../GenPMAX_630/reflector/refl_top.PMAX'      | 2                             |
| ! | *****      | *****                                         | *****                         |
|   | PMAXS_F 3  | '../GenPMAX_630/reflector/refl_rad_S.PMAX'    | 3                             |
|   | PMAXS_F 4  | '../GenPMAX_630/reflector/refl_rad_SE.PMAX'   | 4                             |
|   | PMAXS_F 5  | '../GenPMAX_630/reflector/refl_rad_E.PMAX'    | 5                             |
|   | PMAXS_F 6  | '../GenPMAX_630/reflector/refl_rad_edge.PMAX' | 6                             |
|   | PMAXS_F 7  | '../GenPMAX_630/reflector/refl_rad_diag.PMAX' | 7                             |
| ! | *****      | *****                                         | *****                         |
|   | PMAXS_F 8  | '../GenPMAX_630/spacer_000/2.1.PMAX'          | 8                             |
|   | PMAXS_F 9  | '../GenPMAX_630/spacer_230/2.1.PMAX'          | 9                             |
|   | PMAXS_F 10 | '../GenPMAX_630/spacer_000/2.6.PMAX'          | 10                            |
|   | PMAXS_F 11 | '../GenPMAX_630/spacer_230/2.6.PMAX'          | 11                            |
|   | PMAXS_F 12 | '../GenPMAX_630/spacer_000/2.6_bp16.PMAX'     | 12                            |
|   | PMAXS_F 13 | '../GenPMAX_630/spacer_230/2.6_bp16.PMAX'     | 13                            |
|   | PMAXS_F 14 | '../GenPMAX_630/spacer_000/2.6_bp20.PMAX'     | 14                            |
|   | PMAXS_F 15 | '../GenPMAX_630/spacer_230/2.6_bp20.PMAX'     | 15                            |
|   | PMAXS_F 16 | '../GenPMAX_630/spacer_000/2.6_bp24.PMAX'     | 16                            |
|   | PMAXS_F 17 | '../GenPMAX_630/spacer_230/2.6_bp24.PMAX'     | 17                            |
|   | PMAXS_F 18 | '../GenPMAX_630/spacer_000/3.1.PMAX'          | 18                            |
|   | PMAXS_F 19 | '../GenPMAX_630/spacer_230/3.1.PMAX'          | 19                            |
|   | PMAXS_F 20 | '../GenPMAX_630/spacer_000/3.1_bp08.PMAX'     | 20                            |
|   | PMAXS_F 21 | '../GenPMAX_630/spacer_230/3.1_bp08.PMAX'     | 21                            |
|   | PMAXS_F 22 | '../GenPMAX_630/spacer_000/3.1_bp12.PMAX'     | 22                            |
|   | PMAXS_F 23 | '../GenPMAX_630/spacer_230/3.1_bp12.PMAX'     | 23                            |
|   | PMAXS_F 24 | '../GenPMAX_630/spacer_000/3.1_bp16.PMAX'     | 24                            |
|   | PMAXS_F 25 | '../GenPMAX_630/spacer_230/3.1_bp16.PMAX'     | 25                            |
|   | PMAXS_F 26 | '../GenPMAX_630/spacer_000/3.1_bp24.PMAX'     | 26                            |
|   | PMAXS_F 27 | '../GenPMAX_630/spacer_230/3.1_bp24.PMAX'     | 27                            |
|   | PMAXS_F 28 | '../GenPMAX_630/spacer_000/BLKT.PMAX'         | 28 ! No spacers for this type |
|   | PMAXS_F 29 | '../GenPMAX_630/spacer_000/BLKT_t08.PMAX'     | 29 ! No spacers for this type |
|   | PMAXS_F 30 | '../GenPMAX_630/spacer_000/3.7.PMAX'          | 30                            |
|   | PMAXS_F 31 | '../GenPMAX_630/spacer_230/3.7.PMAX'          | 31                            |
|   | PMAXS_F 32 | '../GenPMAX_630/spacer_000/3.7_i048.PMAX'     | 32                            |
|   | PMAXS_F 33 | '../GenPMAX_630/spacer_230/3.7_i048.PMAX'     | 33                            |
|   | PMAXS_F 34 | '../GenPMAX_630/spacer_000/3.7_i104_t08.PMAX' | 34                            |
|   | PMAXS_F 35 | '../GenPMAX_630/spacer_230/3.7_i104_t08.PMAX' | 35                            |
|   | PMAXS_F 36 | '../GenPMAX_630/spacer_000/3.7_i104_w04.PMAX' | 36                            |
|   | PMAXS_F 37 | '../GenPMAX_630/spacer_230/3.7_i104_w04.PMAX' | 37                            |
|   | PMAXS_F 38 | '../GenPMAX_630/spacer_000/3.7_i104_w08.PMAX' | 38                            |
|   | PMAXS_F 39 | '../GenPMAX_630/spacer_230/3.7_i104_w08.PMAX' | 39                            |
|   | PMAXS_F 40 | '../GenPMAX_630/spacer_000/3.7_i128.PMAX'     | 40                            |
|   | PMAXS_F 41 | '../GenPMAX_630/spacer_230/3.7_i128.PMAX'     | 41                            |
|   | PMAXS_F 42 | '../GenPMAX_630/spacer_000/3.7_t08.PMAX'      | 42 ! No spacer for this type  |
|   | PMAXS_F 43 | '../GenPMAX_630/spacer_000/3.7_w04.PMAX'      | 43 ! No spacer for this type  |
|   | PMAXS_F 44 | '../GenPMAX_630/spacer_000/3.7_w08.PMAX'      | 44 ! No spacer for this type  |
|   | PMAXS_F 45 | '../GenPMAX_630/spacer_000/B5A0I.PMAX'        | 45 ! No spacer for this type  |
|   | PMAXS_F 46 | '../GenPMAX_630/spacer_000/B5A104I.PMAX'      | 46                            |
|   | PMAXS_F 47 | '../GenPMAX_630/spacer_230/B5A104I.PMAX'      | 47                            |
|   | PMAXS_F 48 | '../GenPMAX_630/spacer_000/B5A128I.PMAX'      | 48                            |
|   | PMAXS_F 49 | '../GenPMAX_630/spacer_230/B5A128I.PMAX'      | 49                            |
|   | PMAXS_F 50 | '../GenPMAX_630/spacer_000/B5B0I.PMAX'        | 50 ! No spacer for this type  |
|   | PMAXS_F 51 | '../GenPMAX_630/spacer_000/B5B16I.PMAX'       | 51                            |
|   | PMAXS_F 52 | '../GenPMAX_630/spacer_230/B5B16I.PMAX'       | 52                            |
|   | PMAXS_F 53 | '../GenPMAX_630/spacer_000/B5B128I.PMAX'      | 53                            |
|   | PMAXS_F 54 | '../GenPMAX_630/spacer_230/B5B128I.PMAX'      | 54                            |
|   | PMAXS_F 55 | '../GenPMAX_630/spacer_000/B5B128I_w08.PMAX'  | 55                            |
|   | PMAXS_F 56 | '../GenPMAX_630/spacer_230/B5B128I_w08.PMAX'  | 56                            |
|   | PMAXS_F 57 | '../GenPMAX_630/spacer_000/B5B0I_w08.PMAX'    | 57                            |
|   | PMAXS_F 58 | '../GenPMAX_630/spacer_230/B5B0I_w08.PMAX'    | 58                            |
|   | PMAXS_F 59 | '../GenPMAX_630/spacer_000/BL16I.PMAX'        | 59 ! No spacer for this type  |
|   | PMAXS_F 60 | '../GenPMAX_630/spacer_000/BL104I.PMAX'       | 60 ! No spacer for this type  |
|   | PMAXS_F 61 | '../GenPMAX_630/spacer_000/BL128I.PMAX'       | 61 ! No spacer for this type  |
| ! | *****      | *****                                         | *****                         |

## C.2 PARCS Input for the Assembly Geometry “depl\_assm.geom” File

```

grid_x      1*10.75 8*21.50
neutmesh_x  1*1 8*1
grid_y      1*10.75 8*21.50
neutmesh_y  1*1 8*1

```

|                   |        |       |        |        |        |        |                         |        |              |        |
|-------------------|--------|-------|--------|--------|--------|--------|-------------------------|--------|--------------|--------|
| !                 | Spacer | Refl  | Spacer | Refl   | Spacer | Spacer | Spacer                  | Spacer | Spacer       | Spacer |
| grid_z            | 16.951 | 3.810 | 11.430 | 15.24  | 15.432 | 15.432 | 3.810                   | 16.13  | 16.13        | 16.13  |
| 16.13 3.810 9.126 | 15.24  | 13.97 | 1.27   | 36.226 |        |        |                         |        |              |        |
| boun_cond         | 0      | 2     | 0      | 2      | 2      | 2      |                         |        |              |        |
| assy_type         | 4      | 1*1   | 30*3   | 1*2    | REFL   | !      | S                       | -      | 1-sided refl |        |
| assy_type         | 5      | 1*1   | 30*4   | 1*2    | REFL   | !      | SE                      | -      | 2-sided refl |        |
| assy_type         | 6      | 1*1   | 30*5   | 1*2    | REFL   | !      | E                       | -      | 1-sided refl |        |
| assy_type         | 7      | 1*1   | 30*6   | 1*2    | REFL   | !      | island node on edge     |        |              |        |
| assy_type         | 8      | 1*1   | 30*7   | 1*2    | REFL   | !      | island node on diagonal |        |              |        |
| !                 | 11.95  |       | 27.19  | 42.43  |        |        |                         |        | 347.2        | 362.4  |
| !                 | R      | S     | S      | P      | W      | I      | S                       | S      | S            | S      |
| assy_type         | 10     | 1*1   | 1*8    | 1*8    | 1*8    | 2*8    | 1*9                     | 3*8    | 1*9          | 3*8    |
| assy_type         | 20     | 1*1   | 1*10   | 1*10   | 1*10   | 2*10   | 1*11                    | 3*10   | 1*11         | 3*10   |
| assy_type         | 21     | 1*1   | 1*12   | 1*12   | 1*12   | 2*12   | 1*13                    | 3*12   | 1*13         | 3*12   |
| assy_type         | 22     | 1*1   | 1*14   | 1*14   | 1*14   | 2*14   | 1*15                    | 3*14   | 1*15         | 3*14   |
| assy_type         | 23     | 1*1   | 1*16   | 1*16   | 1*16   | 2*16   | 1*17                    | 3*16   | 1*17         | 3*16   |
| assy_type         | 30     | 1*1   | 1*18   | 1*18   | 1*18   | 2*18   | 1*19                    | 3*18   | 1*19         | 3*18   |
| assy_type         | 31     | 1*1   | 1*20   | 1*20   | 1*20   | 2*20   | 1*21                    | 3*20   | 1*21         | 3*20   |
| assy_type         | 32     | 1*1   | 1*22   | 1*22   | 1*22   | 2*22   | 1*23                    | 3*22   | 1*23         | 3*22   |
| assy_type         | 33     | 1*1   | 1*24   | 1*24   | 1*24   | 2*24   | 1*25                    | 3*24   | 1*25         | 3*24   |
| assy_type         | 34     | 1*1   | 1*26   | 1*26   | 1*26   | 2*26   | 1*27                    | 3*26   | 1*27         | 3*26   |
| assy_type         | 40     | 1*1   | 1*28   | 1*28   | 1*30   | 2*30   | 1*31                    | 3*30   | 1*31         | 3*30   |
| assy_type         | 41     | 1*1   | 1*28   | 1*28   | 1*30   | 2*32   | 1*33                    | 3*32   | 1*33         | 3*32   |
| assy_type         | 42     | 1*1   | 1*29   | 1*29   | 1*42   | 2*34   | 1*35                    | 3*34   | 1*35         | 3*34   |
| assy_type         | 43     | 1*1   | 1*28   | 1*28   | 1*43   | 2*36   | 1*37                    | 3*36   | 1*37         | 3*36   |
| assy_type         | 44     | 1*1   | 1*28   | 1*28   | 1*44   | 2*38   | 1*39                    | 3*38   | 1*39         | 3*38   |
| assy_type         | 45     | 1*1   | 1*28   | 1*28   | 1*30   | 2*40   | 1*41                    | 3*40   | 1*41         | 3*40   |
| assy_type         | 46     | 1*1   | 1*28   | 1*28   | 1*40   | 2*40   | 1*41                    | 3*40   | 1*41         | 3*40   |
| assy_type         | 51     | 1*1   | 1*60   | 1*60   | 1*45   | 2*46   | 1*47                    | 3*46   | 1*47         | 3*46   |
| assy_type         | 52     | 1*1   | 1*61   | 1*61   | 1*45   | 2*48   | 1*49                    | 3*48   | 1*49         | 3*48   |
| assy_type         | 61     | 1*1   | 1*59   | 1*59   | 1*50   | 2*51   | 1*52                    | 3*51   | 1*52         | 3*51   |
| assy_type         | 62     | 1*1   | 1*61   | 1*61   | 1*50   | 2*53   | 1*54                    | 3*53   | 1*54         | 3*53   |
| assy_type         | 63     | 1*1   | 1*61   | 1*61   | 1*57   | 2*55   | 1*56                    | 3*55   | 1*56         | 3*55   |
| PINCL_IOC         | 1      | 2     | 3      | 4      | 5      | 6      | 7                       | 8      | 0            |        |
|                   | 9      | 10    | 11     | 12     | 13     | 14     | 15                      | 16     | 0            |        |
|                   | 17     | 18    | 19     | 20     | 21     | 22     | 23                      | 24     | 0            |        |
|                   | 25     | 26    | 27     | 28     | 29     | 30     | 31                      | 32     | 0            |        |
|                   | 33     | 34    | 35     | 36     | 37     | 38     | 39                      | 0      | 0            |        |
|                   | 40     | 41    | 42     | 43     | 44     | 45     | 46                      | 0      |              |        |
|                   | 47     | 48    | 49     | 50     | 51     | 52     | 0                       | 0      |              |        |
|                   | 53     | 54    | 55     | 56     | 0      | 0      | 0                       |        |              |        |
|                   | 0      | 0     | 0      | 0      | 0      |        |                         |        |              |        |
| DET_Z_IOC         | 30     | 1     |        |        |        |        |                         |        |              |        |
|                   | 2      | 3     | 4      | 5      | 6      | 7      | 8                       | 9      | 10           | 11     |
| DET_Z_WEI         | 1      | 1     | 1      | 1      | 1      | 1      | 1                       | 1      | 1            | 1      |
|                   | 1      | 1     | 1      | 1      | 1      | 1      | 1                       | 1      | 1            | 1      |
| DET_NAME          | 1      | D01   |        |        |        |        |                         |        |              |        |
|                   | 2      | D02   |        |        |        |        |                         |        |              |        |
|                   | 3      | D03   |        |        |        |        |                         |        |              |        |
|                   | 4      | D04   |        |        |        |        |                         |        |              |        |
|                   | 5      | D05   |        |        |        |        |                         |        |              |        |
|                   | 6      | D06   |        |        |        |        |                         |        |              |        |
|                   | 7      | D07   |        |        |        |        |                         |        |              |        |
|                   | 8      | D08   |        |        |        |        |                         |        |              |        |
|                   | 9      | D09   |        |        |        |        |                         |        |              |        |
|                   | 10     | D10   |        |        |        |        |                         |        |              |        |
|                   | 11     | D11   |        |        |        |        |                         |        |              |        |
|                   | 12     | D12   |        |        |        |        |                         |        |              |        |
|                   | 13     | D13   |        |        |        |        |                         |        |              |        |
|                   | 14     | D14   |        |        |        |        |                         |        |              |        |
|                   | 15     | D15   |        |        |        |        |                         |        |              |        |
|                   | 16     | D16   |        |        |        |        |                         |        |              |        |
|                   | 17     | D17   |        |        |        |        |                         |        |              |        |
|                   | 18     | D18   |        |        |        |        |                         |        |              |        |
|                   | 19     | D19   |        |        |        |        |                         |        |              |        |
|                   | 20     | D20   |        |        |        |        |                         |        |              |        |

21 D21  
 22 D22  
 23 D23  
 24 D24  
 25 D25  
 26 D26  
 27 D27  
 28 D28  
 29 D29  
 30 D30  
 31 D31  
 32 D32  
 33 D33  
 34 D34  
 35 D35  
 36 D36  
 37 D37  
 38 D38  
 39 D39  
 40 D40  
 41 D41  
 42 D42  
 43 D43  
 44 D44  
 45 D45  
 46 D46  
 47 D47  
 48 D48  
 49 D49  
 50 D50  
 51 D51  
 52 D52  
 53 D53  
 54 D54  
 55 D55  
 56 D56

## C.4 PARCS Input for the HFP Cycle 1

```
!*****
CASEID WBN1_cycle1_hfp
!*****
CNTL
  core_type      PWR
!
  int_th         T  -1
  xe_sm          1   1
!
  core_power     65.7
!
  bank_pos       A(1) B(2) C(3) D(4) SA(5) SB(6) SC(7) SD(8) BP(9)
                400.0 400.0 400.0 186.0 400.0 400.0 400.0 400.0 0.0
!
  ppm            1285.0
  search         ppm    1.0   1500.0
!
  mult_cyc       T
  detector        T  250.0  2
!-----
  depletion      T  1.0e-4  T
!
  TREE_XS        TREE,nset,adf,xes,ene,jlf,chi,ehd,vel,det,yld,cdf,gff,bet,lam,dht
                T    61  T  T  F  F  F  F  F  F  T  T  F  T  F  F  F
!
  PLOT_OPTS      pow flux th prec xesm pin
                0   0    0   0    0    2
  PIN_POWER      T
!*****
PARAM
  nlupd_ss       3  5  1
  n_iters_ss     15 200
  conv_ss        1.0e-6 1.0e-5 5.0e-5
  cnv_th_ss      0.001
  wielandt       0.3 10.0 1.0
  decusp         2
  cmfd           2
!*****
GEOM
  geo_dim        9  9 32  1  1          !nasyx,nasyy,nz,nzbr,nztr
  rad_conf
  10 22 10 22 10 22 10 32  6
  22 10 23 10 22 10 34 30  6
  10 23 10 22 10 21 10 31  6
```

```

22 10 22 10 22 10 33 30 6
10 22 10 22 20 23 30 5 7
22 10 21 10 23 32 30 6 0
10 34 10 33 30 30 5 8 0
32 30 31 30 5 4 8 0 0
4 4 4 4 7 0 0 0 0

file 'depl_assm.geom'

crb_def      4
              1 0 2 1 0      5.08 101.6 259.08 500.0      ! Four Types
              2 0 1 0      3.81 360.68 500.0      ! Control Rod: None AIC B4C None
              3 0 1 0      15.24 335.28 500.0      ! Pyrex Rods None Pyrx None
              4 1      500.0      ! WABA Rods: None WABA None
              ! TP bars: length active fuel
crb_type      1 1 1 1 1 1 1 2

cr_axinfo      16.591 1.5875      ! All adjustments for BP, WABA done with crb definitions

bank_conf
4 9 1 9 4 9 3 9 0
9 0 9 0 9 6 9 0 0
1 9 3 9 0 9 2 9 0
9 0 9 1 9 7 9 0 0
4 9 0 9 4 9 5 0 0
9 6 9 8 9 9 0 0
3 9 2 9 5 0 0 0
9 0 9 0 0 0 0
0 0 0 0 0

file 'depl_detect.geom'

!*****
FDBK
fa_powpit      17.673575 21.50      ! assembly power(Mw) and pitch(cm)      3411MW/193 assm =
!*****
TH
unif_th      0.74300 600.0 291.85      ! cool den, ftemp(C), ctemp(C)
!
flow_cond      291.67 85.963      ! tin,cmfrfa(Kg/sec)
n_pingt      264 25      ! npin,ngt
pin_dim      4.096 4.750 0.57 0.602      ! pelletrad,cladrad,cladthick,rgt(mm)
hgap      8000.0      ! hgap(w/M^2-C)
n_ring      8      ! number of meshes in pellet
!*****
DEPL
!      INP_HST      'shuf-cycl2-2.dep' 1 2
!      INP_OPT      F F F T
!
HST_OPT      T T T T F      ! LHCR, LHDC, LHPC, LHTF, LHTC
OUT_OPT      T T T T F      ! PPOW, PHST, PTHS, PXESM, PXSS
!      HCR_BPRA      T
!
file 'pmax.dir'

!*****
MCYCLE
bank_def      1 400.0 400.0 400.0 186.0 400.0 400.0 400.0 400.0 0.0
bank_def      2 400.0 400.0 400.0 192.0 400.0 400.0 400.0 400.0 0.0
bank_def      3 400.0 400.0 400.0 219.0 400.0 400.0 400.0 400.0 0.0
bank_def      4 400.0 400.0 400.0 218.0 400.0 400.0 400.0 400.0 0.0
bank_def      5 400.0 400.0 400.0 219.0 400.0 400.0 400.0 400.0 0.0
bank_def      6 400.0 400.0 400.0 215.0 400.0 400.0 400.0 400.0 0.0
bank_def      7 400.0 400.0 400.0 217.0 400.0 400.0 400.0 400.0 0.0
bank_def      8 400.0 400.0 400.0 220.0 400.0 400.0 400.0 400.0 0.0
bank_def      9 400.0 400.0 400.0 220.0 400.0 400.0 400.0 400.0 0.0
bank_def      10 400.0 400.0 400.0 219.0 400.0 400.0 400.0 400.0 0.0
bank_def      11 400.0 400.0 400.0 214.0 400.0 400.0 400.0 400.0 0.0
bank_def      12 400.0 400.0 400.0 216.0 400.0 400.0 400.0 400.0 0.0
bank_def      13 400.0 400.0 400.0 214.0 400.0 400.0 400.0 400.0 0.0
bank_def      14 400.0 400.0 400.0 220.0 400.0 400.0 400.0 400.0 0.0
bank_def      15 400.0 400.0 400.0 218.0 400.0 400.0 400.0 400.0 0.0
bank_def      16 400.0 400.0 400.0 222.0 400.0 400.0 400.0 400.0 0.0
bank_def      17 400.0 400.0 400.0 220.0 400.0 400.0 400.0 400.0 0.0
bank_def      18 400.0 400.0 400.0 222.0 400.0 400.0 400.0 400.0 0.0
bank_def      19 400.0 400.0 400.0 211.0 400.0 400.0 400.0 400.0 0.0
bank_def      20 400.0 400.0 400.0 215.0 400.0 400.0 400.0 400.0 0.0
bank_def      21 400.0 400.0 400.0 211.0 400.0 400.0 400.0 400.0 0.0
bank_def      22 400.0 400.0 400.0 217.0 400.0 400.0 400.0 400.0 0.0
bank_def      23 400.0 400.0 400.0 215.0 400.0 400.0 400.0 400.0 0.0
bank_def      24 400.0 400.0 400.0 220.0 400.0 400.0 400.0 400.0 0.0
bank_def      25 400.0 400.0 400.0 219.0 400.0 400.0 400.0 400.0 0.0

```

```

bank_def 26 400.0 400.0 400.0 202.0 400.0 400.0 400.0 400.0 0.0
bank_def 27 400.0 400.0 400.0 202.0 400.0 400.0 400.0 400.0 0.0
bank_def 28 400.0 400.0 400.0 220.0 400.0 400.0 400.0 400.0 0.0
bank_def 29 400.0 400.0 400.0 220.0 400.0 400.0 400.0 400.0 0.0
bank_def 30 400.0 400.0 400.0 224.0 400.0 400.0 400.0 400.0 0.0
bank_def 31 400.0 400.0 400.0 228.0 400.0 400.0 400.0 400.0 0.0
bank_def 32 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0

cycle_def 1
! bpra_pos 0 0 0 0 0 0 0 0 1
depl_step -0.346 -0.883 -0.691 -0.538 -0.538 -0.565 -0.503 -0.58 -0.495 -0.561 -0.572 -0.726 -0.465
-0.515 -0.514 -0.649 -0.461 -0.742 -0.499 -0.472 -0.672 -0.565 -0.807 -0.975 -0.00001 -0.734 -0.00001 -0.242 -
0.465 -0.495 -0.669
bank_seq 1 2 3 4 5 6 7 8 9 10 11 12 13
14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29
30 31 32
power_lev 0.1 65.7 99.7 98.0 100.0 99.7 99.7 99.8 99.8 99.5 98.0 95.1 94.8
99.8 93.9 100.1 99.7 100.2 95.6 96.4 93.4 99.7 98.0 99.4 99.9 86.9 86.9 99.6
99.6 89.9 78.8 64.5
inlet_ent -564.9 -565.2 -565.5 -565.5 -565.8 -565.8 -565.8 -565.9 -565.6 -565.4 -565.4 -565.4 -565.4 -
565.3 -565.1 -565.4 -565.3 -565.2 -565.4 -565.5 -565.1 -565.1 -565.2 -565.3 -565.3 -564.7 -564.7 -565.4 -
565.4 -564.9 -564.5 -563.7
flow_rate 32*85.93

LOCATION
1 2 3 4 5 6 7 8 0
9 10 11 12 13 14 15 16 0
17 18 19 20 21 22 23 24 0
25 26 27 28 29 30 31 32 0
33 34 35 36 37 38 39 0 0
40 41 42 43 44 45 46 0
47 48 49 50 51 52 0 0
53 54 55 56 0 0 0
0 0 0 0 0

SHUF_MAP 1 1 ! shuf index, shuf mode: 1-individual; 2-batch
1 2 3 4 5 6 7 8 0
9 10 11 12 13 14 15 16 0
17 18 19 20 21 22 23 24 0
25 26 27 28 29 30 31 32 0
33 34 35 36 37 38 39 0 0
40 41 42 43 44 45 46 0
47 48 49 50 51 52 0 0
53 54 55 56 0 0 0
0 0 0 0 0

CYCLE_IND 1 0 1
!
CONV_EC 0.1 1 ! convergence criterion (GWD/T), max number of iteration
!*****
.

```

## C.5 PARCS Input for the HZP Cycle 2 Fuel Shuffling

```

!*****
CASEID WBN1_cycle2_hzp_BOC
!*****
CNTL
core_type PWR
!
int_th F -1
xe_sm 3 3
!
core_power 1.0
! A(1) B(2) C(3) D(4) SA(5) SB(6) SC(7) SD(8) BP(9)
bank_pos 400.0 400.0 400.0 400.0 400.0 400.0 400.0 400.0 400.0
!
! ppm 500.0
! search ppm 1.0 1500.0
!
mult_cyc T
detector T 250.0 2
!-----
depletion T 1.0e-4 F
! TREE,nset,adf,xes,ene,jlf,chi, chd,vel,det,yld,cdf,gff,bet,lam,dht
TREE_XS T 61 T T F F F F F T T F F F F
! pow flux th prec xesm pin
PLOT_OPTS 0 0 0 0 0 2
PIN_POWER T

```

```

!*****
PARAM
  nlupd_ss      3  5  1
  n_iters_ss    15 200
  conv_ss       1.0e-6 1.0e-5 5.0e-5
  cnv_th_ss     0.001
  wielandt      0.3 10.0 1.0
  decusp        2
  cmfd          2
!*****
GEOM
  geo_dim      9  9 32  1  1      !nasyx,nasyy,nz,nzbr,nztr
  rad_conf
    10 22 10 22 10 22 10 32  6
    22 10 23 10 22 10 34 30  6
    10 23 10 22 10 21 10 31  6
    22 10 22 10 22 10 33 30  6
    10 22 10 22 20 23 30  5  7
    22 10 21 10 23 32 30  6  0
    10 34 10 33 30 30  5  8  0
    32 30 31 30  5  4  8  0  0
    4  4  4  4  7  0  0  0  0

  file 'depl_assm.geom'

  crb_def      4
    1  0  2  1  0      5.08 101.6 259.08 500.0 ! Four Types
    2  0  1  0      3.81 360.68 500.0 ! Control Rod: None AIC B4C None
    3  0  1  0      15.24 335.28 500.0 ! Pyrex Rods None Pyrx None
    4  1      500.0 ! WABA Rods: None WABA None
    ! TP bars: length active fuel

  crb_type     1 1 1 1 1 1 1 2

  cr_axinfo    16.591 1.5875 ! All adjustments for BP, WABA done with crb definitions

  bank_conf
    4  9  1  9  4  9  3  9  0
    9  0  9  0  9  6  9  0  0
    1  9  3  9  0  9  2  9  0
    9  0  9  1  9  7  9  0  0
    4  9  0  9  4  9  5  0  0
    9  6  9  8  9  9  0  0
    3  9  2  9  5  0  0  0
    9  0  9  0  0  0  0
    0  0  0  0  0

  file 'depl_detect.geom'

!*****
FDBK
  fa_powpit    17.673575 21.50      ! assembly power(Mw) and pitch(cm) 3411MW/193 assm =
!*****
TH
  unif_th      0.74300 600.0 291.85 ! cool den, ftemp(C), ctemp(C)
!
  flow_cond    291.67 85.963      ! tin,cmfrfa(Kg/sec)
  n_pingt      264 25      ! npin,ngt
  pin_dim      4.096 4.750 0.57 0.602 ! pelletrad,cladrad,cladthick,rgt(mm)
  hgap         8000.0      ! hgap(w/M^2-C)
  n_ring       8      ! number of meshes in pellet
!*****
DEPL
  INP_HST      'WBN1_cycle1_hfp.parcs_cyc-01' 1 32
  INP_OPT      F F F T
!
  HST_OPT      T T T T F      ! LHCR, LHDC, LHPC, LHTF, LHTC
  OUT_OPT      T T T T F      ! PPOW, PHST, PTHS, PXESM, PXSS
!
  HCR_BPRA     T
!
  file 'pmax.dir'

!*****
MCYCLE
  bank_def     1  400.0 400.0 400.0 400.0 400.0 400.0 400.0 400.0 400.0
!
  cycle_def    1
  depl_step    0.0001
  power_lev    2*0.001
  bank_seq     2*1
!
  bpri_pos     9*0
!
  cycle_def    2

```

```

depl_step      0.0001
power_lev      2*0.001
bank_seq       2*1
! bpra_pos      9*0

LOCATION         0
H-8  G-8  F-8  E-8  D-8  C-8  B-8  A-8  0
H-9  G-9  F-9  E-9  D-9  C-9  B-9  A-9  0
H-10 G-10 F-10 E-10 D-10 C-10 B-10 A-10 0
H-11 G-11 F-11 E-11 D-11 C-11 B-11 A-11 0
H-12 G-12 F-12 E-12 D-12 C-12 B-12 0 0
H-13 G-13 F-13 E-13 D-13 C-13 B-13 0 0
H-14 G-14 F-14 E-14 D-14 C-14 0 0
H-15 G-15 F-15 E-15 0 0 0 0
0 0 0 0 0 0 0 0

SHUF_MAP 1 1 ! shuf index, shuf mode: 1-individual; 2-batch
H-14 C-13 -10 A-8 -10 C-8 E-15 F-11 0
C-13 -10 A-9 -10 B-11 -10 -10 C-12 0
-10 G-15 E-15 D-9 -10 B-9 -10 G-10 0
H-15 -10 G-12 -10 C-14 -10 -10 F-13 0
-10 E-14 -10 B-13 A-10 -10 B-12 0 0
H-13 -10 G-14 -10 -10 -10 B-10 0
A-11 -10 -10 -10 D-14 F-14 0 0
E-10 D-13 F-9 C-10 0 0 0
0 0 0 0 0 0 0

CYCLE_IND 1 0 1
CYCLE_IND 2 1 2
!
CONV_EC 0.1 2 ! convergence criterion (GWD/T), max number of iteration
!*****
.

```

## C.6 PARCS Input for the HFP Cycle 2

```

!*****
CASEID WBN1_cycle2_hfp
!*****
CNTL
core_type      PWR
!
int_th         T -1
xe_sm          1 1
!
core_power     100.0
A(1) B(2) C(3) D(4) SA(5) SB(6) SC(7) SD(8) BP(9) TPB(10)
bank_pos       400.0 400.0 400.0 186.0 400.0 400.0 400.0 400.0 0.0 0.0
!
ppm            1285.0
search         ppm 1.0 1500.0
!
mult_cyc       T
detector       T 250.0 2
!-----
depletion      T 1.0e-4 T
TREE_XS        TREE,nset,adf,xes,ene,jlf,chi,ehd,vel,det,yld,cdf,gff,bet,lam,dht
! TREE_XS      T 61 T T F F F F F T T F T F F F
! pow flux th prec xesm pin
PLOT_OPTS      0 0 0 0 0 2
PIN_POWER      T
!*****
PARAM
nlupd_ss       3 5 1
n_iters_ss     15 200
conv_ss        1.0e-6 1.0e-5 5.0e-5
cnv_th_ss      0.001
wielandt       0.3 10.0 1.0
decusp         2
cmfd           2
!*****
GEOM
geo_dim        9 9 32 1 1 !nasyx,nasyy,nz,nzbr,nztr
rad_conf
10 32 46 32 45 22 30 22 6
32 42 30 44 33 46 41 23 6
46 30 30 22 44 34 41 23 6
32 44 22 46 30 45 41 21 6
45 33 44 30 30 43 30 5 7
22 46 34 45 43 40 10 6 0

```



```

30 41 41 41 30 10 5 8 0
22 23 23 21 5 4 8 0 0
4 4 4 4 7 0 0 0 0

file 'depl_assm.geom'

crb_def      4
              1 0 2 1 0      5.08 101.6 259.08 500.0      ! Four Types
              2 0 1 0      3.81 360.68 500.0      ! Control Rod: None AIC B4C None
              3 0 1 0      15.24 335.28 500.0      ! Pyrex Rods: None Pyrx None
              4 1      500.0      ! WABA Rods: None WABA None
                                ! TP bars: length active fuel
crb_type      1 1 1 1 1 1 1 3 4

cr_axinfo      16.591 1.5875      ! All adjustments for BP, WABA done with crb definitions

bank_conf
4 0 1 0 4 0 3 0 0
0 10 0 9 0 6 0 0 0
1 0 3 0 9 0 2 0 0
0 9 0 1 0 7 0 0 0
4 0 9 0 4 9 5 0 0
0 6 0 8 9 0 0 0
3 0 2 0 5 0 0 0
0 0 0 0 0 0 0
0 0 0 0 0

file 'depl_detect.geom'

!*****
FDBK
fa_powpit      17.673575 21.50      ! assembly power(Mw) and pitch(cm)      3411MW/193 assm =
!*****
TH
unif_th      0.74300 600.0 291.85      ! cool den, ftemp(C), ctemp(C)
!
flow_cond      291.67 85.963      ! tin,cmfrfa(Kg/sec)
n_pingt      264 25      ! npin,ngt
pin_dim      4.096 4.750 0.57 0.602      ! pelletrad,cladrad,cladthick,rgt(mm)
hgap      8000.0      ! hgap(w/M^2-C)
n_ring      8      ! number of meshes in pellet
!*****
DEPL
INP_HST      'WBN1_cycle2_hzp_BOC.parcs_cyc-02' 1 2
INP_OPT      F F F F
!
HST_OPT      T T T T F      ! LHCR, LHDC, LHPC, LHTF, LHTC
OUT_OPT      T T T T F      ! PPOW, PHST, PTHS, PXESM, PXSS
!
HCR_BPRA      T
!
file 'pmax.dir'

!*****
MCYCLE
bank_def      1 400.0 400.0 400.0 180.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      1 400.0 400.0 400.0 180.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      2 400.0 400.0 400.0 219.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      3 400.0 400.0 400.0 216.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      4 400.0 400.0 400.0 215.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      5 400.0 400.0 400.0 215.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      6 400.0 400.0 400.0 216.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      7 400.0 400.0 400.0 216.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      8 400.0 400.0 400.0 216.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      9 400.0 400.0 400.0 216.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      10 400.0 400.0 400.0 216.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      11 400.0 400.0 400.0 216.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      12 400.0 400.0 400.0 216.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      13 400.0 400.0 400.0 216.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      14 400.0 400.0 400.0 216.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      15 400.0 400.0 400.0 217.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      16 400.0 400.0 400.0 217.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      17 400.0 400.0 400.0 217.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      18 400.0 400.0 400.0 217.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      19 400.0 400.0 400.0 217.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      20 400.0 400.0 400.0 217.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      21 400.0 400.0 400.0 218.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      22 400.0 400.0 400.0 218.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      23 400.0 400.0 400.0 218.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      24 400.0 400.0 400.0 218.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      25 400.0 400.0 400.0 218.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      26 400.0 400.0 400.0 218.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def      27 400.0 400.0 400.0 217.0 400.0 400.0 400.0 400.0 0.0 0.0

```

```

bank_def 28 400.0 400.0 400.0 217.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def 29 400.0 400.0 400.0 217.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def 30 400.0 400.0 400.0 219.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def 31 400.0 400.0 400.0 219.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def 32 400.0 400.0 400.0 219.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def 33 400.0 400.0 400.0 222.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def 34 400.0 400.0 400.0 223.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def 35 400.0 400.0 400.0 225.0 400.0 400.0 400.0 400.0 0.0 0.0
bank_def 36 400.0 400.0 400.0 225.0 400.0 400.0 400.0 400.0 0.0 0.0

cycle_def 1 !36 steps
! bpra_pos 0 0 0 0 0 0 0 0 0 1 1
depl_step -0.142 -0.379 -0.479 -0.479 -0.586 -0.410 -0.655 -0.449 -0.440 -0.679 -0.463 -0.364 -
0.586 -0.414 -0.621 -0.559 -0.514 -0.575 -0.494 -0.571 -0.498 -0.575 -0.498 -0.532 -0.579 -0.498 -0.498 -0.529
-0.552 -0.597 -0.548 -0.556 -0.582 -0.575 -0.586
bank_seq 1 2 3 4 5 6 7 8 9 10 11 12 13
14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36
power_lev 0.1 98.4 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0
100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0
100.0 100.0 100.0 100.0 89.9 77.3 63.2
inlet_ent 36*-564.81
flow_rate 36*85.93

LOCATION
1 2 3 4 5 6 7 8 0
9 10 11 12 13 14 15 16 0
17 18 19 20 21 22 23 24 0
25 26 27 28 29 30 31 32 0
33 34 35 36 37 38 39 0 0
40 41 42 43 44 45 46 0
47 48 49 50 51 52 0 0
53 54 55 56 0 0 0
0 0 0 0 0

SHUF_MAP 1 1 ! shuf index, shuf mode: 1-individual; 2-batch
1 2 3 4 5 6 7 8 0
9 10 11 12 13 14 15 16 0
17 18 19 20 21 22 23 24 0
25 26 27 28 29 30 31 32 0
33 34 35 36 37 38 39 0 0
40 41 42 43 44 45 46 0
47 48 49 50 51 52 0 0
53 54 55 56 0 0 0
0 0 0 0 0

CYCLE_IND 1 0 1
!
CONV_EC 0.1 1 ! convergence criterion (GWD/T), max number of iteration
!*****
.

```

## C.7 PARCS Input for the HZP Cycle 3 Fuel Shuffling

```

!*****
CASEID WBN1_cycle3_hzp_BOC
!*****
CNTL
core_type PWR
!
int_th F -1
xe_sm 0 0
! xe_sm 3 3
!
core_power 1.0
! A(1) B(2) C(3) D(4) SA(5) SB(6) SC(7) SD(8) BP(9) TPB(10)
bank_pos 400.0 400.0 400.0 400.0 400.0 400.0 400.0 400.0 400.0 400.0
!
ppm 500.0
! search ppm 1.0 1500.0
!
mult_cyc T
detector T 250.0 2
!-----
depletion T 1.0e-4 F
! TREE,nset,adf,xes,ene,jlf,chi, chd,vel,det,yld,cdf,gff,bet,lam,dht
TREE_XS T 61 T T F F F F T T F T F F F
! pow flux th prec xesm pin
PLOT_OPTS 0 0 0 0 0 2
PIN_POWER T
!*****

```

```

PARAM
  nlupd_ss      3 5 1
  n_iters_ss    15 200
  conv_ss       1.0e-6 1.0e-5 5.0e-5
  cnv_th_ss     0.001
  wielandt      0.3 10.0 1.0
  decusp        2
  cmfd          2
!*****
GEOM
  geo_dim      9 9 32 1 1          !nasyx,nasyy,nz,nzbr,nztr
  rad_conf
  31 46 52 22 40 45 51 42 6
  46 62 43 62 41 51 61 44 6
  52 43 41 45 61 44 61 30 6
  46 62 45 61 46 52 63 21 6
  40 41 61 46 41 61 41 5 7
  45 51 44 52 61 63 23 6 0
  51 61 61 63 41 23 5 8 0
  42 44 30 21 5 4 8 0 0
  4 4 4 4 7 0 0 0 0

  file 'depl_assm.geom'

  crb_def      4
                1 0 2 1 0      5.08 101.6 259.08 500.0 ! Four Types
                2 0 1 0      3.81 360.68 500.0 ! Control Rod: None AIC B4C None
                3 0 1 0      15.24 335.28 500.0 ! Pyrex Rods: None Pyrx None
                4 1      500.0 ! WABA Rods: None WABA None
                                ! TP bars: length active fuel
  crb_type     1 1 1 1 1 1 1 3 4

  cr_axinfo    16.591 1.5875 ! All adjustments for BP, WABA done with crb definitions

  bank_conf
  4 0 1 0 4 0 3 0 0
  0 10 0 9 0 6 0 0 0
  1 0 3 0 9 0 2 0 0
  0 9 0 1 0 7 0 0 0
  4 0 9 0 4 9 5 0 0
  0 6 0 8 9 0 0 0
  3 0 2 0 5 0 0 0
  0 0 0 0 0 0 0
  0 0 0 0 0

  file 'depl_detect.geom'

!*****
FDBK
  fa_powpit    17.673575 21.50          ! assembly power(Mw) and pitch(cm) 3411MW/193 assm =
!*****
TH
  unif_th      0.74300 600.0 291.85      ! cool den, ftemp(C), ctemp(C)
!
  flow_cond    291.67 85.963          ! tin,cmfrfa(Kg/sec)
  n_pingt      264 25          ! npin,ngt
  pin_dim      4.096 4.750 0.57 0.602 ! pelletrad,cladrad,cladthick,rgt(mm)
  hgap         8000.0          ! hgap(w/M^2-C)
  n_ring       8          ! number of meshes in pellet
!*****
DEPL
!   INP_HST     'depl-cycl1.parcs_cyc-01' 1 30
!   INP_OPT     F F F T
!
  HST_OPT      T T T T F          ! LHCR, LHDC, LHPC, LHTF, LHTC
  OUT_OPT      T T T T F          ! PPOW, PHST, PTHS, PXESM, PXSS
!   HCR_BPRA    T
!
  file 'pmax.dir'

!*****
MCYCLE
  bank_def     1 400.0 400.0 400.0 400.0 400.0 400.0 400.0 400.0 400.0
!   bpra_pos    10*0

  cycle_def    1
  depl_step    0.0001
  power_lev    2*0.01
  bank_seq     2*1

  prev_cyc     2
                'cyc2.lay' './WBN1_cycle2_hfp.parcs_cyc-01' 1 36

```

```

'cycl1.lay' './WBN1_cycle1_hfp.parc3_cyc-01' 1 30

LOCATION      1
  2      1      0      1      1      1      0      1      0
  1      0      1      0      1      0      0      1      0
  0      1      1      1      0      1      0      1      0
  1      0      1      0      1      0      0      1      0
  1      1      0      1      1      0      1      0      0
  1      0      1      0      0      0      1      0      0
  0      0      0      0      1      1      0      0      0
  1      1      1      1      0      0      0      0      0
  0      0      0      0      0      0      0      0      0

SHUF_MAP      1      1      ! shuf index, shuf mode: 1-individual; 2-batch
13124 24617 -52 24628 24045 24505 -51 24210 0
24603 -62 24344 -62 24115 -51 -61 24421 0
-52 24338 24150 24530 -61 24426 -61 13051 0
24628 -62 24543 -61 24641 -52 -63 12142 0
24045 24148 -61 24614 24150 -61 24123 0 0
24533 -51 24412 -52 -61 -63 12344 0
-51 -61 -61 -63 24149 12338 0 0
24210 24435 13039 12122 0 0 0
0 0 0 0 0 0 0

CYCLE_IND      1      0      1
CYCLE_IND      2      1      1

!
CONV_EC      0.1      2      ! convergence criterion (GWD/T), max number of iteration
! *****
.

```

## C.8 PARCS Input for the HFP Cycle 3

```

! *****
CASEID WBN1_cycle3_hfp
! *****
CNTL
  core_type      PWR
!
  int_th      T -1
  xe_sm      1 1
!
  core_power      100.0
!
  bank_pos      A(1) B(2) C(3) D(4) SA(5) SB(6) SC(7) SD(8) BP(9)
  400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
!
  ppm      1285.0
  search      ppm 1.0 1800.0
!
  mult_cyc      T
  detector      T 250.0 2
!-----
  depletion      T 1.0e-4 T
!
  TREE_XS      TREE,nset,adf,xes,ene,jlf,chi, chd,vel,det,yld,cdf,gff,bet,lam,dht
  T 61 T T F F F F T T F T F F F
!
  pow flux th prec xesm pin
PLOT_OPTS      0 0 0 0 0 2
PIN_POWER      T
! *****
PARAM
  nlupd_ss      3 5 1
  n_iters_ss      15 200
  conv_ss      1.0e-6 1.0e-5 5.0e-5
  cnv_th_ss      0.001
  wielandt      0.3 10.0 1.0
  decusp      2
  cmfd      2
! *****
GEOM
  geo_dim      9 9 32 1 1      !nasyx,nasyy,nz,nzbr,nztr
  rad_conf
  31 46 52 46 40 45 51 42 6
  46 63 43 63 41 51 62 44 6
  52 43 41 45 62 44 62 30 6
  46 63 45 62 46 52 61 21 6
  40 41 62 46 41 62 41 5 7
  45 51 44 52 62 61 23 6 0
  51 62 62 61 41 23 5 8 0
  42 44 30 21 5 4 8 0 0
  4 4 4 4 7 0 0 0 0

```

```

file 'depl_assm.geom'

crb_def      4
             1 0 2 1 0      5.08 101.6 259.08 500.0      ! Four Types
             2 0 1 0      3.81 360.68 500.0      ! Control Rod: None AIC B4C None
             3 0 1 0      15.24 335.28 500.0      ! Pyrex Rods None Pyrx None
             4 1      500.0      ! WABA Rods: None WABA None
                                     ! TP bars: length active fuel
crb_type      1 1 1 1 1 1 1 3

cr_axinfo      16.591 1.5875      ! All adjustments for BP, WABA done with crb definitions

bank_conf
4 0 1 0 4 0 3 0 0
0 9 0 9 0 6 0 0 0
1 0 3 0 0 0 2 0 0
0 9 0 1 0 7 0 0 0
4 0 0 0 4 0 5 0 0
0 6 0 8 0 0 0 0
3 0 2 0 5 0 0 0
0 0 0 0 0 0 0
0 0 0 0 0

file 'depl_detect.geom'

!*****
FDBK
fa_powpit      17.673575 21.50      ! assembly power(Mw) and pitch(cm)      3411MW/193 assm =
!*****
TH
unif_th      0.74300 600.0 291.85      ! cool den, ftemp(C), ctemp(C)
!
flow_cond      291.67 85.963      ! tin,cmfrfa(Kg/sec)
n_pingt      264 25      ! npin,ngt
pin_dim      4.096 4.750 0.57 0.602      ! pelletrad,cladrad,cladthick,rgt(mm)
hgap      8000.0      ! hgap(w/M^2-C)
n_ring      8      ! number of meshes in pellet
!*****
DEPL
INP_HST      'WBN1_cycle3_hzp_BOC.parcs_cyc-02' 1 2
INP_OPT      F F F F
!
HST_OPT      T T T T F      ! LHCR, LHDC, LHPC, LHTF, LHTC
OUT_OPT      T T T T F      ! PPOW, PHST, PTHS, PXESM, PXSS
!
HCR_BPRA      T
!
file 'pmax.dir'

!*****
MCYCLE
bank_def      1 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def      2 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def      3 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def      4 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def      5 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def      6 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def      7 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def      8 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def      9 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     10 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     11 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     12 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     13 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     14 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     15 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     16 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     17 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     18 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     19 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     20 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     21 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     22 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     23 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     24 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     25 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     26 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     27 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     28 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     29 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     30 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def     31 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0

```

```

bank_def 32 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def 33 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def 34 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def 35 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def 36 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def 37 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def 38 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def 39 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0
bank_def 40 400.0 400.0 400.0 230.0 400.0 400.0 400.0 400.0 0.0

cycle_def 1 !40 steps
! bpra_pos 0 0 0 0 0 0 0 0 0 1
depl_step -0.195 -0.497 -0.474 -0.532 -0.547 -0.524 -0.535 -0.115 -0.092 -0.447 -0.394 -0.558 -
0.547 -0.367 -0.612 -0.623 -0.524 -0.551 -0.520 -0.547 -0.524 -0.551 -0.520 -0.550 -0.524 -0.539 -0.536 -0.543
-0.524 -0.547 -0.527 -0.536 -0.539 -0.539 -0.532 -0.516 -0.394 -0.646 -0.536
bank_seq 1 2 3 4 5 6 7 8 9 10 11 12 13
14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39 40
power_lev 0.1 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0
100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0
100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 85.9 75.8
inlet_ent 40*-564.81
flow_rate 40*85.963

LOCATION
1 2 3 4 5 6 7 8 0
9 10 11 12 13 14 15 16 0
17 18 19 20 21 22 23 24 0
25 26 27 28 29 30 31 32 0
33 34 35 36 37 38 39 0 0
40 41 42 43 44 45 46 0
47 48 49 50 51 52 0 0
53 54 55 56 0 0 0
0 0 0 0 0

SHUF_MAP 1 1 ! shuf index, shuf mode: 1-individual; 2-batch
1 2 3 4 5 6 7 8 0
9 10 11 12 13 14 15 16 0
17 18 19 20 21 22 23 24 0
25 26 27 28 29 30 31 32 0
33 34 35 36 37 38 39 0 0
40 41 42 43 44 45 46 0
47 48 49 50 51 52 0 0
53 54 55 56 0 0 0
0 0 0 0 0

CYCLE_IND 1 0 1
!
CONV_EC 0.1 1 ! convergence criterion (GWD/T), max number of iteration
!*****
.

```



# APPENDIX D. PROGRAM TO COMPARE FLUX MAPS FOR PWRs

## D.1 Program Source

```
program DetectorResponse
! This is for the PWR detector response
implicit none
!
character(10) :: ss10
character(100) :: ctitle, file1, file2
character(400) :: ssw
integer :: nmeasure, nburn, naxial, ndets, nassembly, nxa, nya, numdet
integer :: i, j, k, nx, nid, ix, j1, j2, k2, idparcs, kid
integer, allocatable :: nidmeasure(:), numx(:), mapaxial(:, :), mapradial(:)
character(3), allocatable :: cDET(:), mDET(:), config(:, :)
real, allocatable :: weight(:), axialmesh(:)
real :: tempr, sumfa, totsum, sum0, sum1, sum2, sum3, diff, diff2, zpoint, x1, x2, y1, y2, yout
real :: mao, cao, dz
logical :: kr
!
type meas_type
integer :: nid, ndets, naxial, ndets_active
real :: burnup, efpd
character(3), allocatable :: mDET(:)
real, allocatable :: axialmesh(:)
real, allocatable :: mpownodal(:, :), mpowradial(:), mpowaxial(:)
real, allocatable :: mpownodal2(:, :), mpowradial2(:), mpowaxial2(:)
real, allocatable :: cpownodal(:, :), cpowradial(:), cpowaxial(:)
logical, allocatable :: ldata(:)
end type
type(meas_type), allocatable :: measdata(:)
!
type calc_type
character(3), allocatable :: cDET(:)
real, allocatable :: pownodal(:, :), axialmesh(:)
end type
type(calc_type), allocatable :: calcddata(:)
real :: rms_all, rms_radial, rms_axial
real :: max_all, max_radial, max_axial, min_all, min_radial, min_axial
!
! read input file
open(1, file='DetRes.in', status='old')
open(21, file='DetRes.out', status='unknown')
read(1, *) !%TITLE
read(1, *) ctitle
read(1, *) !%MEASURED
read(1, *) ss10, file1
read(1, *) ss10, nmeasure, ndets
allocate(nidmeasure(nmeasure))
read(1, *) (nidmeasure(i), i=1, nmeasure)
read(1, *) !%CALCULATED
read(1, *) ss10, file2
read(1, *) ss10, nburn, naxial, nassembly
allocate(axialmesh(0:naxial))
axialmesh(0)=0.0
do i=1, naxial
read(1, *) axialmesh(i)
enddo
read(1, *) !%INDEX
allocate(mDET(ndets), cDET(ndets), weight(ndets))
do i=1, ndets
read(1, *) ix, mDET(i), cDET(i) !Detector names in meas. and calc.
enddo
read(1, *) !%CORE
read(1, *) nxa, nya
allocate(config(nxa, nya))
do j=1, nya
read(1, *) (config(i, j), i=1, nxa)
enddo
!
! read calculated data
open(3, file=file2, status='old')
read(3, *)
allocate(calcddata(nburn))
do i=1, nburn
allocate(calcddata(i)%cDET(nassembly))
allocate(calcddata(i)%pownodal(naxial, nassembly))
read(3, *) ss10, ss10, ss10, ss10, (calcddata(i)%cDET(k), k=1, nassembly)
```



```

do j=naxial,1,-1
    read(3,*) tempr,tempr,tempr,tempr,(calcddata(i)%pownodal(j,k),k=1,nassembly)
enddo
enddo
!
!
read measured data
open(2,file=file1,status='old')
allocate(measdata(nmeasure))
read(2,*) ssl0,nx
if (nx .ne. nmeasure) then
    write(6,*) "Error: # of cases do not match in the measured data"
    stop
endif
do i=1,nmeasure
    idparcs=nidmeasure(i)

    read(2,*) measdata(i)%nid,measdata(i)%ndets,measdata(i)%naxial
    read(2,*)
    read(2,*) measdata(i)%burnup, measdata(i)%efpd
    allocate(measdata(i)%mDET(measdata(i)%ndets))
    allocate(measdata(i)%mpownodal(measdata(i)%naxial,measdata(i)%ndets))
    allocate(measdata(i)%axialmesh(measdata(i)%naxial))
    read(2,*) ssl0,(measdata(i)%mDET(k),k=1,measdata(i)%ndets)
    do j=1,measdata(i)%naxial
        read(2,*) measdata(i)%axialmesh(j),(measdata(i)%mpownodal(j,k),k=1,measdata(i)%ndets)
    enddo
    read(2,*,end=200)
    continue
    !sum over radial & axial and normalize
    allocate(measdata(i)%mpowradial(measdata(i)%ndets))
    allocate(measdata(i)%ldata(measdata(i)%ndets))
    allocate(measdata(i)%mpowaxial(measdata(i)%naxial))
    measdata(i)%mpowradial(:)=0.0
    measdata(i)%mpowaxial(:)=0.0
    sum0=0.0
    do j=1,measdata(i)%naxial
        do k=1,measdata(i)%ndets
            measdata(i)%mpowradial(k)=measdata(i)%mpowradial(k)+measdata(i)%mpownodal(j,k)
            measdata(i)%mpowaxial(j)=measdata(i)%mpowaxial(j)+measdata(i)%mpownodal(j,k)
            sum0=sum0+measdata(i)%mpownodal(j,k)
        enddo
    enddo
    !Sort out the detectors without data
    measdata(i)%ldata(:)=.false.
    numdet=0
    do k=1,measdata(i)%ndets
        if (measdata(i)%mpowradial(k) > 1.0e-5) then
            measdata(i)%ldata(k)=.true.
            numdet=numdet+1
        endif
    enddo
    measdata(i)%ndets_active=numdet
    !normalization
    do k=1,measdata(i)%ndets
        measdata(i)%mpowradial(k)=measdata(i)%mpowradial(k)/sum0*float(numdet)
    enddo
    do j=1,measdata(i)%naxial
        measdata(i)%mpowaxial(j)=measdata(i)%mpowaxial(j)/sum0*float(measdata(i)%naxial)
    enddo
    do j=1,measdata(i)%naxial
        do k=1,measdata(i)%ndets
            measdata(i)%mpownodal(j,k)=measdata(i)%mpownodal(j,k)/sum0*float(numdet*measdata(i)%naxial)
        enddo
    enddo
    !
    !convert the calculated data into the measured format
    allocate(measdata(i)%cpownodal(naxial,measdata(i)%ndets))
    allocate(measdata(i)%cpowradial(measdata(i)%ndets))
    allocate(measdata(i)%cpowaxial(naxial))
    allocate(measdata(i)%mpownodal2(naxial,measdata(i)%ndets))
    allocate(measdata(i)%mpowradial2(measdata(i)%ndets))
    allocate(measdata(i)%mpowaxial2(naxial))
    allocate(mapradial(measdata(i)%ndets))
    allocate(mapaxial(naxial,2))
    !axial mesh mapping
    mapaxial(:, :)=0
    do j=1,naxial
        zpoint=0.5*(axialmesh(j-1)+axialmesh(j))
        do j2=1,measdata(i)%naxial-1
            if ((zpoint-measdata(i)%axialmesh(j2))*(zpoint-measdata(i)%axialmesh(j2+1)) < 0.0) then
                nid=j2
            enddo
        enddo
    enddo

```

```

        !write(6,*) j,j2,measdata(i)%axialmesh(j),axialmesh(j2),axialmesh(j2+1)
        exit
    elseif ((zpoint-measdata(i)%axialmesh(j2))*(zpoint-measdata(i)%axialmesh(j2+1)) == 0.0) then
        if (zpoint == measdata(i)%axialmesh(j2)) then
            nid=j2
        else
            nid=j2+1
        endif
        exit
    endif
enddo
if (nid > measdata(i)%naxial) nid=nid-1
mapaxial(j,1)=nid
mapaxial(j,2)=nid+1
enddo
!radial mesh mapping
mapradial(:)=0
do k=1,measdata(i)%ndets
    if (.not.measdata(i)%ldata(k)) cycle
    kr=.false.
    do k2=1,ndets
        if (measdata(i)%mDET(k) == mDET(k2)) then
            !write(6,*) k,mDET(k2)," ",cDET(k2)
            kr=.true.
            kid=k2
            exit
        endif
    enddo
    if (.not.kr) then
        write(6,*) "Error: there is no matching detector #1!"
    endif
    kr=.false.
    do k2=1,nassembly
        if (calcddata(idparcs)%cDET(k2) == cDET(kid)) then
            kr=.true.
            mapradial(k)=k2
            exit
        endif
    enddo
    if (.not.kr) then
        write(6,*) "Error: there is no matching detector #2!"
    endif
endif
enddo
!Data re-assignment for the calculated
measdata(i)%cpownodal(:,:)=0.0
do j=1,naxial
    do k=1,measdata(i)%ndets
        if (.not.measdata(i)%ldata(k)) cycle
        k2=mapradial(k)
        measdata(i)%cpownodal(j,k)=calcddata(idparcs)%pownodal(j,k2)
    enddo
enddo
measdata(i)%cpowradial(:)=0.0
measdata(i)%cpowaxial(:)=0.0
sum0=0.0
do j=1,naxial
    do k=1,measdata(i)%ndets
        measdata(i)%cpowradial(k)=measdata(i)%cpowradial(k)+measdata(i)%cpownodal(j,k)
        measdata(i)%cpowaxial(j)=measdata(i)%cpowaxial(j)+measdata(i)%cpownodal(j,k)
        sum0=sum0+measdata(i)%cpownodal(j,k)
    enddo
enddo
!normalization
do k=1,measdata(i)%ndets
    measdata(i)%cpowradial(k)=measdata(i)%cpowradial(k)/sum0*float(measdata(i)%ndets_active)
enddo
do j=1,naxial
    measdata(i)%cpowaxial(j)=measdata(i)%cpowaxial(j)/sum0*float(naxial)
enddo
do j=1,naxial
    do k=1,measdata(i)%ndets
        measdata(i)%cpownodal(j,k)=measdata(i)%cpownodal(j,k)/sum0*float(measdata(i)%ndets_active*naxial)
    enddo
enddo
!Data re-assignment for the measured
measdata(i)%mpownodal2(:,:)=0.0
do j=1,naxial
    zpoint=0.5*(axialmesh(j-1)+axialmesh(j))
    j1=mapaxial(j,1)
    j2=mapaxial(j,2)
    x1=measdata(i)%axialmesh(j1)

```

```

x2=measdata(i)%axialmesh(j2)
do k=1,measdata(i)%ndets
  if (.not.measdata(i)%ldata(k)) cycle
  y1=measdata(i)%mpownodal(j1,k)
  y2=measdata(i)%mpownodal(j2,k)
  if (j1 == j2) then
    yout=measdata(i)%mpownodal(j1,k)
  else
    yout=(y2-y1)*zpoint/(x2-x1)+(y1*x2-y2*x1)/(x2-x1)
  endif
  measdata(i)%mpownodal2(j,k)=yout
enddo
enddo
measdata(i)%mpowradial2(:)=0.0
measdata(i)%mpowaxial2(:)=0.0
sum0=0.0
do j=1,naxial
  do k=1,measdata(i)%ndets
    measdata(i)%mpowradial2(k)=measdata(i)%mpowradial2(k)+measdata(i)%mpownodal2(j,k)
    measdata(i)%mpowaxial2(j)=measdata(i)%mpowaxial2(j)+measdata(i)%mpownodal2(j,k)
    sum0=sum0+measdata(i)%mpownodal2(j,k)
  enddo
enddo
!normalization
do k=1,measdata(i)%ndets
  measdata(i)%mpowradial2(k)=measdata(i)%mpowradial2(k)/sum0*float(measdata(i)%ndets_active)
enddo
do j=1,naxial
  measdata(i)%mpowaxial2(j)=measdata(i)%mpowaxial2(j)/sum0*float(naxial)
enddo
do j=1,naxial
  do k=1,measdata(i)%ndets
    measdata(i)%mpownodal2(j,k)=measdata(i)%mpownodal2(j,k)/sum0*float(measdata(i)%ndets_active*naxial)
  enddo
enddo
!
!Data processing for %RMSs
rms_all=0.0
rms_radial=0.0
rms_axial=0.0
max_all=0.0
max_radial=0.0
max_axial=0.0
min_all=0.0
min_radial=0.0
min_axial=0.0
do j=1,naxial
  do k=1,measdata(i)%ndets
    if (.not.measdata(i)%ldata(k)) cycle
    diff=measdata(i)%cpownodal(j,k)-measdata(i)%mpownodal2(j,k)
    diff2=diff**2
    rms_all=rms_all+diff2
    if (diff > max_all) max_all=diff
    if (diff < min_all) min_all=diff
  enddo
enddo
rms_all=sqrt(rms_all/float(measdata(i)%ndets_active*naxial))*100.0
do k=1,measdata(i)%ndets
  if (.not.measdata(i)%ldata(k)) cycle
  diff=measdata(i)%cpowradial(k)-measdata(i)%mpowradial2(k)
  diff2=diff**2
  rms_radial=rms_radial+diff2
  if (diff > max_radial) max_radial=diff
  if (diff < min_radial) min_radial=diff
enddo
rms_radial=sqrt(rms_radial/float(measdata(i)%ndets_active))*100.0
do j=1,naxial
  diff=measdata(i)%cpowaxial(j)-measdata(i)%mpowaxial2(j)
  diff2=diff**2
  rms_axial=rms_axial+diff2
  if (diff > max_axial) max_axial=diff
  if (diff < min_axial) min_axial=diff
enddo
rms_axial=sqrt(rms_axial/float(naxial))*100.0
mao=0.0
cao=0.0
sum1=0.0
sum2=0.0
zpoint=0.5*axialmesh(naxial)
do j=1,naxial
  dz=axialmesh(j)-axialmesh(j-1)

```

```

        sum1=sum1+measdata(i)%mpowaxial2(j)*dz
        sum2=sum2+measdata(i)%cpowaxial(j)*dz
        if (zpoint > axialmesh(j)) then
            mao=mao+measdata(i)%mpowaxial2(j)*dz
            cao=cao+measdata(i)%cpowaxial(j)*dz
        elseif (zpoint < axialmesh(j-1)) then
            cycle
        elseif (zpoint > axialmesh(j-1) .and. zpoint < axialmesh(j) ) then
            dz=abs(zpoint-axialmesh(j-1))
            mao=mao+measdata(i)%mpowaxial2(j)*dz
            cao=cao+measdata(i)%cpowaxial(j)*dz
        endif
    enddo
    mao=(sum1-2.0*mao)/sum1*100.0
    cao=(sum2-2.0*cao)/sum2*100.0
    !
    if (i==1) write(21,*) ctitle
    write(21,*)
    write(21,' (@Case =           ",i5)') i
    write(21,' ("Burnup (MWD/kgU)",f10.3)') measdata(i)%burnup
    write(21,' ("% RMS for 3D           ",f10.3,i5)') rms_all,measdata(i)%ndets_active*naxial
    write(21,' ("% RMS for radial",f10.3,i5)') rms_radial,measdata(i)%ndets_active
    write(21,' ("% RMS for axial ",f10.3,i5)') rms_axial,naxial
    !write(21,' ("% axial offsets ",2f10.3)') mao,cao
    write(21,*)
    do j=1,naxial
        diff=(measdata(i)%cpowaxial(j)-measdata(i)%mpowaxial2(j))*100.0
        zpoint=0.5*(axialmesh(j-1)+axialmesh(j))
        write(21,' (i5,4f10.3)') j,zpoint,measdata(i)%mpowaxial2(j),measdata(i)%cpowaxial(j),diff
    enddo
!
        deallocate(mapaxial)
        deallocate(mapradial)
    enddo
100 continue
!
        close(21)
!
        stop
    end

```

## D.2 Cycle 1 Input

```

%TITLE
'WBN1 cycle-1'
%MEASURED
file WBN1_cyl.dat
dim 13 58 !# of parc burnup points, # of detectors
4 6 7 9 11 12 13 17 19 21 24 29 31
%CALCULATED
file WBN1_cycle1_hfp.parcs_det
dim 32 30 56 4 !# of burnups / # of axials / # of assemblies / symmetry (1:full, 4:quarter
3.810 !1 3.810 bottom to top
15.240 !2 11.430
30.480 !3 15.240
45.912 !4 15.432
61.344 !5 15.432
65.154 !6 3.810
81.284 !7 16.130
97.414 !8 16.130
113.544 !9 16.130
117.354 !10 3.810
133.484 !11 16.130
149.614 !12 16.130
165.744 !13 16.130
169.554 !14 3.810
185.684 !15 16.130
201.814 !16 16.130
217.944 !17 16.130
221.754 !18 3.810
237.884 !19 16.130
254.014 !20 16.130
270.144 !21 16.130
273.954 !22 3.810
290.084 !23 16.130
306.214 !24 16.130
322.344 !25 16.130
326.154 !26 3.810
335.280 !27 9.126

```

```

350.520 !28 15.240
364.490 !29 13.970
365.760 !30 1.270
%INDEX 58
1 1 D54 ! J01 For BEAVRS
2 2 D55 ! F01
3 3 D52 ! N02
4 4 D49 ! K02
5 5 D47 ! H02
6 6 D40 ! H03
7 7 D42 ! F03
8 8 D44 ! D03
9 9 D46 ! B03
10 10 D39 ! P04
11 11 D38 ! N04
12 12 D33 ! H04
13 13 D28 ! L05
14 14 D26 ! G05
15 15 D28 ! E05
16 16 D30 ! C05
17 17 D24 ! R06
18 18 D22 ! N06
19 19 D19 ! K06
20 20 D17 ! H06
21 21 D23 ! B06
22 22 D13 ! M07
23 23 D10 ! J07
24 24 D11 ! F07
25 25 D14 ! C07
26 26 D08 ! R08
27 27 D06 ! N08
28 28 D04 ! L08
29 29 D02 ! J08
30 30 D03 ! F08
31 31 D05 ! D08
32 32 D06 ! C08
33 33 D07 ! B08
34 34 D15 ! P09
35 35 D10 ! G09
36 36 D12 ! E09
37 37 D16 ! A09
38 38 D20 ! L10
39 39 D18 ! J10
40 40 D21 ! D10
41 41 D32 ! R11
42 42 D28 ! L11
43 43 D25 ! H11
44 44 D28 ! E11
45 45 D32 ! A11
46 46 D35 ! K12
47 47 D34 ! G12
48 48 D37 ! D12
49 49 D45 ! N13
50 50 D43 ! L13
51 51 D40 ! H13
52 52 D46 ! B13
53 53 D52 ! N14
54 54 D48 ! J14
55 55 D49 ! F14
56 56 D51 ! D14
57 57 D56 ! L15
58 58 D53 ! H15
%CORE
15 15
0 0 0 0 0 0 1 0 0 2 0 0 0 0 0
0 0 3 0 0 4 0 5 0 0 0 0 0 0 0
0 0 0 0 0 0 0 6 0 7 0 8 0 9 0
0 10 11 0 0 0 0 12 0 0 0 0 0 0
0 0 0 0 13 0 0 0 14 0 15 0 16 0 0
17 0 18 0 0 19 0 20 0 0 0 0 21 0
0 0 0 22 0 0 23 0 0 24 0 0 25 0 0
26 0 27 0 28 0 29 0 0 30 0 31 32 33 0
0 34 0 0 0 0 0 0 35 0 36 0 0 0 37
0 0 0 0 38 0 39 0 0 0 0 40 0 0
41 0 0 0 42 0 0 43 0 0 44 0 0 0 45
0 0 0 0 0 46 0 0 47 0 0 48 0 0
0 0 49 0 50 0 0 51 0 0 0 0 52 0
0 0 53 0 0 0 54 0 0 55 0 56 0 0
0 0 0 0 57 0 0 58 0 0 0 0 0 0
%END

```

## D.3 Cycle 2 Input

```
%TITLE
'WBN1 cycle-2'
%MEASURED
file WBN1_cy2.dat
dim 13 58 !# of parc burnup points, # of detectors
4 6 8 10 12 14 16 18 20 22 24 26 28
%CALCULATED
file WBN1_cycle2_hfp.parc_det
dim 36 30 56 4 !# of burnups / # of axials / # of assemblies / symmetry (1:full, 4:quarter
3.810 !1 3.810 bottom to top
15.240 !2 11.430
30.480 !3 15.240
45.912 !4 15.432
61.344 !5 15.432
65.154 !6 3.810
81.284 !7 16.130
97.414 !8 16.130
113.544 !9 16.130
117.354 !10 3.810
133.484 !11 16.130
149.614 !12 16.130
165.744 !13 16.130
169.554 !14 3.810
185.684 !15 16.130
201.814 !16 16.130
217.944 !17 16.130
221.754 !18 3.810
237.884 !19 16.130
254.014 !20 16.130
270.144 !21 16.130
273.954 !22 3.810
290.084 !23 16.130
306.214 !24 16.130
322.344 !25 16.130
326.154 !26 3.810
335.280 !27 9.126
350.520 !28 15.240
364.490 !29 13.970
365.760 !30 1.270
%INDEX 58
1 1 D54 ! J01 For BEAVRS
2 2 D55 ! F01
3 3 D52 ! N02
4 4 D49 ! K02
5 5 D47 ! H02
6 6 D40 ! H03
7 7 D42 ! F03
8 8 D44 ! D03
9 9 D46 ! B03
10 10 D39 ! P04
11 11 D38 ! N04
12 12 D33 ! H04
13 13 D28 ! L05
14 14 D26 ! G05
15 15 D28 ! E05
16 16 D30 ! C05
17 17 D24 ! R06
18 18 D22 ! N06
19 19 D19 ! K06
20 20 D17 ! H06
21 21 D23 ! B06
22 22 D13 ! M07
23 23 D10 ! J07
24 24 D11 ! F07
25 25 D14 ! C07
26 26 D08 ! R08
27 27 D06 ! N08
28 28 D04 ! L08
29 29 D02 ! J08
30 30 D03 ! F08
31 31 D05 ! D08
32 32 D06 ! C08
33 33 D07 ! B08
34 34 D15 ! P09
35 35 D10 ! G09
36 36 D12 ! E09
37 37 D16 ! A09
```

```

38 38 D20 ! L10
39 39 D18 ! J10
40 40 D21 ! D10
41 41 D32 ! R11
42 42 D28 ! L11
43 43 D25 ! H11
44 44 D28 ! E11
45 45 D32 ! A11
46 46 D35 ! K12
47 47 D34 ! G12
48 48 D37 ! D12
49 49 D45 ! N13
50 50 D43 ! L13
51 51 D40 ! H13
52 52 D46 ! B13
53 53 D52 ! N14
54 54 D48 ! J14
55 55 D49 ! F14
56 56 D51 ! D14
57 57 D56 ! L15
58 58 D53 ! H15

%CORE
15 15
0 0 0 0 0 0 1 0 0 2 0 0 0 0 0
0 0 3 0 0 4 0 5 0 0 0 0 0 0 0
0 0 0 0 0 0 0 6 0 7 0 8 0 9 0
0 10 11 0 0 0 0 12 0 0 0 0 0 0 0
0 0 0 0 13 0 0 0 14 0 15 0 16 0 0
17 0 18 0 0 19 0 20 0 0 0 0 0 21 0
0 0 0 22 0 0 23 0 0 24 0 0 25 0 0
26 0 27 0 28 0 29 0 0 30 0 31 32 33 0
0 34 0 0 0 0 0 0 35 0 36 0 0 0 37
0 0 0 0 38 0 39 0 0 0 0 40 0 0 0
41 0 0 0 42 0 0 43 0 0 44 0 0 0 45
0 0 0 0 0 46 0 0 47 0 0 48 0 0 0
0 0 49 0 50 0 0 51 0 0 0 0 0 52 0
0 0 53 0 0 0 54 0 0 55 0 56 0 0 0
0 0 0 0 57 0 0 58 0 0 0 0 0 0 0

%END

```

## D.4 Cycle 3 Input

```

%TITLE
'WBN1 cycle-3'
%MEASURED
file WBN1_cy3.dat
dim 19 58 !# of parc burnup points, # of detectors
4 6 8 10 12 14 15 17 19 21 23 25 27 29 31 33 35 37 39
%CALCULATED
file WBN1_cycle3_hfp.parcs_det
dim 40 30 56 4 !# of burnups / # of axials / # of assemblies / symmetry (1:full, 4:quarter
3.810 !1 3.810 bottom to top
15.240 !2 11.430
30.480 !3 15.240
45.912 !4 15.432
61.344 !5 15.432
65.154 !6 3.810
81.284 !7 16.130
97.414 !8 16.130
113.544 !9 16.130
117.354 !10 3.810
133.484 !11 16.130
149.614 !12 16.130
165.744 !13 16.130
169.554 !14 3.810
185.684 !15 16.130
201.814 !16 16.130
217.944 !17 16.130
221.754 !18 3.810
237.884 !19 16.130
254.014 !20 16.130
270.144 !21 16.130
273.954 !22 3.810
290.084 !23 16.130
306.214 !24 16.130
322.344 !25 16.130
326.154 !26 3.810
335.280 !27 9.126
350.520 !28 15.240

```

```

364.490 !29 13.970
365.760 !30 1.270
%INDEX 58
1 1 D54 ! J01 For BEAVRS
2 2 D55 ! F01
3 3 D52 ! N02
4 4 D49 ! K02
5 5 D47 ! H02
6 6 D40 ! H03
7 7 D42 ! F03
8 8 D44 ! D03
9 9 D46 ! B03
10 10 D39 ! P04
11 11 D38 ! N04
12 12 D33 ! H04
13 13 D28 ! L05
14 14 D26 ! G05
15 15 D28 ! E05
16 16 D30 ! C05
17 17 D24 ! R06
18 18 D22 ! N06
19 19 D19 ! K06
20 20 D17 ! H06
21 21 D23 ! B06
22 22 D13 ! M07
23 23 D10 ! J07
24 24 D11 ! F07
25 25 D14 ! C07
26 26 D08 ! R08
27 27 D06 ! N08
28 28 D04 ! L08
29 29 D02 ! J08
30 30 D03 ! F08
31 31 D05 ! D08
32 32 D06 ! C08
33 33 D07 ! B08
34 34 D15 ! P09
35 35 D10 ! G09
36 36 D12 ! E09
37 37 D16 ! A09
38 38 D20 ! L10
39 39 D18 ! J10
40 40 D21 ! D10
41 41 D32 ! R11
42 42 D28 ! L11
43 43 D25 ! H11
44 44 D28 ! E11
45 45 D32 ! A11
46 46 D35 ! K12
47 47 D34 ! G12
48 48 D37 ! D12
49 49 D45 ! N13
50 50 D43 ! L13
51 51 D40 ! H13
52 52 D46 ! B13
53 53 D52 ! N14
54 54 D48 ! J14
55 55 D49 ! F14
56 56 D51 ! D14
57 57 D56 ! L15
58 58 D53 ! H15
%CORE
15 15
0 0 0 0 0 0 1 0 0 2 0 0 0 0 0
0 0 3 0 0 4 0 5 0 0 0 0 0 0 0
0 0 0 0 0 0 0 6 0 7 0 8 0 9 0
0 10 11 0 0 0 0 12 0 0 0 0 0 0 0
0 0 0 0 13 0 0 0 14 0 15 0 16 0 0
17 0 18 0 0 19 0 20 0 0 0 0 0 21 0
0 0 0 22 0 0 23 0 0 24 0 0 25 0 0
26 0 27 0 28 0 29 0 0 30 0 31 32 33 0
0 34 0 0 0 0 0 35 0 36 0 0 0 37
0 0 0 0 38 0 39 0 0 0 0 40 0 0 0
41 0 0 0 42 0 0 43 0 0 44 0 0 0 45
0 0 0 0 46 0 0 47 0 0 48 0 0 0
0 0 49 0 50 0 0 51 0 0 0 0 52 0
0 0 53 0 0 0 54 0 0 55 0 56 0 0 0
0 0 0 0 57 0 0 58 0 0 0 0 0 0
%END

```



CUE-1 7  
CUE-1 13  
9 58 61  
MND/CIUEE

[illegible][illegible]



















[illegible]

|   |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |         |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|---|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 0 | 0.3600 | 0.2803 | 0.2332 | 0.6214 | 0.5401 | 0.7053 | 0.6639 | 0.7196 | 0.2302 | 0.3689 | 0.7237 | 0.0000 | 0.7272 | 0.7527 | 0.7356 | 0.7381 | 0.2735 | 0.6646 | 0.6659 | 0.7252 | 0.6103 | 0.7091 | 0.6814 | 0.7254 | 0.0000 | 0.2945 | 0.6223 | 0.9516 | 0.6113 | 0.7336 | 0.7536 | 0.7228 | 0.5798 | 0.5929 | 0.6734 | 0.74675 | 0.3706 | 0.6573 | 0.7441 | 0.7613 | 0.2223 | 0.7239 | 0.7514 | 0.7319 | 0.2635 | 0.7510 | 0.6608 | 0.7454 | 0.4717 | 0.7732 | 0.6639 | 0.2199 | 0.2401 | 0.5956 | 0.0000 | 0.5193 | 0.2113 | 0.2731 |
|---|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|

|       |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 6.106 | 0.3324 | 0.2028 | 0.2629 | 0.6201 | 0.6902 | 0.9886 | 0.7863 | 0.6722 | 0.2633 | 0.4106 | 0.5876 | 0.0000 | 0.8859 | 0.9474 | 0.1070 | 0.9388 | 0.8398 | 0.7482 | 0.7471 | 0.8483 | 0.6877 | 0.7172 | 0.8159 | 0.8895 | 0.0000 | 0.3024 | 0.7704 | 0.6601 | 0.7652 | 0.8780 | 0.8632 | 0.7478 | 0.7255 | 0.6274 | 0.6969 | 0.7635 | 0.2223 | 0.7073 | 0.7836 | 0.9007 | 0.2782 | 0.6837 | 0.9446 | 0.1030 | 0.2525 | 0.8880 | 0.9107 | 0.9818 | 0.6236 | 0.8013 | 0.7307 | 0.2469 | 0.3328 | 0.8953 | 0.0000 | 0.3720 | 0.2263 | 0.4377 | 0.3477 |
|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|

24.384 0.5503 0.5477 0.4264 1.1559 1.0984 1.1703 1.3354 1.2341 0.4255 0.6923 1.2415 0.0000 1.1817 1.2776 1.2449 1.3659 0.5825 1.3224 1.3001 1.1880 1.1801 1.2394 1.1555 1.3732 0.0000 0.5346 1.2987 1.3004 1.2101 1.2077 1.3659 1.2875 1.1519 1.1105 1.1329 1.2081 0.5594 1.2708 1.3180 1.2588 0.4308 1.1897 1.4467 1.2318 0.4236 1.2416 1.3671 1.4217 0.9637 1.3807 1.2404 0.4444 0.4635 1.1594 0.0000 0.7076 0.4068 0.5455

[illegible]

|       |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 54.86 | 0.6890 | 0.6866 | 0.5963 | 1.3217 | 1.394  | 1.594  | 1.6419 | 1.3212 | 0.9488 | 0.8702 | 1.2384 | 0.0000 | 1.4006 | 1.2017 | 1.4088 | 1.4945 | 0.7048 | 1.6265 | 1.5689 | 1.5940 | 1.4754 | 1.6147 | 1.2700 | 1.6480 | 0.0000 | 0.6654 | 1.3839 | 1.6882 | 1.4779 | 1.5281 | 1.488  | 1.5965 | 1.3811 | 1.2430 | 1.2886 | 1.3480 | 0.7067 | 1.5005 | 1.6160 | 1.3309 | 0.5339 | 1.4787 | 1.6103 | 1.4686 | 0.5296 | 1.3318 | 1.6598 | 1.6636 | 1.3831 | 1.6461 | 1.5459 | 0.5736 | 0.5252 | 1.3622 | 0.0000 | 0.9327 | 0.5162 | 0.6674 | 0.6734 |
| 54.96 | 0.6655 | 0.6699 | 0.5963 | 1.3217 | 1.7734 | 1.5954 | 1.6562 | 1.3605 | 0.5400 | 0.8662 | 1.7584 | 0.0000 | 1.4007 | 1.2017 | 1.4463 | 1.4047 | 0.6641 | 1.6265 | 1.5267 | 1.6550 | 1.6348 | 1.6148 | 1.2170 | 1.5920 | 0.0000 | 0.6657 | 1.5046 | 1.7270 | 1.4616 | 1.4880 | 1.4733 | 1.5534 | 1.2950 | 1.2886 | 1.3400 | 0.6917 | 1.5615 | 1.6167 | 1.2729 | 0.5370 | 1.4642 | 1.6104 | 1.4686 | 0.5296 | 1.3318 | 1.6598 | 1.6636 | 1.3831 | 1.6461 | 1.5459 | 0.5736 | 0.5252 | 1.3622 | 0.0000 | 0.9327 | 0.5162 | 0.6674 | 0.6734 |        |

|        |        |        |        |       |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |       |        |        |        |        |
|--------|--------|--------|--------|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------|--------|--------|--------|--------|
| 73.152 | 0.7554 | 0.5983 | 0.5593 | 0.422 | 1.3864 | 1.5994 | 1.7232 | 1.3812 | 0.5782 | 0.9461 | 1.7169 | 0.0000 | 1.5236 | 1.5476 | 1.5985 | 1.5476 | 0.7315 | 1.6146 | 1.6338 | 1.5850 | 1.6354 | 1.6787 | 1.3168 | 1.6739 | 0.0000 | 0.6851 | 1.6549 | 1.7592 | 1.5417 | 1.5787 | 1.5225 | 1.6522 | 1.4657 | 1.4072 | 1.2782 | 1.7363 | 0.7448 | 1.6244 | 1.6797 | 1.3346 | 0.5638 | 1.5230 | 1.7703 | 1.2272 | 0.5754 | 1.3300 | 1.6673 | 1.7636 | 1.2294 | 1.5544 | 1.5988 | 0.5953 | 0.5736 | 1.427 | 0.0000 | 0.2887 | 0.5382 | 0.6993 |
|--------|--------|--------|--------|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------|--------|--------|--------|--------|

|      |        |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 9.44 | 0.7243 | 0.725 | 0.565 | 1.426 | 1.435 | 1.694 | 1.743 | 1.433 | 0.595 | 0.926 | 1.433 | 0.000 | 1.506 | 1.433 | 1.599 | 1.570 | 1.755 | 1.615 | 1.630 | 1.560 | 1.484 | 1.705 | 1.343 | 1.678 | 0.000 | 0.684 | 1.629 | 1.795 | 1.500 | 1.580 | 1.552 | 1.686 | 1.635 | 1.412 | 1.312 | 1.340 | 1.768 | 1.658 | 1.705 | 1.458 | 0.563 | 1.530 | 1.705 | 1.545 | 0.563 | 1.429 | 1.654 | 1.786 | 1.250 | 1.580 | 1.634 | 0.597 | 0.578 | 1.459 | 0.000 | 0.949 | 0.508 | 0.703 |
|------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|

D-19



D-21

D-21

| Location | 1      | 2      | 3      | 4      | 5      | 6      | 7      | 8      | 9      | 10     | 11     | 12     | 13     | 14     | 15     | 16     | 17     | 18     | 19     | 20     | 21     | 22     | 23     | 24     | 25     | 26     | 27     | 28     | 29     | 30     | 31     | 32     | 33     | 34     | 35     | 36     | 37     | 38     | 39     | 40     | 41     | 42     | 43     | 44     | 45     | 46     | 47     | 48     | 49     | 50     | 51     | 52     | 53     | 54     | 55     | 56     | 57     |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |
|----------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.0225   | 0.0256 | 0.0294 | 0.0336 | 0.0384 | 0.0432 | 0.0480 | 0.0528 | 0.0576 | 0.0624 | 0.0672 | 0.0720 | 0.0768 | 0.0816 | 0.0864 | 0.0912 | 0.0960 | 0.1008 | 0.1056 | 0.1104 | 0.1152 | 0.1200 | 0.1248 | 0.1296 | 0.1344 | 0.1392 | 0.1440 | 0.1488 | 0.1536 | 0.1584 | 0.1632 | 0.1680 | 0.1728 | 0.1776 | 0.1824 | 0.1872 | 0.1920 | 0.1968 | 0.2016 | 0.2064 | 0.2112 | 0.2160 | 0.2208 | 0.2256 | 0.2304 | 0.2352 | 0.2400 | 0.2448 | 0.2496 | 0.2544 | 0.2592 | 0.2640 | 0.2688 | 0.2736 | 0.2784 | 0.2832 | 0.2880 | 0.2928 | 0.2976 | 0.3024 | 0.3072 | 0.3120 | 0.3168 | 0.3216 | 0.3264 | 0.3312 | 0.3360 | 0.3408 | 0.3456 | 0.3504 | 0.3552 | 0.3600 | 0.3648 | 0.3696 | 0.3744 | 0.3792 | 0.3840 | 0.3888 | 0.3936 | 0.3984 | 0.4032 | 0.4080 | 0.4128 | 0.4176 | 0.4224 | 0.4272 | 0.4320 | 0.4368 | 0.4416 | 0.4464 | 0.4512 | 0.4560 | 0.4608 | 0.4656 | 0.4704 | 0.4752 | 0.4800 | 0.4848 | 0.4896 | 0.4944 | 0.4992 | 0.5040 | 0.5088 | 0.5136 | 0.5184 | 0.5232 | 0.5280 | 0.5328 | 0.5376 | 0.5424 | 0.5472 | 0.5520 | 0.5568 | 0.5616 | 0.5664 | 0.5712 | 0.5760 | 0.5808 | 0.5856 | 0.5904 | 0.5952 | 0.6000 | 0.6048 | 0.6096 | 0.6144 | 0.6192 | 0.6240 | 0.6288 | 0.6336 | 0.6384 | 0.6432 | 0.6480 | 0.6528 | 0.6576 | 0.6624 | 0.6672 | 0.6720 | 0.6768 | 0.6816 | 0.6864 | 0.6912 | 0.6960 | 0.7008 | 0.7056 | 0.7104 | 0.7152 | 0.7200 | 0.7248 | 0.7296 | 0.7344 | 0.7392 | 0.7440 | 0.7488 | 0.7536 | 0.7584 | 0.7632 | 0.7680 | 0.7728 | 0.7776 | 0.7824 | 0.7872 | 0.7920 | 0.7968 | 0.8016 | 0.8064 | 0.8112 | 0.8160 | 0.8208 | 0.8256 | 0.8304 | 0.8352 | 0.8400 | 0.8448 | 0.8496 | 0.8544 | 0.8592 | 0.8640 | 0.8688 | 0.8736 | 0.8784 | 0.8832 | 0.8880 | 0.8928 | 0.8976 | 0.9024 | 0.9072 | 0.9120 | 0.9168 | 0.9216 | 0.9264 | 0.9312 | 0.9360 | 0.9408 | 0.9456 | 0.9504 | 0.9552 | 0.9600 | 0.9648 | 0.9696 | 0.9744 | 0.9792 | 0.9840 | 0.9888 | 0.9936 | 0.9984 | 1.0032 | 1.0080 | 1.0128 | 1.0176 | 1.0224 | 1.0272 | 1.0320 | 1.0368 | 1.0416 | 1.0464 | 1.0512 | 1.0560 | 1.0608 | 1.0656 | 1.0704 | 1.0752 | 1.0800 | 1.0848 | 1.0896 | 1.0944 | 1.0992 | 1.1040 | 1.1088 | 1.1136 | 1.1184 | 1.1232 | 1.1280 | 1.1328 | 1.1376 | 1.1424 | 1.1472 | 1.1520 | 1.1568 | 1.1616 | 1.1664 | 1.1712 | 1.1760 | 1.1808 | 1.1856 | 1.1904 | 1.1952 | 1.2000 | 1.2048 | 1.2096 | 1.2144 | 1.2192 | 1.2240 | 1.2288 | 1.2336 | 1.2384 | 1.2432 | 1.2480 | 1.2528 | 1.2576 | 1.2624 | 1.2672 | 1.2720 | 1.2768 | 1.2816 | 1.2864 | 1.2912 | 1.2960 | 1.3008 | 1.3056 | 1.3104 | 1.3152 | 1.3200 | 1.3248 | 1.3296 | 1.3344 | 1.3392 | 1.3440 | 1.3488 | 1.3536 | 1.3584 | 1.3632 | 1.3680 | 1.3728 | 1.3776 | 1.3824 | 1.3872 | 1.3920 | 1.3968 | 1.4016 | 1.4064 | 1.4112 | 1.4160 | 1.4208 | 1.4256 | 1.4304 | 1.4352 | 1.4400 | 1.4448 | 1.4496 | 1.4544 | 1.4592 | 1.4640 | 1.4688 | 1.4736 | 1.4784 | 1.4832 | 1.4880 | 1.4928 | 1.4976 | 1.5024 | 1.5072 | 1.5120 | 1.5168 | 1.5216 | 1.5264 | 1.5312 | 1.5360 | 1.5408 | 1.5456 | 1.5504 | 1.5552 | 1.5600 | 1.5648 | 1.5696 | 1.5744 | 1.5792 | 1.5840 | 1.5888 | 1.5936 | 1.5984 | 1.6032 | 1.6080 | 1.6128 | 1.6176 |







[illegible]

D-25









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|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------|
| 4.39   | 1.74  | 2.65  | 3.61  | 4.59  | 5.58  | 6.56  | 7.54  | 8.52  | 9.50  | 10.48 | 11.46 | 12.44 | 13.42 | 14.40 | 15.38 | 16.36 | 17.34 | 18.32 | 19.30 | 20.28 | 21.26 | 22.24 | 23.22 | 24.20 | 25.18 | 26.16 | 27.14 | 28.12 | 29.10 | 30.08 | 31.06 | 32.04 | 33.02 | 34.00 | 35.00 | 36.00 | 37.00 | 38.00 | 39.00 | 40.00 | 41.00 | 42.00 | 43.00 | 44.00 | 45.00 | 46.00 | 47.00 | 48.00 | 49.00 | 50.00 | 51.00 | 52.00 | 53.00 | 54.00 | 55.00 | 56.00 | 57.00 | 58.00 | 59.00 | 60.00 |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |       |
| 0.2389 | 0.265 | 0.294 | 0.321 | 0.352 | 0.382 | 0.412 | 0.442 | 0.472 | 0.501 | 0.531 | 0.561 | 0.591 | 0.621 | 0.651 | 0.681 | 0.711 | 0.741 | 0.771 | 0.801 | 0.831 | 0.861 | 0.891 | 0.921 | 0.951 | 0.981 | 1.011 | 1.041 | 1.071 | 1.101 | 1.131 | 1.161 | 1.191 | 1.221 | 1.251 | 1.281 | 1.311 | 1.341 | 1.371 | 1.401 | 1.431 | 1.461 | 1.491 | 1.521 | 1.551 | 1.581 | 1.611 | 1.641 | 1.671 | 1.701 | 1.731 | 1.761 | 1.791 | 1.821 | 1.851 | 1.881 | 1.911 | 1.941 | 1.971 | 2.001 | 2.031 | 2.061 | 2.091 | 2.121 | 2.151 | 2.181 | 2.211 | 2.241 | 2.271 | 2.301 | 2.331 | 2.361 | 2.391 | 2.421 | 2.451 | 2.481 | 2.511 | 2.541 | 2.571 | 2.601 | 2.631 | 2.661 | 2.691 | 2.721 | 2.751 | 2.781 | 2.811 | 2.841 | 2.871 | 2.901 | 2.931 | 2.961 | 2.991 | 3.021 | 3.051 | 3.081 | 3.111 | 3.141 | 3.171 | 3.201 | 3.231 | 3.261 | 3.291 | 3.321 | 3.351 | 3.381 | 3.411 | 3.441 | 3.471 | 3.501 | 3.531 | 3.561 | 3.591 | 3.621 | 3.651 | 3.681 | 3.711 | 3.741 | 3.771 | 3.801 | 3.831 | 3.861 | 3.891 | 3.921 | 3.951 | 3.981 | 4.011 | 4.041 | 4.071 | 4.101 | 4.131 | 4.161 | 4.191 | 4.221 | 4.251 | 4.281 | 4.311 | 4.341 | 4.371 | 4.401 | 4.431 | 4.461 | 4.491 | 4.521 | 4.551 | 4.581 | 4.611 | 4.641 | 4.671 | 4.701 | 4.731 | 4.761 | 4.791 | 4.821 | 4.851 | 4.881 | 4.911 | 4.941 | 4.971 | 5.001 | 5.031 | 5.061 | 5.091 | 5.121 | 5.151 | 5.181 | 5.211 | 5.241 | 5.271 | 5.301 | 5.331 | 5.361 | 5.391 | 5.421 | 5.451 | 5.481 | 5.511 | 5.541 | 5.571 | 5.601 | 5.631 | 5.661 | 5.691 | 5.721 | 5.751 | 5.781 | 5.811 | 5.841 | 5.871 | 5.901 | 5.931 | 5.961 | 5.991 | 6.021 | 6.051 | 6.081 | 6.111 | 6.141 | 6.171 | 6.201 | 6.231 | 6.261 | 6.291 | 6.321 | 6.351 | 6.381 | 6.411 | 6.441 | 6.471 | 6.501 | 6.531 | 6.561 | 6.591 | 6.621 | 6.651 | 6.681 | 6.711 | 6.741 | 6.771 | 6.801 | 6.831 | 6.861 | 6.891 | 6.921 | 6.951 | 6.981 | 7.011 | 7.041 | 7.071 | 7.101 | 7.131 | 7.161 | 7.191 | 7.221 | 7.251 | 7.281 | 7.311 | 7.341 | 7.371 | 7.401 | 7.431 | 7.461 | 7.491 | 7.521 | 7.551 | 7.581 | 7.611 | 7.641 | 7.671 | 7.701 | 7.731 | 7.761 | 7.791 | 7.821 | 7.851 | 7.881 | 7.911 | 7.941 | 7.971 | 8.001 | 8.031 | 8.061 | 8.091 | 8.121 | 8.151 | 8.181 | 8.211 | 8.241 | 8.271 | 8.301 | 8.331 | 8.361 | 8.391 | 8.421 | 8.451 | 8.481 | 8.511 | 8.541 | 8.571 | 8.601 | 8.631 | 8.661 | 8.691 | 8.721 | 8.751 | 8.781 | 8.811 | 8.841 | 8.871 | 8.901 | 8.931 | 8.961 | 8.991 | 9.021 | 9.051 | 9.081 | 9.111 | 9.141 | 9.171 | 9.201 | 9.231 | 9.261 | 9.291 | 9.321 | 9.351 | 9.381 | 9.411 | 9.441 | 9.471 | 9.501 | 9.531 | 9.561 | 9.591 | 9.621 | 9.651 | 9.681 | 9.711 | 9.741 | 9.771 | 9.801 | 9.831 | 9.861 | 9.891 | 9.921 | 9.951 | 9.981 | 10.011 | 10.041 | 10.071 | 10.101 | 10.131 | 10.161 | 10.191 | 10.221 | 10.251 | 10.281 | 10.311 | 10.341 | 10.371 | 10.401 | 10.431 | 10.461 | 10.491 | 10.521 | 10.551 | 10.581 | 10.61 |























